

Time Series Analysis of GDP, Exports, and Population in the Central African Republic (1960–2017)

STA 137 Final Project Report

Caitlyn Koyabu

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University of California, Davis
Instructor: Professor Ding

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1 Introduction

The Central African Republic (CAR) [1] is a resource-rich country whose economy depends on natural resource exports, such as timber and diamonds. These exports play a crucial role in generating foreign income and supporting economic stability. Alongside exports, GDP and population dynamics are key indicators of the country's economic health, influencing and reflecting overall development.

This study aims to analyze the time series of GDP, exports, and population to uncover underlying trends, patterns, and potential relationships among these variables. By applying time series analysis, we can better understand how changes in exports relate to economic growth and population dynamics, providing valuable insights for policymakers and stakeholders seeking to strengthen the CAR's economic planning and resource management.

2 Exploratory Data Analysis

2.1 Raw Data Visualization

We first examine the raw GDP, Exports, and Population series (1960–2017) to identify trends and volatility. Since the series are annual, no seasonal patterns are present.

2.1.1 GDP

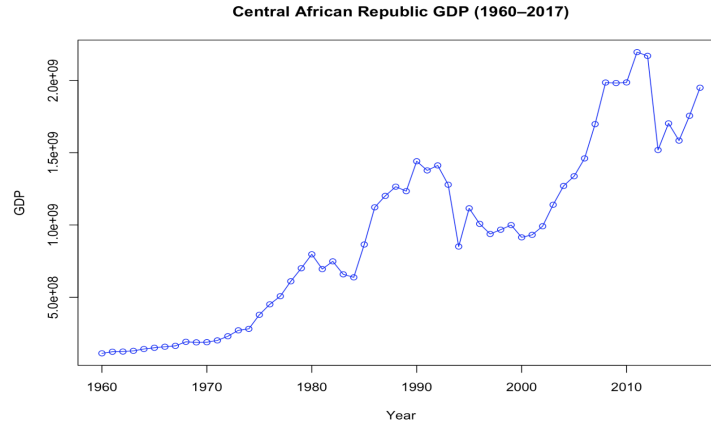


Figure 1: GDP Time Series, 1960–2017

The GDP series exhibits a clear upward trend with fluctuations in the 1980s-1990s and early 2010s, indicating periods of instability. Overall, the series is non-stationary due to its persistent growth trend and varying variance over time.

2.1.2 Exports



Figure 2: Exports Time Series, 1960–2017

Exports are more volatile, showing sharp spikes and drops. The trend is slightly downward, with structural breaks suggesting sensitivity to external shocks. This series is also non-stationary.

2.1.3 Population

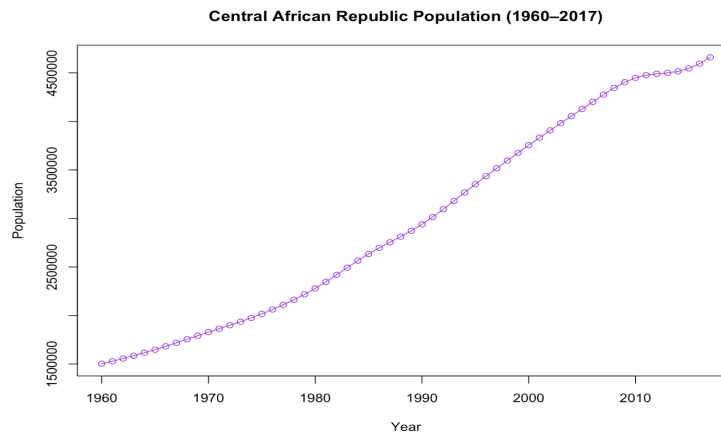
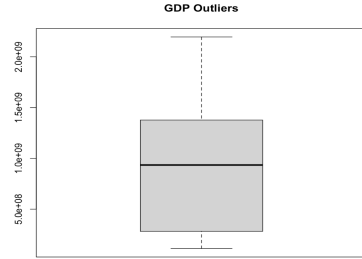


Figure 3: Population Time Series, 1960–2017

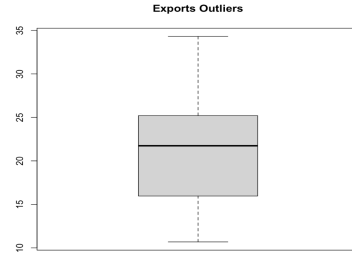
The Population series grows smoothly, with no abrupt changes and a consistent upwards trend, indicating non-stationarity.

2.2 Outliers Assessment

Boxplots and standardized z -scores were used to detect extreme values.



(a) GDP: Distribution and potential outliers from 1960–2017.



(b) Exports: Observed variation and extreme values over time.



(c) Population: Boxplot showing population trends and anomalies.

Figure 4: Boxplots for GDP, Exports, and Population to identify potential outliers in the Central African Republic (1960–2017).

No statistical outliers were detected in any series, suggesting the observed fluctuations represent true economic movements rather than data irregularities.

2.3 Raw ACF and PACF:

We examine autocorrelation (ACF) and partial autocorrelation (PACF) plots to evaluate whether transformations are needed.

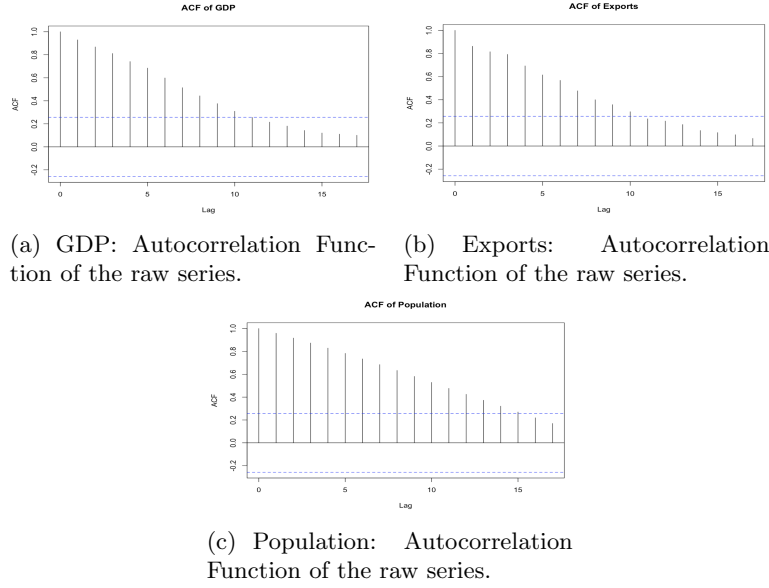


Figure 5: ACF plots for GDP, Exports, and Population raw series (1960–2017) showing correlation patterns across lags.

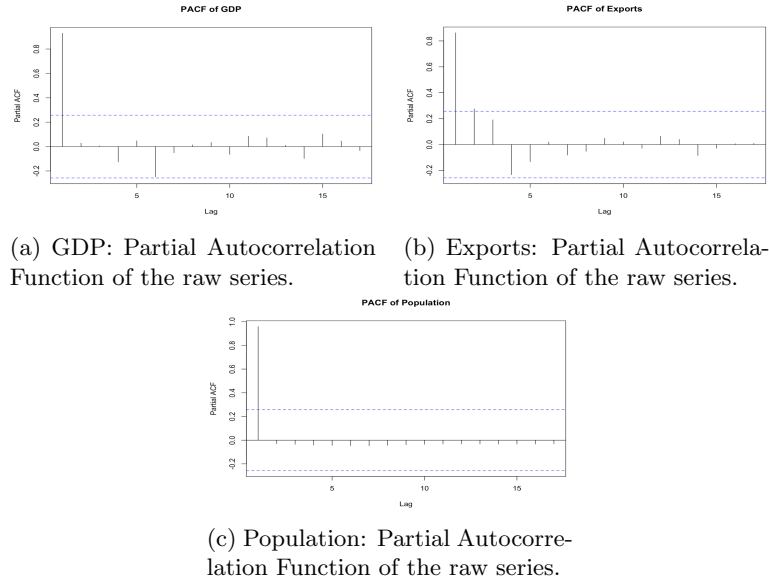


Figure 6: PACF plots for GDP, Exports, and Population raw series (1960–2017) showing lag-specific correlation after controlling for intermediate lags.

Our ACF and PACF looks good, but we want to see if we can do better transforming our data.

3 Methodology

3.1 Data Transformation

All three series were non-stationary with clear trends, so GDP, Exports, and Population were log-transformed to stabilize variance and interpret changes in percentage terms, consistent with economic growth. Box–Cox results supported the log transformation as near-optimal.

3.1.1 Log-Transformed Series ACF/PACF

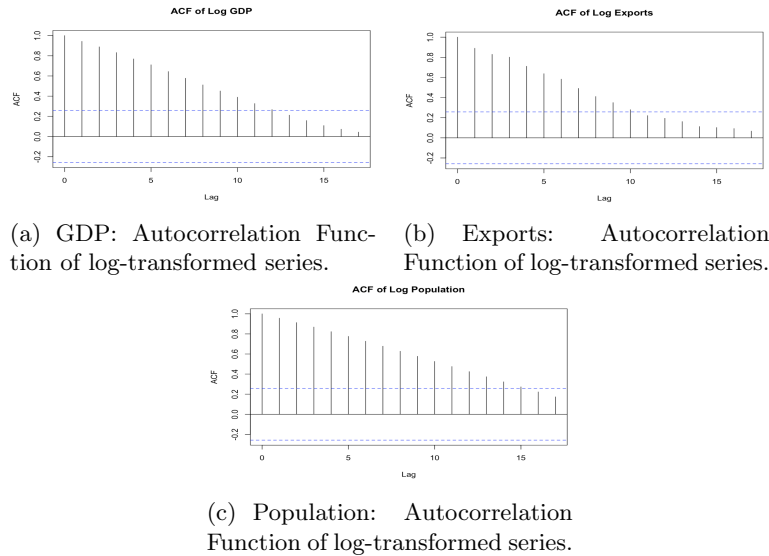


Figure 7: ACF plots for log-transformed GDP, Exports, and Population series (1960–2017), showing correlation patterns across lags.

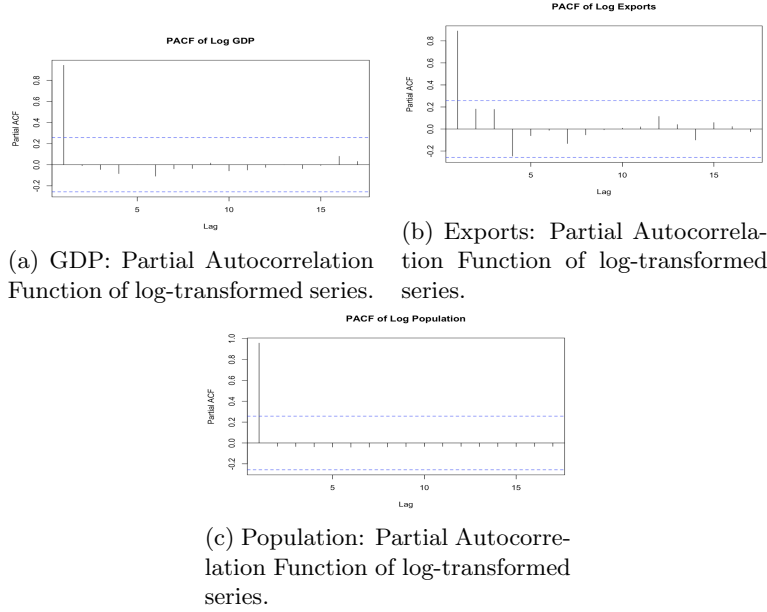


Figure 8: PACF plots for log-transformed GDP, Exports, and Population series (1960–2017), highlighting lag-specific correlation after controlling for intermediate lags.

The log transformation improved stationarity patterns in the ACF and PACF plots.

3.2 Differencing for Stationarity

After applying log transformation, GDP and Population exhibit smooth upward trends, while Exports remain volatile with a gradual long-term decline. Because ARIMA modeling requires stationarity, we apply differencing to remove these deterministic trends. The differenced series are then re-examined visually and through ACF/PACF plots to verify that stationarity has been achieved before fitting ARIMA models.

First-order differencing was applied to remove deterministic trends before ARIMA modeling. Visual inspection and ACF/PACF plots confirm improved stationarity.

3.2.1 Differenced Series ACF/PACF

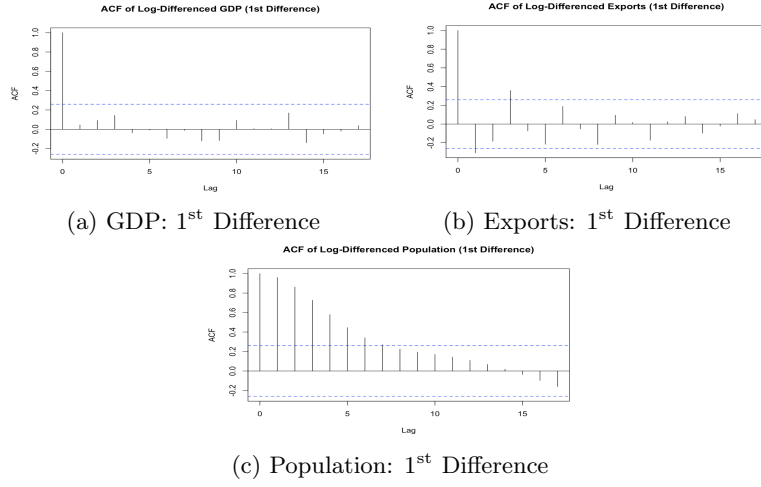


Figure 9: **ACF of First-Differenced Log-Transformed Series (1960–2017)**. These plots show autocorrelation patterns after log-differencing, helping identify AR terms for ARIMA modeling.

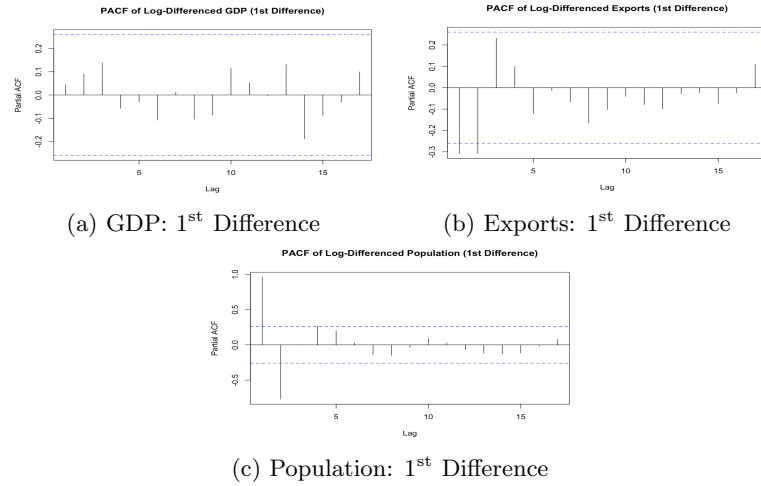


Figure 10: **PACF of First-Differenced Log-Transformed Series (1960–2017)**. These plots highlight significant lags for AR terms in ARIMA modeling after differencing.

After log transformation and first-order differencing, the univariate time series for all three variables appear more stable, with reduced trends and more con-

sistent variance. The ACF and PACF plots also indicate improved stationarity, supporting the use of ARIMA modeling on these transformed series. Because univariate ARIMA models only require each series to be individually stationary, the first-differenced log series were appropriate and sufficient for the univariate analysis based on our ACF and PACF plots.

4 Multivariate Analysis

Although the primary objective of this project is to model each variable separately using univariate ARIMA methods, economic variables rarely evolve independently. To explore whether joint dynamics might inform or supplement the univariate results, we extend the analysis by examining cross-correlation functions (CCFs) among GDP, Exports, and Population.

4.1 Cross Correlation Function (CCF)

The CCF shows whether movements in one variable lead or lag another, capturing interrelationships that provide insight into potential economic linkages within the Central African Republic. All series were log-transformed and differenced to ensure stationarity before computing the CCF. It is also important to note that significant correlations at lag 0 suggest contemporaneous relationships rather than predictive power.

4.1.1 GDP vs. Exports

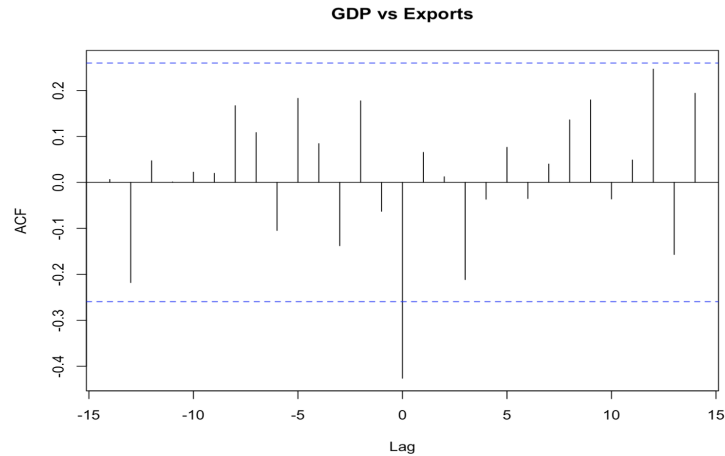


Figure 11: GDP vs. Exports Correlation

A strong negative peak at lag 0 indicates that GDP and Exports move in opposite directions in the same year. This contemporaneous correlation suggests

that when GDP increases, Exports tend to decrease and vice versa, but it does not imply that past GDP can predict future Exports.

4.1.2 GDP vs. Population

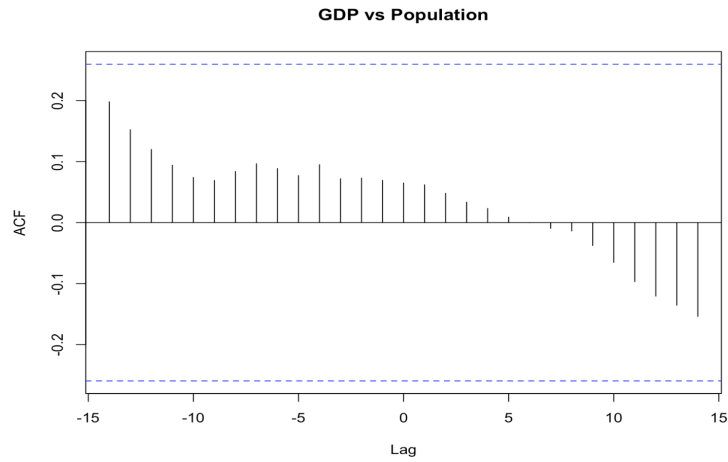


Figure 12: GDP vs. Population Correlation

No peaks exceed the significance bounds, indicating no significant correlation between GDP and Population at any lag. After log transformation and differencing, GDP fluctuations appear largely independent of population changes over time.

4.1.3 Exports vs. Population

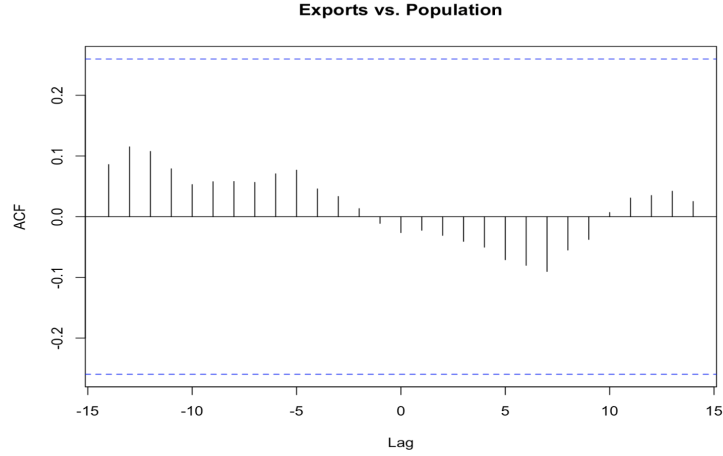


Figure 13: Exports vs. Population Correlation

Similarly, there is no significant correlation between Exports and Population at any lag. Export performance seems driven by factors other than population growth, such as global demand or resource production, rather than exhibiting a contemporaneous relationship with population changes.

5 Model Selection

5.1 Candidate Univariate Models

After applying logarithmic transformation and differencing to achieve stationarity, we examine the ACF and PACF plots to identify potential ARIMA models for each series. These plots provide guidance on selecting appropriate autoregressive p and moving average q orders.

From Figure 9 and Figure 10

- **GDP:** The log-transformed, first-differenced series shows a clear cutoff at lag 1 in the ACF, suggesting an MA model is more appropriate than an AR model.
- **Exports:** Cutoffs at lag 2 in both the ACF and PACF indicate that MA, AR, or ARMA models could be suitable.
- **Population:** A clear cutoff at lag 2 in the PACF suggests an AR model is reasonable.

These candidate models will be further evaluated using information criteria (AIC/BIC) and diagnostic checks to select the best-fitting ARIMA model for each series.

5.2 Multivariate VAR Model

Before estimating a Vector Autoregressive (VAR) [8] model, the three series must be jointly stationary. Stationarity was formally assessed using both the Augmented Dickey–Fuller (ADF) [9] and Kwiatkowski–Phillips–Schmidt–Shin (KPSS) [10] tests, using $\alpha = 0.05$.

Augmented Dickey–Fuller (ADF) Test

H_0 : The series has a unit root (non-stationary) H_A : The series is stationary

The results for our differenced log-transformed series are as follows with $\alpha = 0.05$:

- Log-GDP (second difference): p-value = 0.01 $\Rightarrow$ stationary
- Log-Exports (second difference): p-value = 0.01 $\Rightarrow$ stationary
- Log-Population (third difference): p-value = 0.01 $\Rightarrow$ stationary

Kwiatkowski–Phillips–Schmidt–Shin (KPSS) Test

H_0 : The series is stationary H_A : The series has a unit root (non-stationary)

For the log-transformed Population series, the KPSS test results were:

- Raw series: KPSS = 1.545, p = 0.01 $\Rightarrow$ non-stationary
- First difference: KPSS = 0.598, p = 0.023 $\Rightarrow$ borderline stationary
- Second difference: KPSS = 0.144, p = 0.1 $\Rightarrow$ stationary

Although the ADF test suggested that Population required a third difference, the KPSS results and residual diagnostics indicated that applying a third difference would over-difference the series. Thus, while the univariate models use first differences, the multivariate setting requires a different approach.

For the VAR model, we use the second-differenced log series for all three variables. This level of differencing achieves joint stationarity across GDP, Exports, and Population while avoiding unnecessary loss of long-term structure.

5.3 Selecting Univariate Models using AIC/BIC

Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) [2] are common methods for comparing the fit of time series models while penalizing model complexity. BIC penalizes model complexity more heavily than AIC, generally favoring simpler models to avoid overfitting. Therefore, we will chose our model using BIC to prevent overfitting and ensure more parsimonious models. We will also use an alpha of 0.1 to check if coefficients are significant

GDP (log, differenced once) Two candidate models were considered: ARIMA(1,1,0) and ARIMA(0,1,2). The first model had a BIC of -57.52 , while the second had a BIC of -54.35 . Based on the lower BIC, ARIMA(1,1,0) was selected as the best-fitting model.

$$\text{AR}(1) = 0.1642 \quad (\text{s.e.} = 0.1306, p\text{-value} = 0.2087)$$

Substituting into the AR(1) equation gives:

$$\begin{aligned} \Delta \log(\text{GDP})_t &= \phi_1 \Delta \log(\text{GDP})_{t-1} + W_t, \\ W_t &\sim N(0, \sigma_{\text{GDP}}^2) \end{aligned}$$

- $\phi_1 = 0.1642$ (not statistically significant)
- $\sigma_{\text{GDP}}^2 = 0.01885$

Although the AR(1) coefficient is not statistically significant with $\alpha = 0.10$, non-significant ARIMA coefficients may be retained for forecasting purposes to improve predictive accuracy [3]. Thus, the final ARIMA(1,1,0) model for GDP is:

$$\begin{aligned} \Delta \log(\text{GDP})_t &= 0.1642 \Delta \log(\text{GDP})_{t-1} + W_t, \\ W_t &\sim N(0, 0.01885) \end{aligned}$$

Exports (log, differenced once) Two candidate models were considered: ARIMA(0,1,2) and ARIMA(2,1,0). The ARIMA(0,1,2) model had a BIC of -66.79 , while the ARIMA(2,1,0) had a lower BIC of -69.71 . Thus, ARIMA(2,1,0) was selected as the best-fitting model.

$$\begin{aligned} \text{AR}(1) &= -0.3816 \quad (\text{s.e.} = 0.1269, p\text{-value} = 0.00263), \\ \text{AR}(2) &= -0.2829 \quad (\text{s.e.} = 0.1255, p\text{-value} = 0.02415) \end{aligned}$$

Both autoregressive coefficients are statistically significant when $\alpha = 0.10$. The fitted AR(2) model for the differenced log Exports is:

$$\begin{aligned} \Delta \log(\text{Exports})_t &= \phi_1 \Delta \log(\text{Exports})_{t-1} + \phi_2 \Delta \log(\text{Exports})_{t-2} + W_t, \\ W_t &\sim N(0, \sigma_{\text{Exports}}^2) \end{aligned}$$

- $\phi_1 = -0.3816$
- $\phi_2 = -0.2829$
- $\sigma_{\text{Exports}}^2 = 0.01437$

Substituting the coefficients produces the final ARIMA(2,1,0) model:

$$\Delta \log(Exports)_t = -0.3816 \Delta \log(Exports)_{t-1} - 0.2829 \Delta \log(Exports)_{t-2} + W_t, \\ W_t \sim N(0, 0.01437)$$

Population (log, differenced once) Two candidate models were considered: ARIMA(2,1,0) and ARIMA(1,1,0). The ARIMA(2,1,0) model had a much lower BIC (-636.64) compared to ARIMA(1,1,0) (-559.78), so ARIMA(2,1,0) was selected.

$$\text{AR}(1) = 1.6583 \quad (\text{s.e.} = 0.0561, p\text{-value} < 0.001), \\ \text{AR}(2) = -0.9190 \quad (\text{s.e.} = 0.0566, p\text{-value} < 0.001)$$

At the $\alpha = 0.10$ significance level, both autoregressive coefficients are highly significant.

The fitted AR(2) model for the differenced log Population series is:

$$\Delta \log(\text{Population})_t = \phi_1 \Delta \log(\text{Population})_{t-1} + \phi_2 \Delta \log(\text{Population})_{t-2} + W_t, \\ W_t \sim N(0, \sigma_{\text{Population}}^2)$$

- $\phi_1 = 1.6583$
- $\phi_2 = -0.9190$
- $\sigma_{\text{Population}}^2 = 4.275 \times 10^6$

Substituting these coefficients, the final fitted model is:

$$\Delta \log(\text{Population})_t = 1.6583 \Delta \log(\text{Population})_{t-1} - 0.9190 \Delta \log(\text{Population})_{t-2} + W_t, \\ W_t \sim N(0, 4.275 \times 10^6)$$

5.4 Selecting Multivariate Models using AIC/BIC

Multivariate time series modeling differs from the univariate case because we cannot rely on ACF and PACF plots to choose lag orders. Instead, we use the `VARselect` function, which evaluates candidate models using the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC) [2]. Both criteria balance model fit and complexity, but BIC imposes a stronger penalty on larger models. Therefore, we select the BIC criterion to avoid overfitting and favor a simpler, more parsimonious model.

In our analysis, AIC suggested a model with $p = 10$ lags, whereas BIC selected a more concise model with $p = 2$ lags. Since BIC helps reduce the risk of overfitting and generally performs better in identifying the true lag order in VAR models, we selected a VAR(2) specification for GDP, Exports, and Population.

Significance and Model Structure. In multivariate VAR modeling, including all lagged predictors is often beneficial for forecasting, even when some coefficients are not statistically significant. This helps capture the joint dynamics of the system and improves predictive accuracy. Below, we summarize the statistically significant coefficients and present the full fitted equations for interpretability.

5.4.1 GDP Equation

Significant coefficients:

$$\begin{aligned} \text{GDP}_{t-1} &: -0.6874 \quad (p < 0.001), \\ \text{GDP}_{t-2} &: -0.2938 \quad (p = 0.0671). \end{aligned}$$

Full fitted VAR(2) equation for GDP:

$$\begin{aligned} \text{GDP}_t = & -0.6874 \text{GDP}_{t-1} - 0.2938 \text{GDP}_{t-2} + 0.0891 \text{Exports}_{t-1} + 0.1638 \text{Exports}_{t-2} \\ & + 2.7574 \text{Population}_{t-1} + 3.1328 \text{Population}_{t-2} + 0.0028 + u_t^{(GDP)} \end{aligned}$$

where $u_t^{(GDP)}$ is the residual (error term) for the GDP equation.

Interpretation. GDP dynamics are primarily driven by its own past values. Lagged Exports and Population coefficients are not statistically significant, but are retained for forecasting purposes.

5.4.2 Exports Equation

Significant coefficients:

$$\begin{aligned} \text{Exports}_{t-1} &: -0.9894 \quad (p < 0.001), \\ \text{Exports}_{t-2} &: -0.6775 \quad (p < 0.001), \\ \text{GDP}_{t-1} &: -0.2227 \quad (p = 0.117, \text{mildly influential}). \end{aligned}$$

Full fitted VAR(2) equation for Exports:

$$\begin{aligned} \text{Exports}_t = & -0.2227 \text{GDP}_{t-1} - 0.0716 \text{GDP}_{t-2} - 0.9894 \text{Exports}_{t-1} - 0.6775 \text{Exports}_{t-2} \\ & - 0.2310 \text{Population}_{t-1} + 0.2181 \text{Population}_{t-2} - 0.00008 + u_t^{(Exports)} \end{aligned}$$

where $u_t^{(Exports)}$ is the residual for the Exports equation.

Interpretation. Exports are strongly influenced by their own past values, with GDP providing a mild negative lag effect. Population lags do not have significant predictive power.

5.4.3 Population Equation

Significant coefficients:

$$\begin{aligned}\text{Population}_{t-1} &: 1.705 \quad (p < 0.001), \\ \text{Population}_{t-2} &: -0.975 \quad (p < 0.001).\end{aligned}$$

Full fitted VAR(2) equation for Population:

$$\begin{aligned}\text{Population}_t = & 1.705 \text{Population}_{t-1} - 0.975 \text{Population}_{t-2} - 0.000191 \text{GDP}_{t-1} + 0.000065 \text{GDP}_{t-2} \\ & + 0.000157 \text{Exports}_{t-1} + 0.000174 \text{Exports}_{t-2} - 0.000042 + u_t^{(Pop)}\end{aligned}$$

where $u_t^{(Pop)}$ is the residual for the Population equation.

Interpretation. Population evolves almost entirely as an autoregressive process. Lagged GDP and Exports have little to no influence on Population dynamics.

6 Diagnostics

6.1 Univariate Model Diagnostics

Before selecting the final ARIMA models, we evaluated the model diagnostics for each candidate model using the residual time series plot, ACF of residuals, Q-Q plot, and the Ljung-Box plot. A well-specified model should produce residuals that behave like white noise: no visible structure, no significant autocorrelations, and approximately normal distribution.

6.1.1 GDP: ARIMA(1,1,0)

For the ARIMA(1,1,0) model, the residual plot shows random fluctuations around zero with no visible trend or pattern. The ACF of the residuals displays no significant spikes, indicating that all autocorrelations fall within the 95% bounds. The Q-Q plot shows that most points fall along the reference line, with only mild deviations in the tails, which are not severe enough to indicate strong non-normality. Additionally, the p-values from the Ljung-Box test are all above the 0.05 threshold (blue dashed line), indicating no evidence of remaining autocorrelation in the residuals.

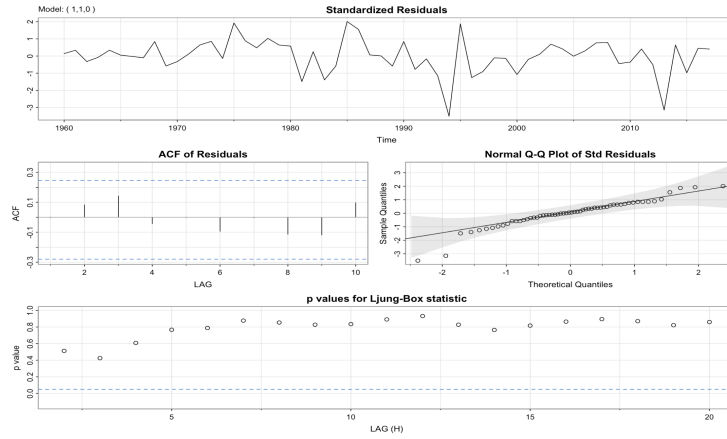


Figure 14: Residuals of the GDP ARIMA(1,1,0) model.

ARIMA(0,2,2) passes all diagnostic checks, so it is selected as the final model for GDP.

6.1.2 Exports: ARIMA(2,1,0)

The residual plot shows random fluctuations around zero, and the ACF indicates no significant autocorrelations. The Q-Q plot shows points lying close to the reference line with only minor tail deviations. The p-values from the Ljung-Box plot are all above the 0.05 threshold (blue dashed line), indicating no evidence of remaining autocorrelation in the residuals.

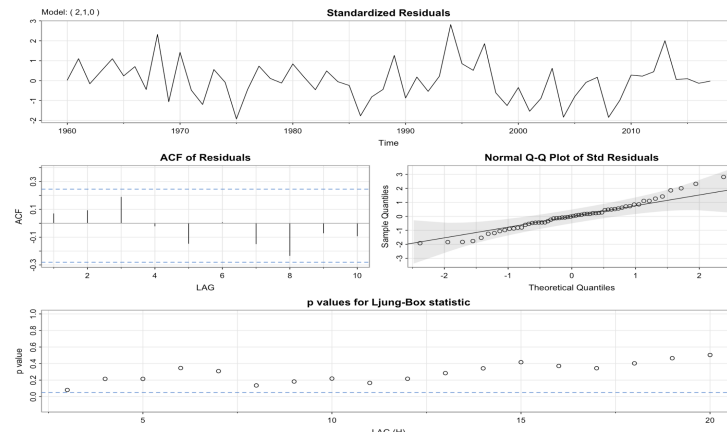


Figure 15: Residuals of the Exports ARIMA(2,1,0) model.

Thus, ARIMA(2,1,0) passes all diagnostic tests and is selected as the final model for Exports.

6.1.3 Population: ARIMA(2,1,0)

For the ARIMA(2,1,0) model, the residual plot displays a sharp drop at the very beginning of the sample, but after 1962 the residuals fluctuate randomly around zero with stable variance. The ACF plot shows one marginally significant spike at lag 1, but all remaining autocorrelations lie within the 95% bounds. The Q-Q plot shows that nearly all points lie on the reference line, with only one outlier in the upper tail. The p-values from the Ljung-Box plot are all above the 0.05 threshold (blue dashed line), indicating no evidence of remaining autocorrelation in the residuals.

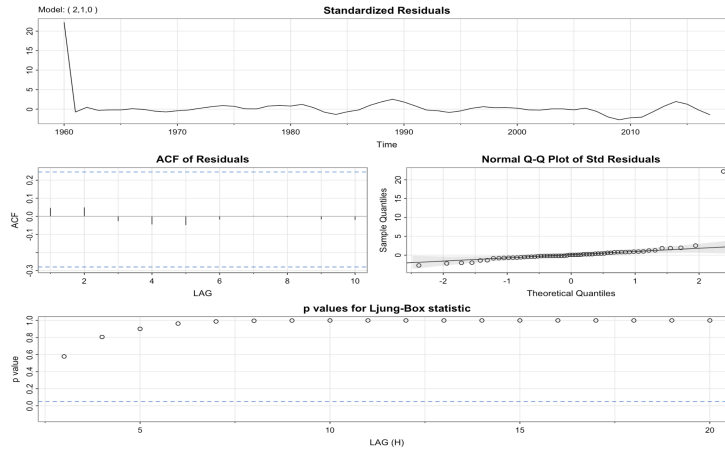


Figure 16: Residuals of the Population ARIMA(2,1,0) model.

Thus, ARIMA(2,1,0) passes all diagnostic tests and is selected as the final model for Population.

6.2 Multivariate Model Diagnostics

After estimating the VAR(2) model, we evaluated the model diagnostics. A good VAR model should show no remaining serial correlation and should have approximately normally distributed residuals.

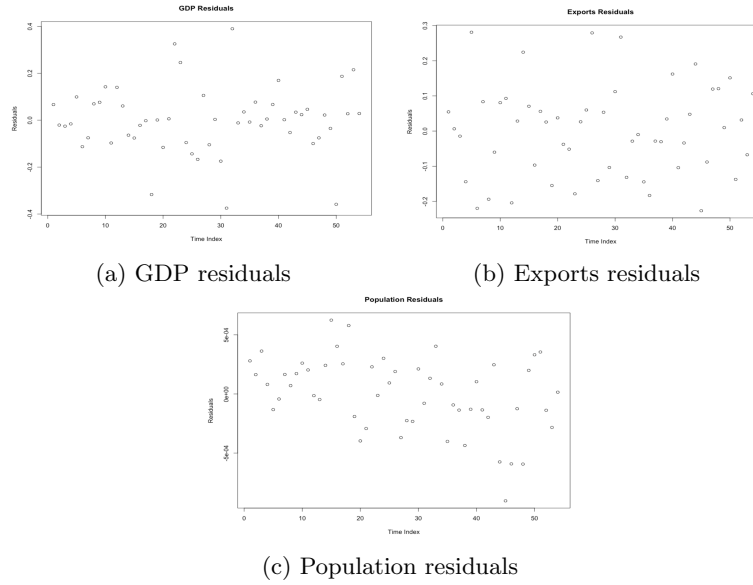


Figure 17: **Residual plots for the VAR(2) model (1960–2017).** These plots show residual patterns for each variable, helping assess model fit and identify potential autocorrelation or heteroskedasticity issues.

Portmanteau Test for Serial Correlation The multivariate Ljung-Box (Portmanteau) test checks for remaining autocorrelation in the residuals:

$$H_0 : \text{No autocorrelation in the residuals} \quad H_A : \text{Residuals exhibit autocorrelation}$$

Since the p-value (0.5712) is above 0.05, we fail to reject the null hypothesis. This indicates that the residuals show no significant autocorrelation, meaning the VAR(2) model has captured the dynamic structure of the series.

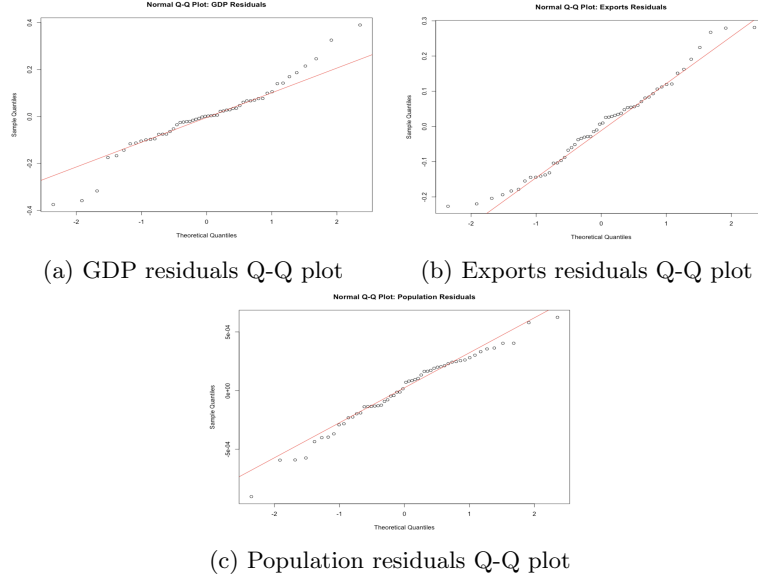


Figure 18: **Normal Q-Q plots for the residuals of the VAR(2) model (1960–2017)**. These plots help assess whether the residuals of each variable are approximately normally distributed, which is an assumption of VAR modeling.

Multivariate Jarque–Bera Test for Normality This test evaluates whether residuals are normally distributed:

$$H_0 : \text{Residuals are multivariate normal} \quad H_A : \text{Residuals are not multivariate normal}$$

The p-value (0.2887) is above 0.05, so we fail to reject the null. This suggests that the residuals do not significantly deviate from multivariate normality.

Conclusion Overall, the residual diagnostics indicate that the VAR(2) model is well-specified: residuals show no significant autocorrelation and approximately multivariate normal.

7 Multivariate Impulse Response Function (IRF)

Impulse Response Functions [\[11\]](#) trace the effect of a one-time shock to one variable on the current and future values of all variables in the VAR model. We compute orthogonalized IRFs for each variable in the VAR(2) model to understand the dynamic interactions between GDP, Exports, and Population.

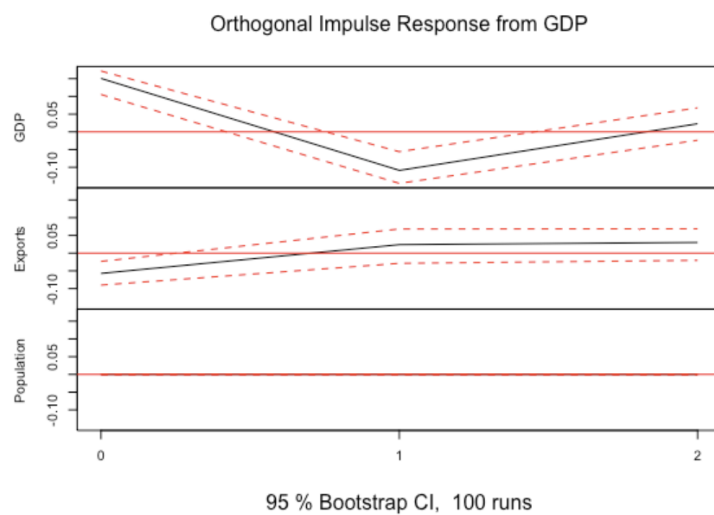


Figure 19: Orthogonalized Impulse Response Functions from GDP for the VAR(2) model over 2 years.

7.1 Response to a GDP Shock

- **GDP:** Initially increases by 0.15, then decreases to -0.10 by the end of year 1. In year 2, it starts at -0.10 and rises slightly to 0.02.
- **Exports:** Starts at -0.05, increases to 0.02 by year 1, and remains stable at 0.02 in year 2.
- **Population:** Remains largely unaffected by GDP shocks.

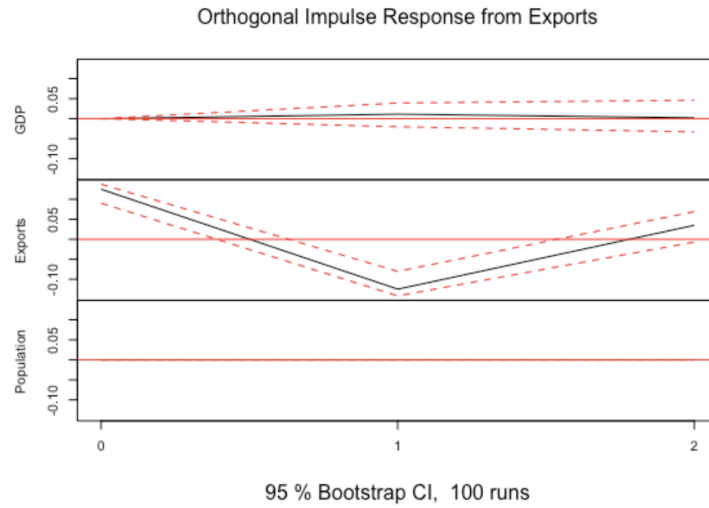


Figure 20: Orthogonalized Impulse Response Functions from Exports for the VAR(2) model over 2 years.

7.2 Response to an Exports Shock

- **GDP:** Minimal effect, starting at 0, rising slightly to 0.01, then returning to 0 by year 2.
- **Exports:** Begins at 0.12, decreases to -0.12 by year 1, and partially recovers to 0.05 by year 2.
- **Population:** No significant response.

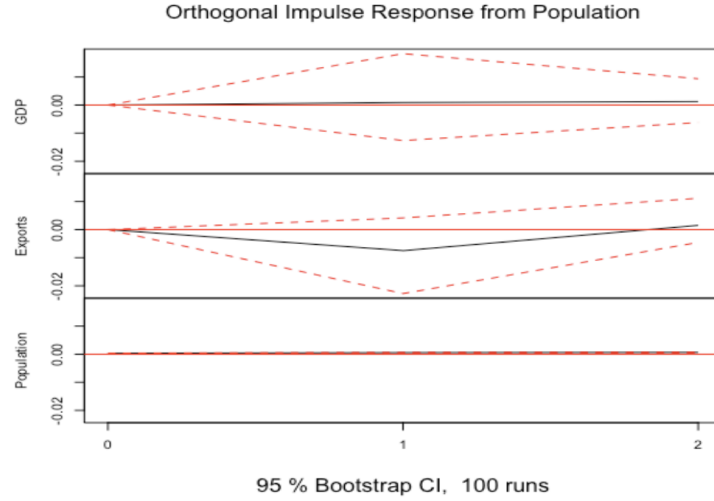


Figure 21: Orthogonalized Impulse Response Functions from Population for the VAR(2) model over 2 years.

7.3 Response to a Population Shock

- All variables show negligible response. GDP, Exports, and Population remain approximately zero over the two-year horizon.

Conclusion: The IRFs suggest that short-term GDP shocks have the strongest impact on GDP itself, with minor effects on Exports and negligible effects on Population. Exports shocks mainly influence Exports and only mildly affect GDP. Population appears mostly independent in this system over a two-year horizon, consistent with the VAR(2) estimation results where Population mainly depends on its own past values and not on GDP or Exports.

8 Forecasting

We performed 2-year forecasts using the ARIMA models selected for each series. The forecasts include 95% confidence intervals, capturing the uncertainty around the predictions.

8.1 Univariate GDP Forecast: 2 Year Ahead (2018-2019)

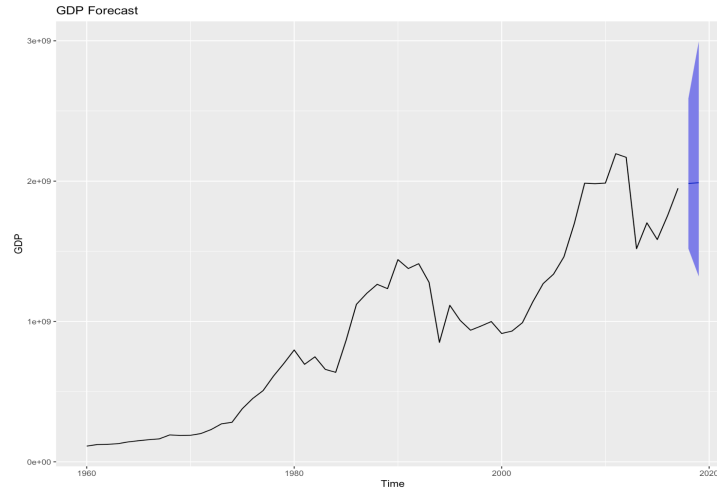


Figure 22: GDP Forecast for 2018–2019 with 95% Confidence Intervals

Table 1: GDP Forecast for 2018–2019 (ARIMA(1,1,0)) with 95% Confidence Intervals

Year	Point Forecast	Lo 95%	Hi 95%
2018	1,983,237,590	1,519,009,254	2,589,339,944
2019	1,988,846,394	1,320,875,915	2,994,611,328

The forecasts indicate a slight increase in GDP over the two-year period. The wide confidence intervals reflect the uncertainty inherent in long-term forecasting. According to the World Bank Group [\[4\]](#), the actual GDP in 2018 and 2019 was approximately 2.22 billion, showing that our forecast was reasonably accurate and fell within the confidence intervals.

8.2 Univariate Exports Forecast: 2 Year Ahead (2018-2019)

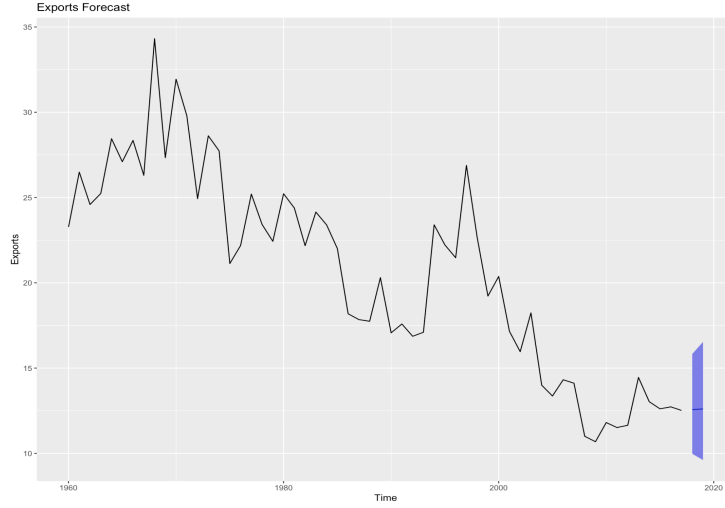


Figure 23: Exports Forecast for 2018–2019 with 95% Confidence Intervals

Table 2: Exports Forecast for 2018–2019 (ARIMA(2,1,0)) with 95% Confidence Intervals

Year	Point Forecast	Lo 95%	Hi 95%
2018	12.56527	9.97549	15.82738
2019	12.60671	9.61049	16.53704

The point forecasts for Exports are in terms of per capita values, showing a slight increase from 12.57 in 2018 to 12.61 in 2019, which indicates a relatively stable trend. To compare our forecasts with actual total exports, we can compute the total exports using the forecasted per capita values multiplied by the actual population for each year:

$$\text{Total Exports}_t = \text{Per Capita Exports}_t \times \text{Population}_t$$

Using the actual population in 2018 and 2019 [6], the estimated total exports become:

$$\text{Total Exports}_{2018} = 12.57 \times 4,878,657 \approx 61,358,000,$$

$$\text{Total Exports}_{2019} = 12.61 \times 4,944,703 \approx 62,359,000.$$

According to the World Integrated Trade Solution [5], the actual total exports were 54,264 in 2018 and 123,970 in 2019. While the forecasts correctly

predicted an upward trend in per capita exports, they underestimated the magnitude of the increase in total exports, particularly in 2019, highlighting the limitations of the ARIMA model for capturing sharp changes in absolute export values.

8.3 Univariate Population Forecast: 2 Year Ahead (2018-2019)

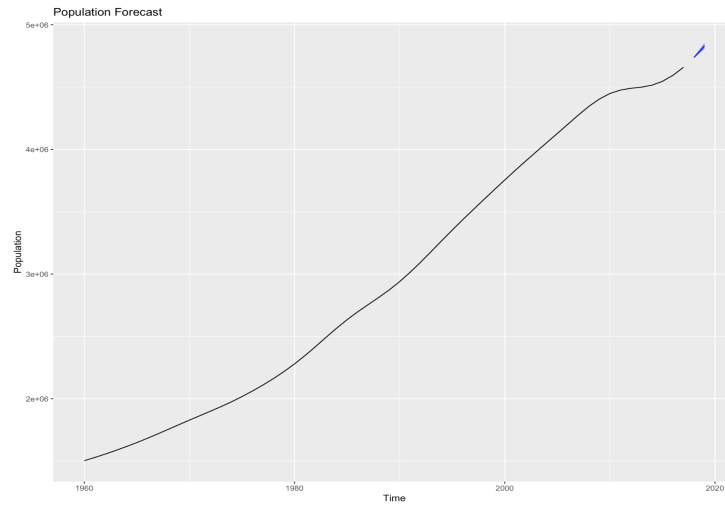


Figure 24: Population Forecast for 2018–2019 with 95% Confidence Intervals

Table 3: Population Forecast for 2018–2019 (ARIMA(2,1,0)) with 95% Confidence Intervals

Year	Point Forecast	Lo 95%	Hi 95%
2018	4,737,936	4,730,878	4,745,005
2019	4,829,829	4,807,935	4,851,823

The forecasted population shows a gradual increase from 4,737,436 in 2018 to 4,827,744 in 2019. The narrow 95% confidence intervals indicate relatively precise predictions. According to Macrotrends website [6], the actual population in 2018 was 4,878,657 and in 2019 it was 4,944,703, suggesting that our forecasts were slightly conservative but still captured the overall upward trend accurately.

9 Limitations and Potential Future Work

One limitation of this study is that our analysis focused primarily on stationary time series data, which may overlook potential insights from non-stationary patterns. Additionally, the study considered primarily univariate forecasting for each variable individually. Future work could explore modeling non-stationary time series directly, as well as multivariate forecasting approaches that capture interdependencies between GDP, Exports, and Population. Incorporating more advanced models, such as vector autoregression (VAR) or machine learning time series models, could provide deeper insights and improve forecast accuracy. Expanding the dataset and including additional economic indicators may enhance the robustness and applicability of the results.

10 Conclusion

In this study, we performed time series analysis of GDP, Exports, and Population in the Central African Republic from 1960 to 2017. Through exploratory data analysis, we observed non-stationary behavior for all three time series with GDP and Population following upward trends and Exports showing higher volatility and structural fluctuations. No statistical outliers were detected, suggesting that the observed variability reflects the true economic behaviors.

After log transformation and differencing, stationarity was achieved, enabling the fitting of univariate ARIMA models and a multivariate VAR(2) model. Using AIC and BIC, we selected ARIMA(1,1,0) for GDP, ARIMA(2,1,0) for Exports, and ARIMA(2,1,0) for Population. Our diagnostics checks confirmed that the residuals approximated white noise and normality. The VAR(2) model indicated that GDP and Exports are influenced primarily by their own past values, while Population evolves almost independently. The impulse response analysis revealed that GDP shocks mainly affect GDP itself, Exports shocks slightly influenced GDP but not Population, and Population shockshad minimal effects on the other variables.

We forecasted for 2018-2019 using the selected ARIMA models and predicted modest GDP growth and a relatively stable Population, while Exports remained volatile. These forecasts, along with their confidence intervals, provide reasonable approximations consistent with historical trends, offering actionable insights for economic planning.

Overall, this analysis suggests that the Central African Republic's economic growth and export performance are driven by intrinsic trends rather than interactions with Population. From this analysis, policymakers can use these insights for resource management, economic forecasting, and strategic planning, recognizing that interventions targeting GDP and Exports may have limited impact on demographic trends, and vice versa.

References

- [1] The Observatory of Economic Complexity. (n.d.). *Central African Republic (CAF) Exports, Imports and Trade Partners*. Retrieved December 5, 2025, from <https://oec.world/en/profile/country/caf>
- [2] GeeksforGeeks. (2025, July 23). *Bayesian Information Criterion (BIC)*. Retrieved December 5, 2025, from <https://www.geeksforgeeks.org/machine-learning/bayesian-information-criterion-bic/>
- [3] rsx. (2022). *Should I include non-significant coefficient for forecasting with ARIMA model?* Cross Validated (StackExchange). Retrieved December 5, 2025, from <https://stats.stackexchange.com/questions/565314/should-i-include-non-significant-coefficient-for-forecasting-with-arma-model>
- [4] World Bank. (n.d.). *GDP (current US\$) – Central African Republic (CAF)*. World Development Indicators. Retrieved December 5, 2025, from <https://data.worldbank.org/indicator/NY.GDP.MKTP.CD?locations=CF>
- [5] World Integrated Trade Solution (WITS). (n.d.). *Central African Republic – Country Profile*. Retrieved December 5, 2025, from <https://wits.worldbank.org/CountryProfile/en/Country/CAF/Year/2018/Summary>
- [6] Macrotrends. (n.d.). *Central African Republic Population (1950–2025)*. Retrieved December 5, 2025, from <https://www.macrotrends.net/global-metrics/countries/caf/central-african-republic/population>
- [7] GeeksforGeeks. (n.d.). *Multivariate Time Series Modelling in R*. Retrieved December 5, 2025, from <https://www.geeksforgeeks.org/r-language/multivariate-time-series-modelling-in-r/>
- [8] GeeksforGeeks. (n.d.). *Vector Autoregression (VAR) for Multivariate Time Series*. Retrieved December 5, 2025, from <https://www.geeksforgeeks.org/machine-learning/vector-autoregression-var-for-multivariate-time-series/>
- [9] GeeksforGeeks. (n.d.). *Augmented Dickey-Fuller (ADF) Test*. Retrieved December 5, 2025, from <https://www.geeksforgeeks.org/machine-learning/augmented-dickey-fuller-adf/>
- [10] GeeksforGeeks. (n.d.). *Kwiatkowski-Phillips-Schmidt-Shin (KPSS) Test for Stationarity*. Retrieved December 5, 2025, from <https://www.geeksforgeeks.org/python/kiwiatkowski-phillips-schmidt-shin-kpss/>
- [11] R-Econometrics. (n.d.). *Impulse Response Functions for Time Series / VAR Models*. Retrieved December 5, 2025, from <https://www.r-econometrics.com/timeseries/irf/>

A Appendix

The full source code is on the next page.

STA137FinalProject

Caitlyn Koyabu

2025-11-22

```
library(forecast)
```

```
## Warning: package 'forecast' was built under R version 4.3.3
```

```
## Registered S3 method overwritten by 'quantmod':
```

```
##   method      from
```

```
## as.zoo.data.frame zoo
```

```
load("/Users/caitlynkoyabu/Downloads/Data_File_finalproject.Rdata")
```

```
head(finalPro_data)
```

```
## # A tibble: 6 x 9
```

##	Country	Code	Year	GDP Growth	CPI	Imports	Exports	Population	
##	<fct>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	
## 1	Central African Re~	CAF	1960	1.12e8	NA	NA	34.2	23.3	1503508
## 2	Central African Re~	CAF	1961	1.23e8	4.95	NA	35.8	26.5	1529227
## 3	Central African Re~	CAF	1962	1.24e8	-3.71	NA	37.7	24.6	1556661
## 4	Central African Re~	CAF	1963	1.29e8	-0.707	NA	38.5	25.2	1585763
## 5	Central African Re~	CAF	1964	1.42e8	2.08	NA	40.8	28.4	1616516
## 6	Central African Re~	CAF	1965	1.51e8	0.948	NA	37.7	27.1	1648833

```
summary(finalPro_data)
```

##	Country	Code	Year	GDP		
##	Central African Republic:58	CAF	:58	Min. :1960	Min. :1.122e+08	
##	Afghanistan	: 0	ABW	: 0	1st Qu.:1974	1st Qu.:3.057e+08
##	Albania	: 0	AFG	: 0	Median :1988	Median :9.348e+08
##	Algeria	: 0	AGO	: 0	Mean :1988	Mean :9.394e+08
##	American Samoa	: 0	ALB	: 0	3rd Qu.:2003	3rd Qu.:1.367e+09
##	Andorra	: 0	AND	: 0	Max. :2017	Max. :2.196e+09
##	(Other)	: 0	(Other):	0		
##	Growth	CPI	Imports	Exports		
##	Min. : -36.69995	Min. : 36.77	Min. :18.00	Min. :10.68		
##	1st Qu.: -0.00087	1st Qu.: 49.63	1st Qu.:25.03	1st Qu.:16.19		
##	Median : 2.05413	Median : 71.42	Median :29.68	Median :21.74		
##	Mean : 1.19893	Mean : 73.58	Mean :30.83	Mean :20.66		
##	3rd Qu.: 4.60753	3rd Qu.: 86.50	3rd Qu.:36.58	3rd Qu.:25.14		
##	Max. : 9.48173	Max. :186.86	Max. :46.90	Max. :34.31		
##	NA's :1	NA's :22				

```
##      Population
## Min.      :1503508
## 1st Qu.:1985984
## Median :2842456
## Mean      :2974790
## 3rd Qu.:3963152
## Max.      :4659080
##
```

No missing data and no duplicate rows.

Plot the raw data

```
# Extract variables
gdp <- ts(finalPro_data$GDP, start = 1960, frequency = 1)
exports <- ts(finalPro_data$Exports, start = 1960, frequency = 1)
population <- ts(finalPro_data$Population, start = 1960, frequency = 1)

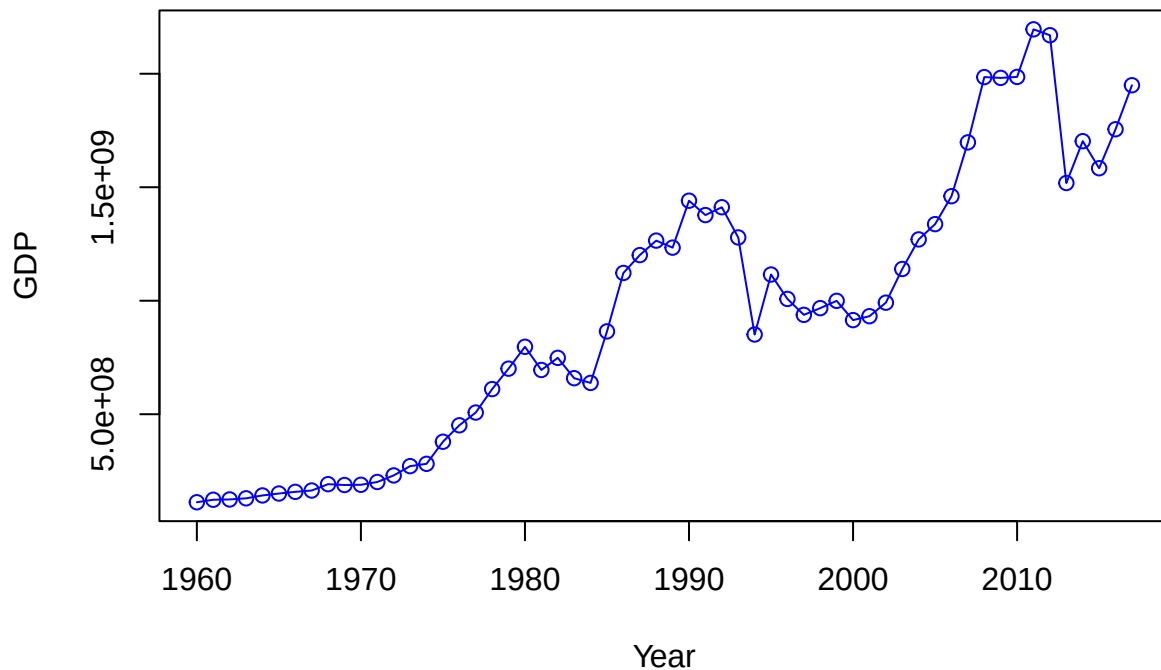
plot(gdp,
     type = "o",
     col = "blue",
     ylab = "GDP",
     xlab = "Year",
     main = "Central African Republic GDP (1960-2017)")
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Central African Republic GDP (1960-2017)' in 'mbcsToSbcs': dot
## substituted for <e2>
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Central African Republic GDP (1960-2017)' in 'mbcsToSbcs': dot
## substituted for <80>
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Central African Republic GDP (1960-2017)' in 'mbcsToSbcs': dot
## substituted for <93>
```

Central African Republic GDP (1960...2017)



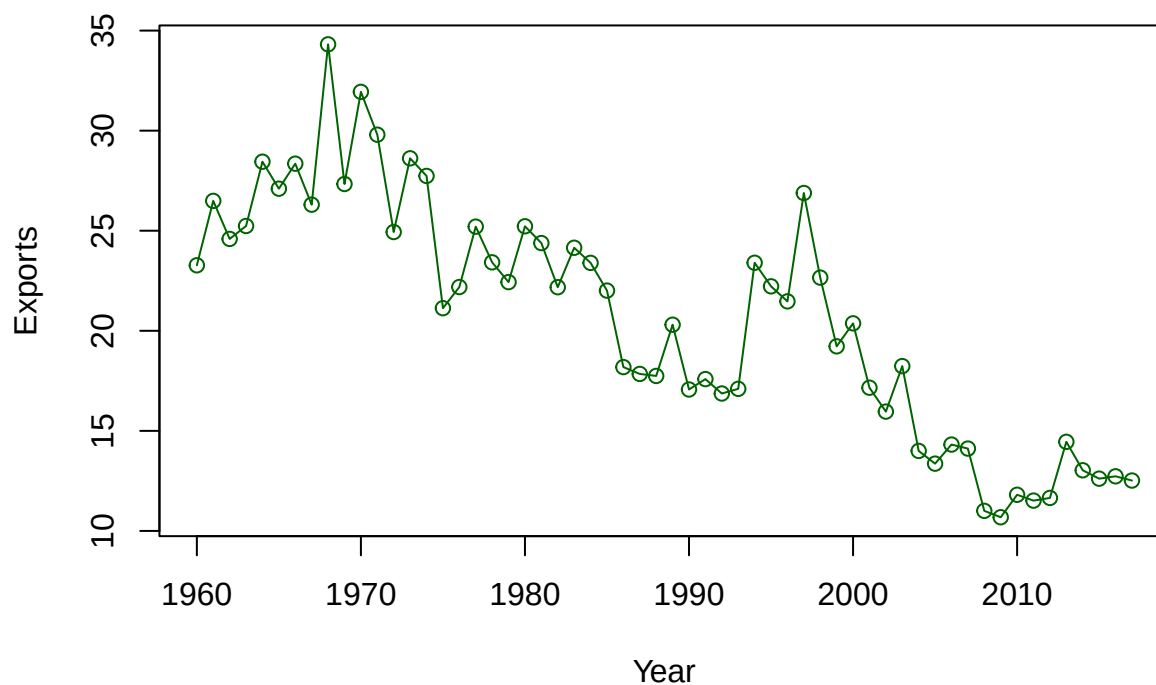
```
plot(exports,  
      type = "o",  
      col = "darkgreen",  
      ylab = "Exports",  
      xlab = "Year",  
      main = "Central African Republic Exports (1960-2017)")
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion  
## failure on 'Central African Republic Exports (1960-2017)' in 'mbcsToSbcs': dot  
## substituted for <e2>
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion  
## failure on 'Central African Republic Exports (1960-2017)' in 'mbcsToSbcs': dot  
## substituted for <80>
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion  
## failure on 'Central African Republic Exports (1960-2017)' in 'mbcsToSbcs': dot  
## substituted for <93>
```

Central African Republic Exports (1960...2017)



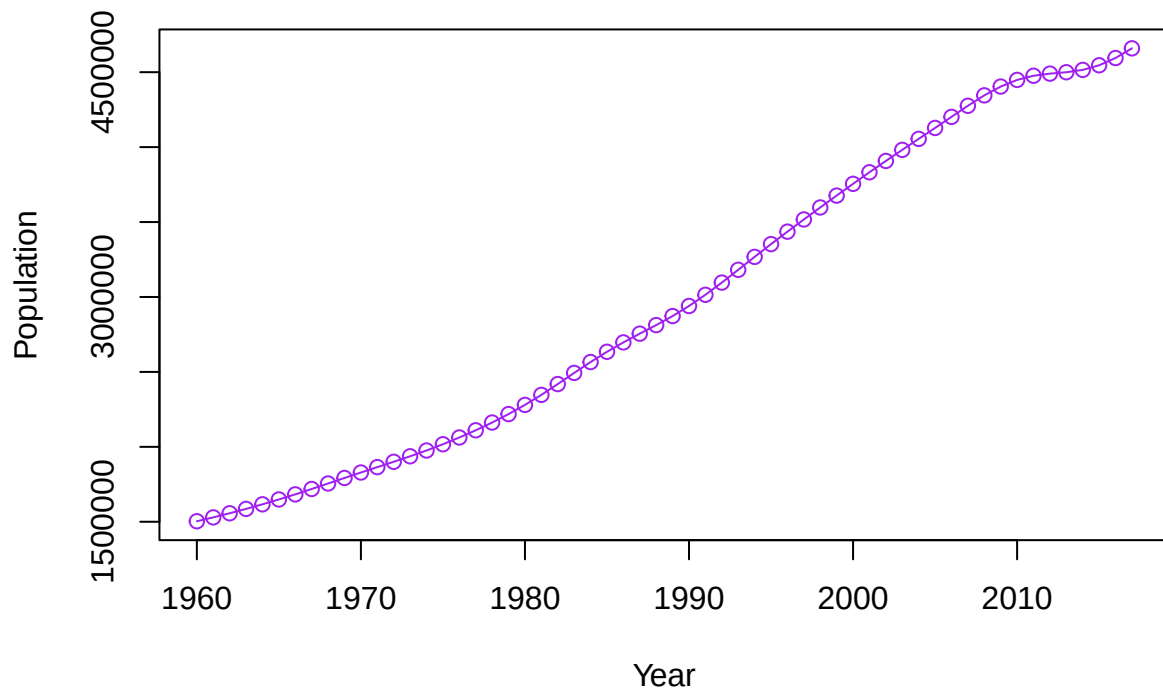
```
plot(population,
      type = "o",
      col = "purple",
      ylab = "Population",
      xlab = "Year",
      main = "Central African Republic Population (1960-2017)")
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Central African Republic Population (1960-2017)' in 'mbcsToSbcs':
## dot substituted for <e2>
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Central African Republic Population (1960-2017)' in 'mbcsToSbcs':
## dot substituted for <80>
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Central African Republic Population (1960-2017)' in 'mbcsToSbcs':
## dot substituted for <93>
```

Central African Republic Population (1960...2017)



Check Outliers - (Z-score and Boxplots)

```
gdp_z <- scale(gdp)
which(abs(gdp_z) > 3)
```

```
## integer(0)
```

```
exports_z <- scale(exports)
which(abs(exports_z) > 3)
```

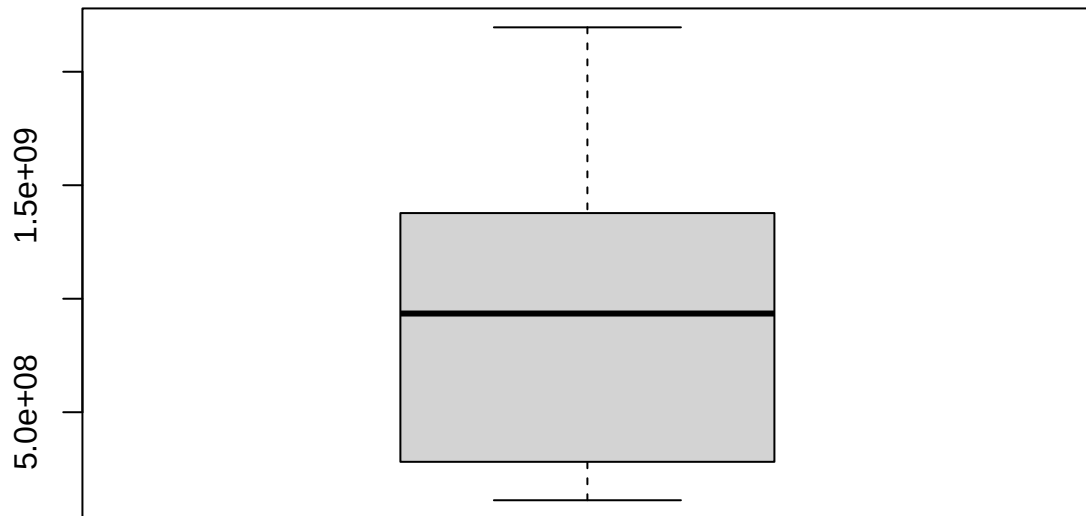
```
## integer(0)
```

```
pop_z <- scale(population)
which(abs(pop_z) > 3)
```

```
## integer(0)
```

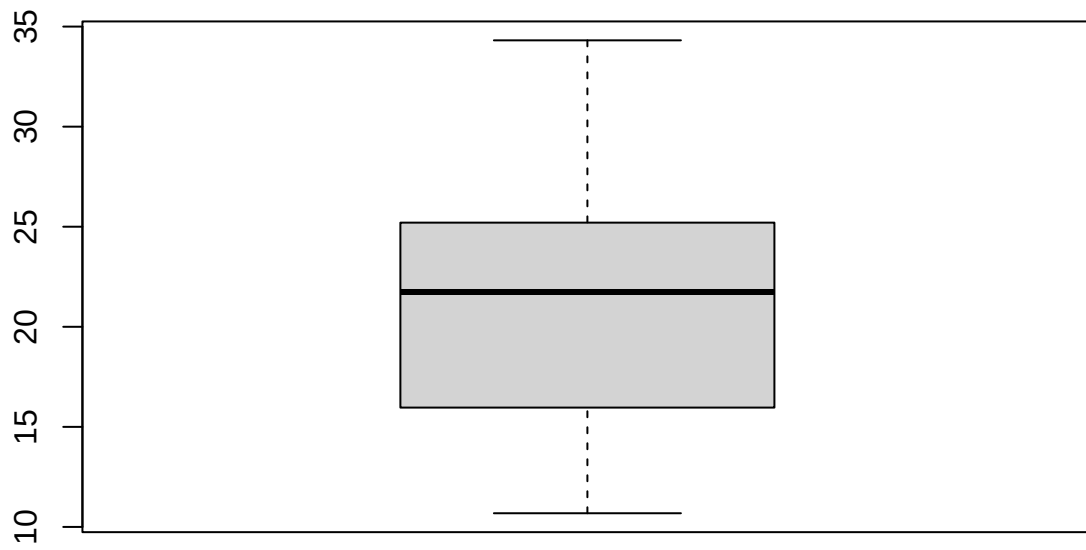
```
boxplot(gdp, main="GDP Outliers")
```

GDP Outliers



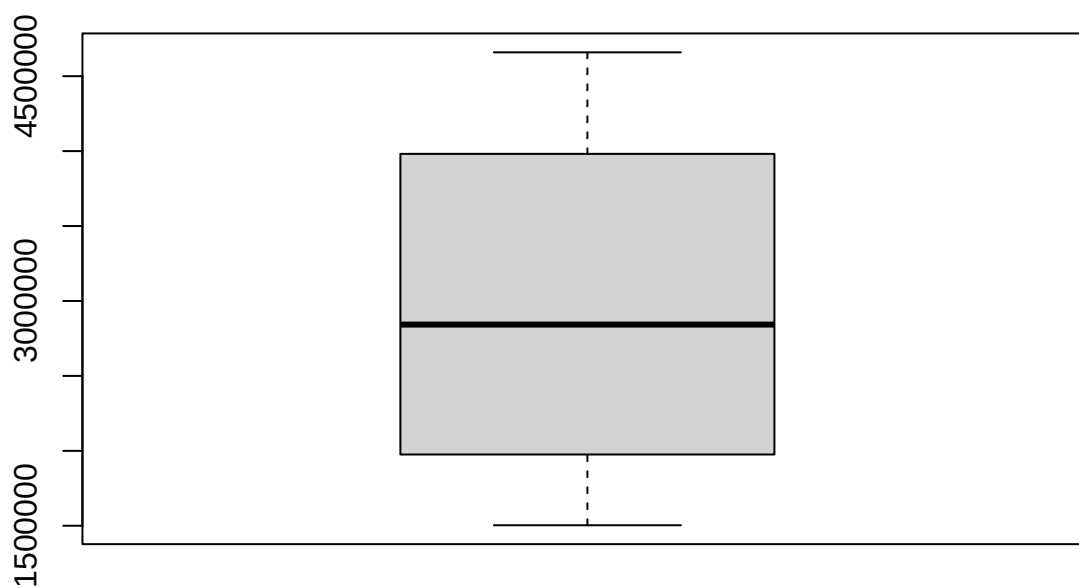
```
boxplot(exports, main="Exports Outliers")
```

Exports Outliers



```
boxplot(population, main=" Outliers")
```

Outliers



```
tsoutliers(gdp)
```

```
## $index  
## integer(0)  
##  
## $replacements  
## numeric(0)
```

```
tsoutliers(exports)
```

```
## $index  
## integer(0)  
##  
## $replacements  
## numeric(0)
```

```
tsoutliers(population)
```

```
## $index  
## integer(0)  
##  
## $replacements  
## numeric(0)
```

GDP Analysis: The Central African Republic's GDP displays an upward trend from 1960 to 2017, with noticeable fluctuations during the 1980s to the mid 1990s and the 2010s to mid 2010s. There is no seasonality trend to be observed because the data we observe is annual. Overall, the trend of the data is relatively smooth with the occasional drops and spikes in the plot, but overall, the Central African Republic's GDP increases over time.

Export Analysis: The Central African Republic's Exports is more volatile than the GDP plot and doesn't show as smooth as a trend. There are several sharp spike and drops, indicating that there could be uncertainty and instability in trade activity. However, it still seems to be having a downward trend when plotting the data. Therefore, the Central African Republic's Exports decreases over time.

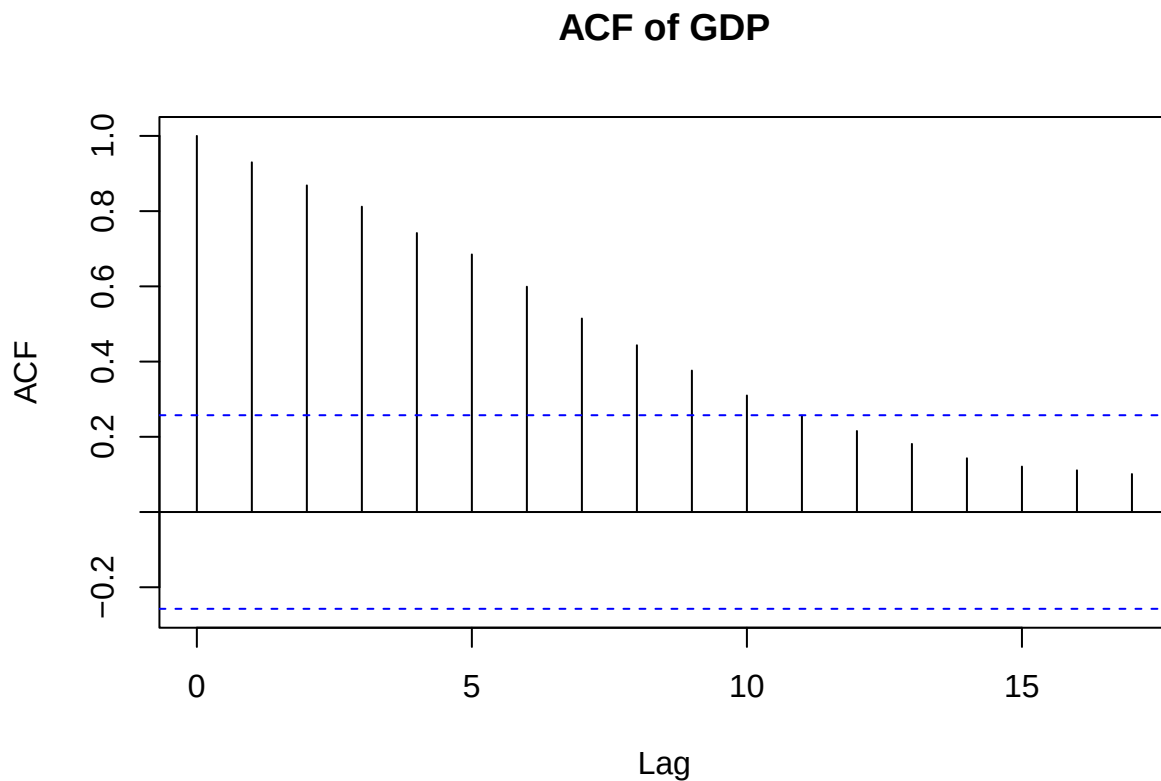
Population Analysis: The Central African Republic's Population displays a smooth consistent increasing trend. Over time, the Central African Republic's Population increases.

Since there are trends for The Central African Republic's GDP, Exports, and Population data, this means that we have non-stationary data. We want to get rid of the trends, so we can work with stationary data. Therefore, we will take first differences of the data until the data is stationary.

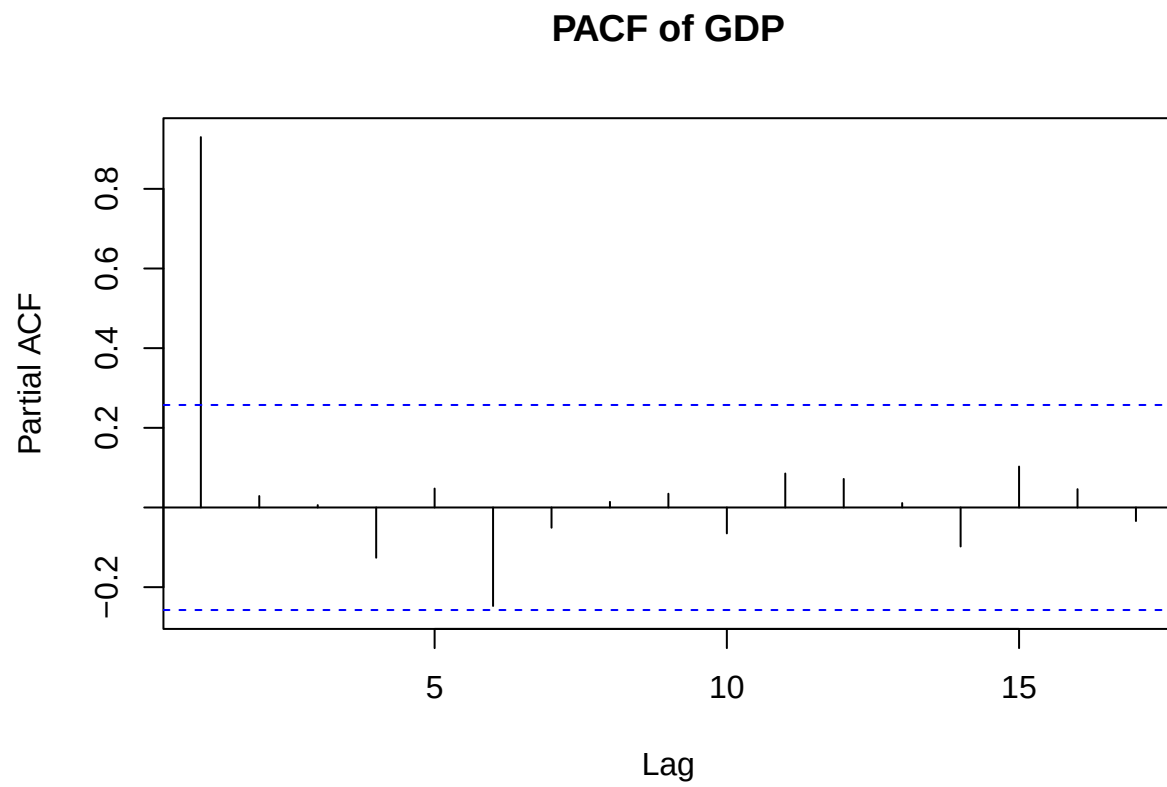
Checking for seasonality in the plots, we can look repeating patterns at regular intervals (yearly cycles) and we can see that there's no regular pattern occurring as we observe each year from 1960s to 2017s. Therefore, we can conclude that there is no seasonality for GDP, exports, and population plots.

ACF/PACF - Raw

```
acf(gdp, main = "ACF of GDP")
```

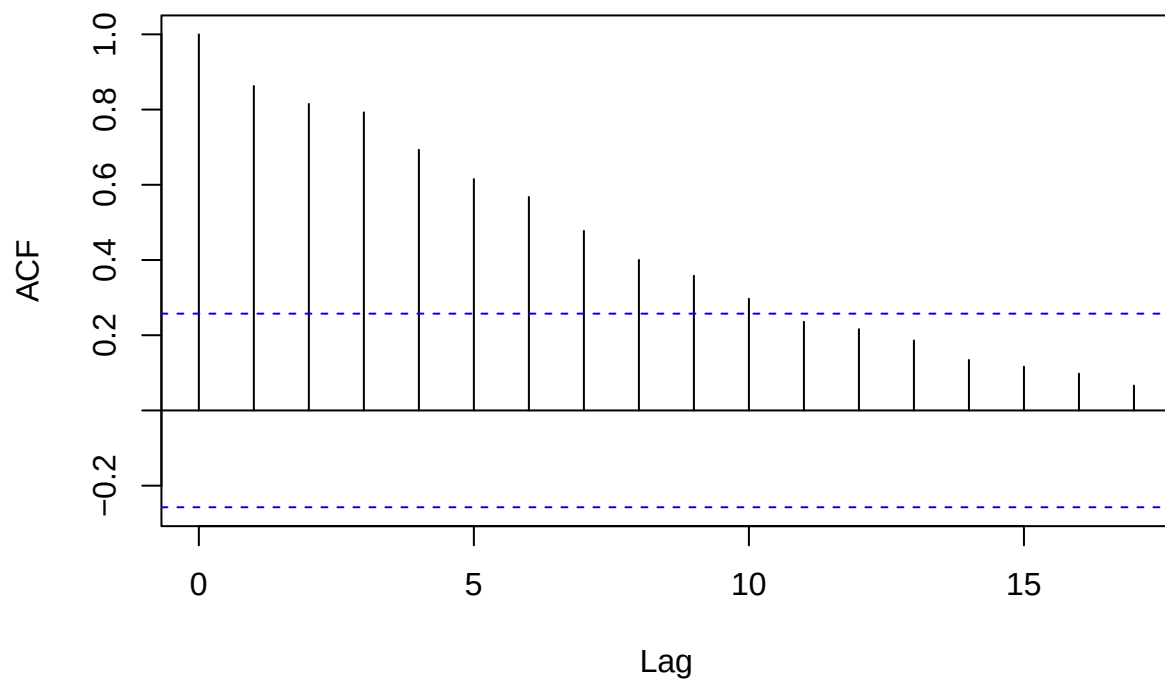


```
pacf(gdp, main = "PACF of GDP")
```



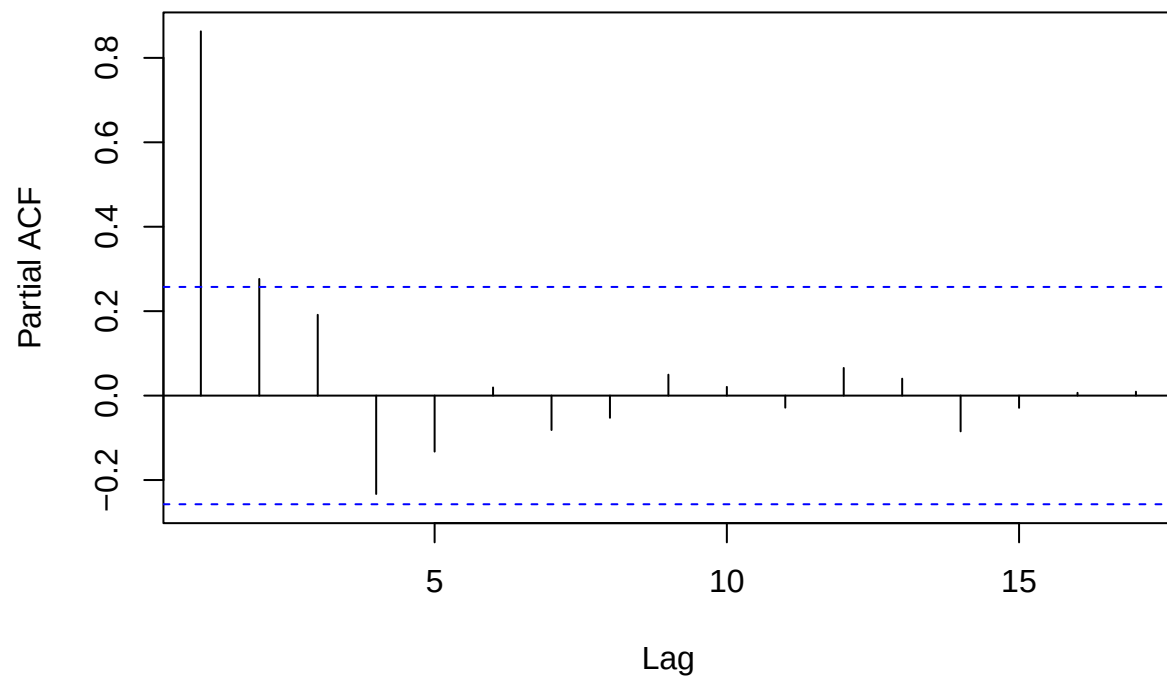
```
acf(exports, main = "ACF of Exports")
```

ACF of Exports



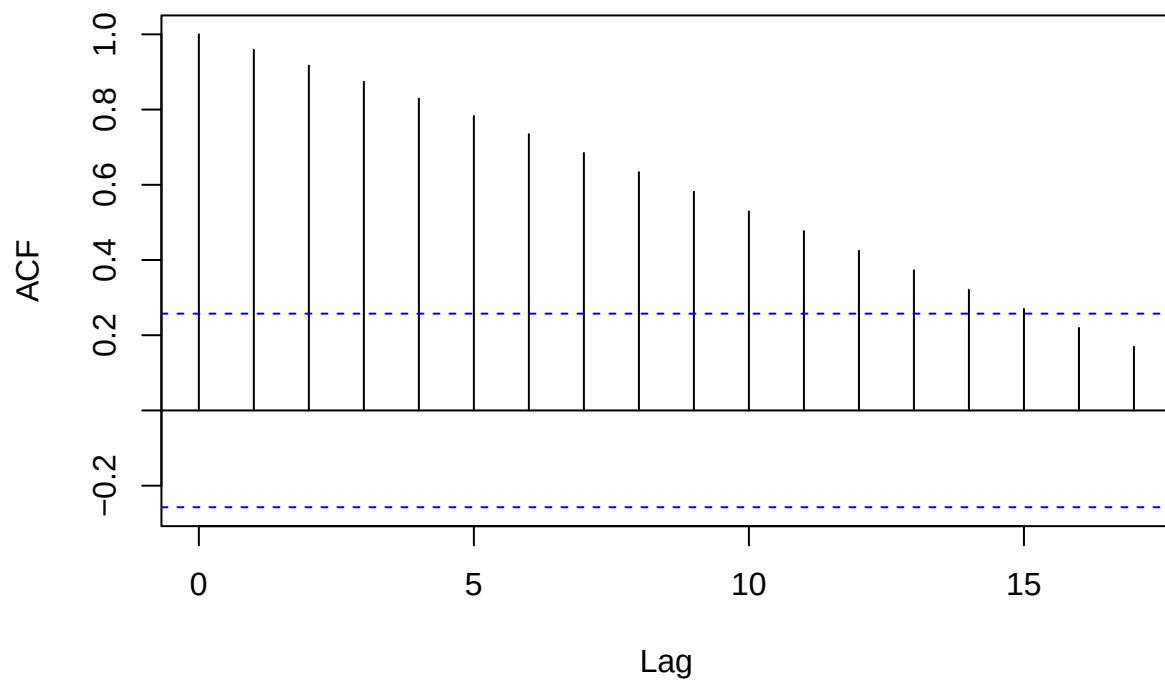
```
pacf(exports, main = "PACF of Exports")
```

PACF of Exports



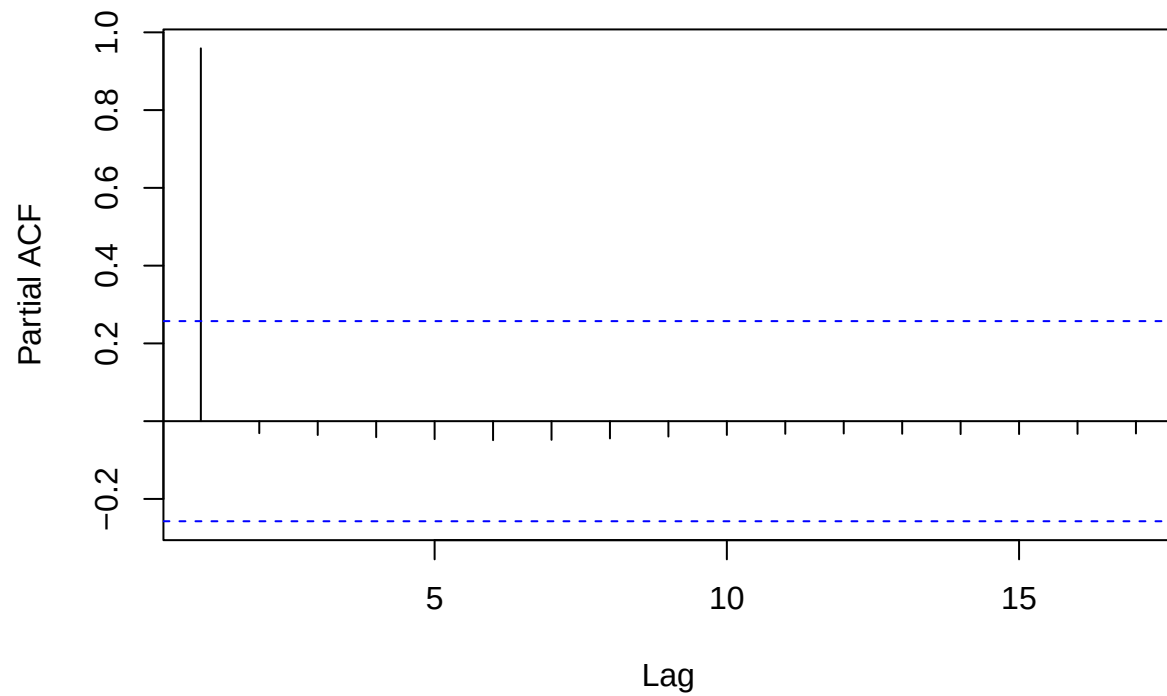
```
acf(population, main = "ACF of Population")
```

ACF of Population



```
pacf(population, main = "PACF of Population")
```

PACF of Population



Box-Cox Transform Data

```
lambda_gdp <- BoxCox.lambda(gdp)
lambda_exports <- BoxCox.lambda(exports)
lambda_population <- BoxCox.lambda(population)

gdp_bc <- BoxCox(gdp, lambda_gdp)
exports_bc <- BoxCox(exports, lambda_exports)
population_bc <- BoxCox(population, lambda_population)

lambda_gdp
```

```
## [1] -0.2054777
```

```
lambda_exports
```

```
## [1] -0.3650884
```

```
lambda_population
```

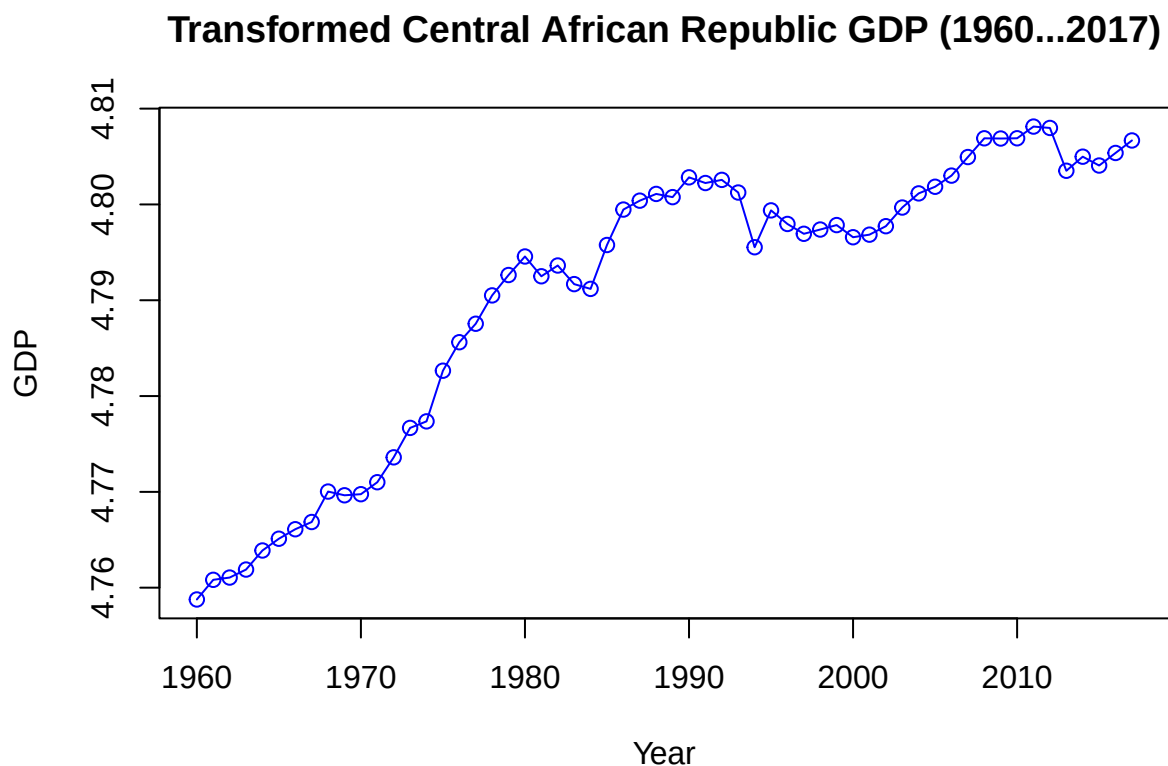
```
## [1] 0.2206428
```

```
plot(gdp_bc,
     type = "o",
     col = "blue",
     ylab = "GDP",
     xlab = "Year",
     main = "Transformed Central African Republic GDP (1960-2017)")
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Transformed Central African Republic GDP (1960-2017)' in
## 'mbcsToSbcs': dot substituted for <e2>
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Transformed Central African Republic GDP (1960-2017)' in
## 'mbcsToSbcs': dot substituted for <80>
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Transformed Central African Republic GDP (1960-2017)' in
## 'mbcsToSbcs': dot substituted for <93>
```

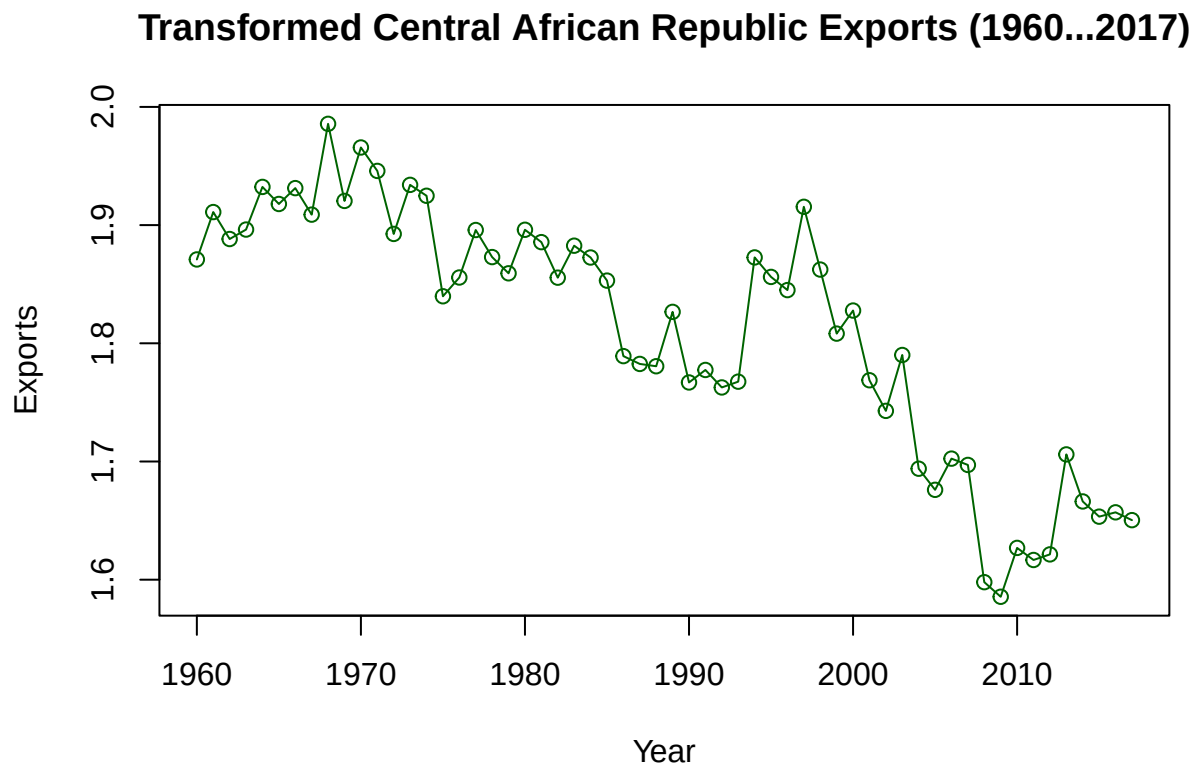


```
plot(exports_bc,
     type = "o",
     col = "darkgreen",
     ylab = "Exports",
     xlab = "Year",
     main = "Transformed Central African Republic Exports (1960-2017)")
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Transformed Central African Republic Exports (1960-2017)' in
## 'mbcsToSbcs': dot substituted for <e2>
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Transformed Central African Republic Exports (1960-2017)' in
## 'mbcsToSbcs': dot substituted for <80>
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Transformed Central African Republic Exports (1960-2017)' in
## 'mbcsToSbcs': dot substituted for <93>
```

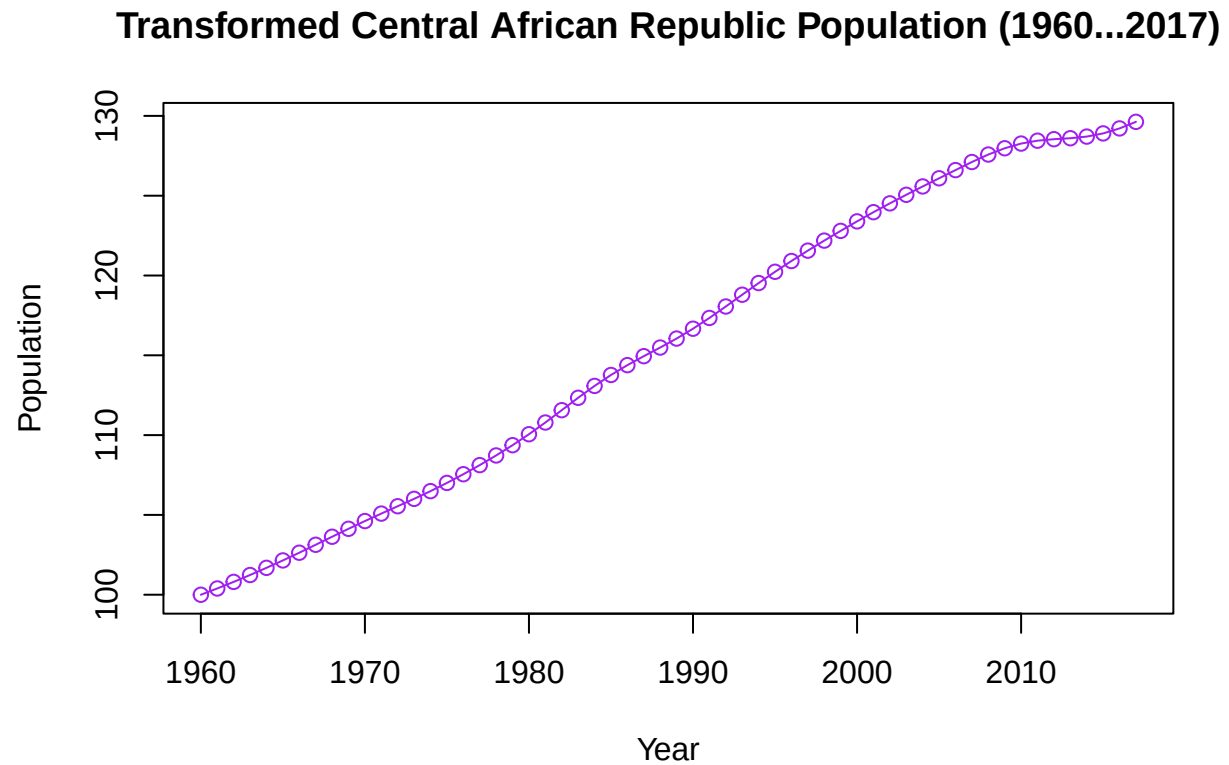


```
plot(population_bc,
     type = "o",
     col = "purple",
     ylab = "Population",
     xlab = "Year",
     main = "Transformed Central African Republic Population (1960-2017)")
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Transformed Central African Republic Population (1960-2017)' in
## 'mbcsToSbcs': dot substituted for <e2>
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Transformed Central African Republic Population (1960-2017)' in
## 'mbcsToSbcs': dot substituted for <80>
```

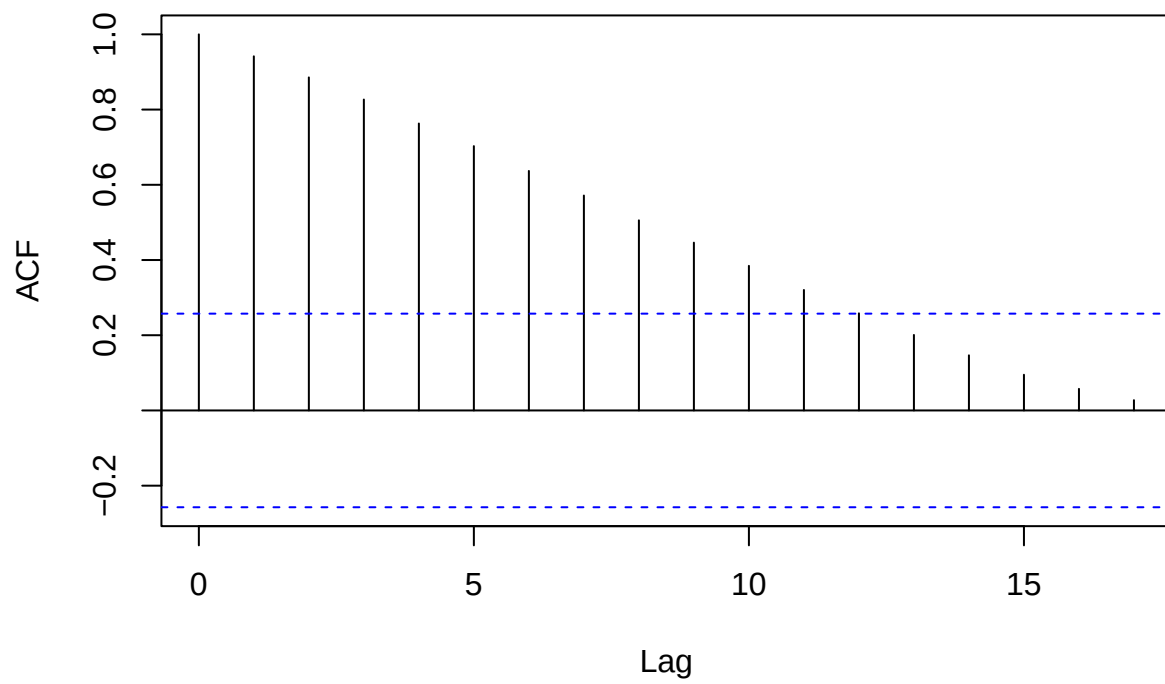
```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Transformed Central African Republic Population (1960-2017)' in
## 'mbcsToSbcs': dot substituted for <93>
```



ACF/PACF Box-Cox Transform Data

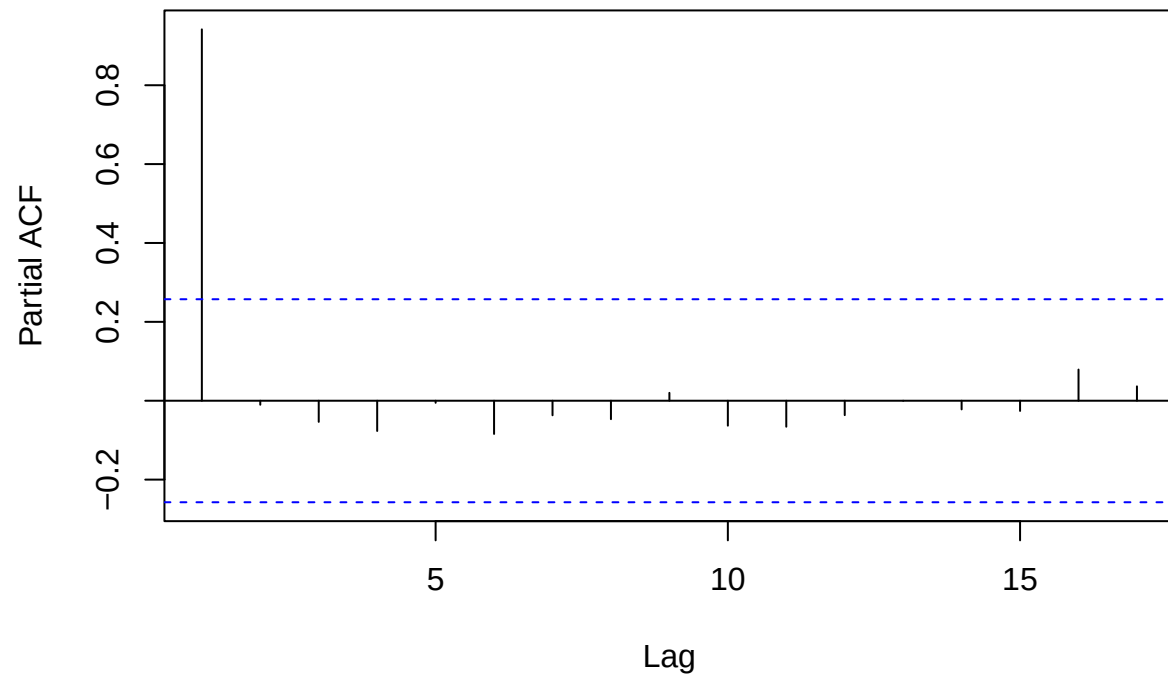
```
acf(gdp_bc)
```

Series gdp_bc



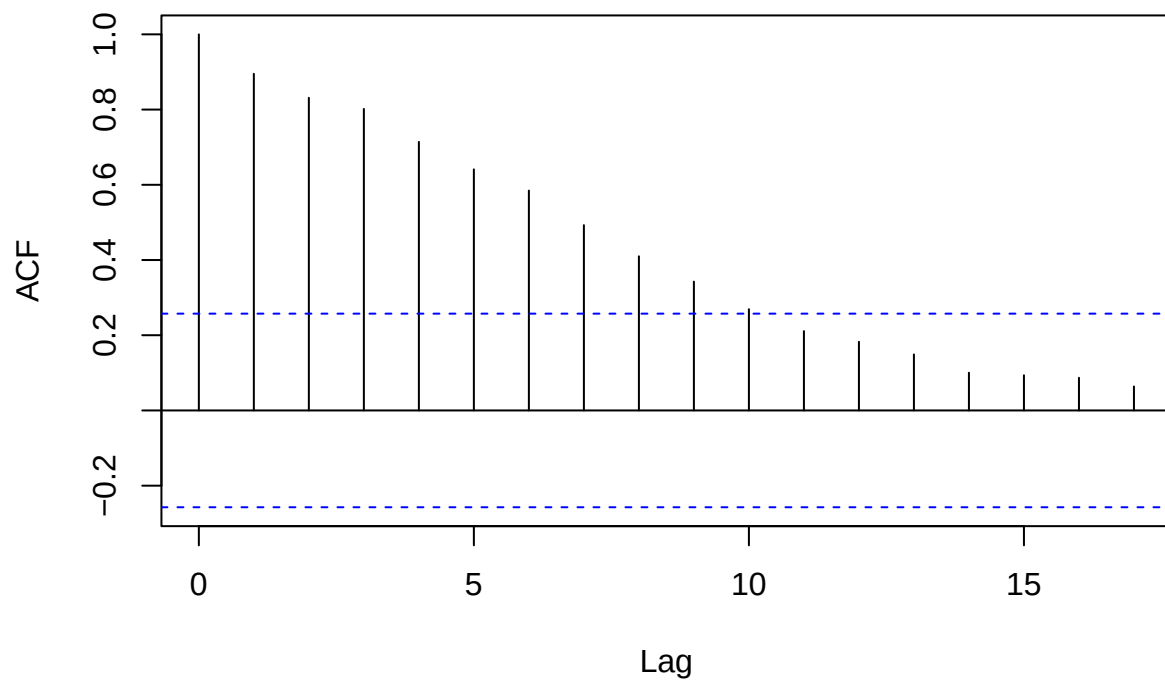
```
pacf(gdp_bc)
```

Series gdp_bc



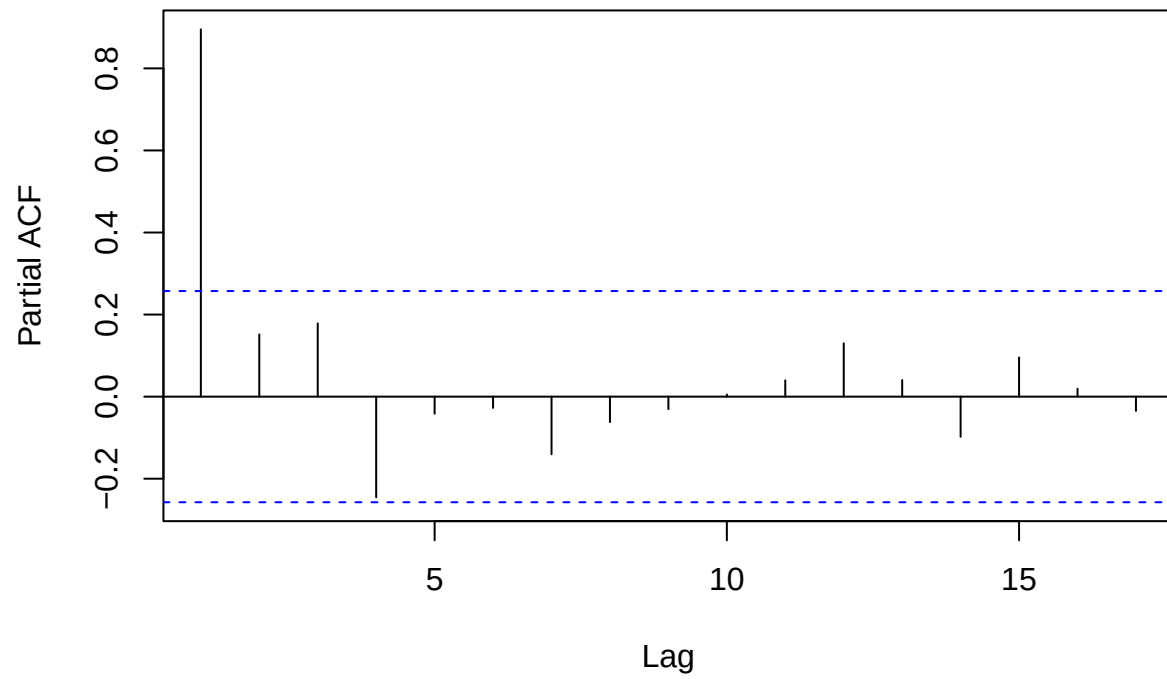
```
acf(exports_bc)
```

Series exports_bc

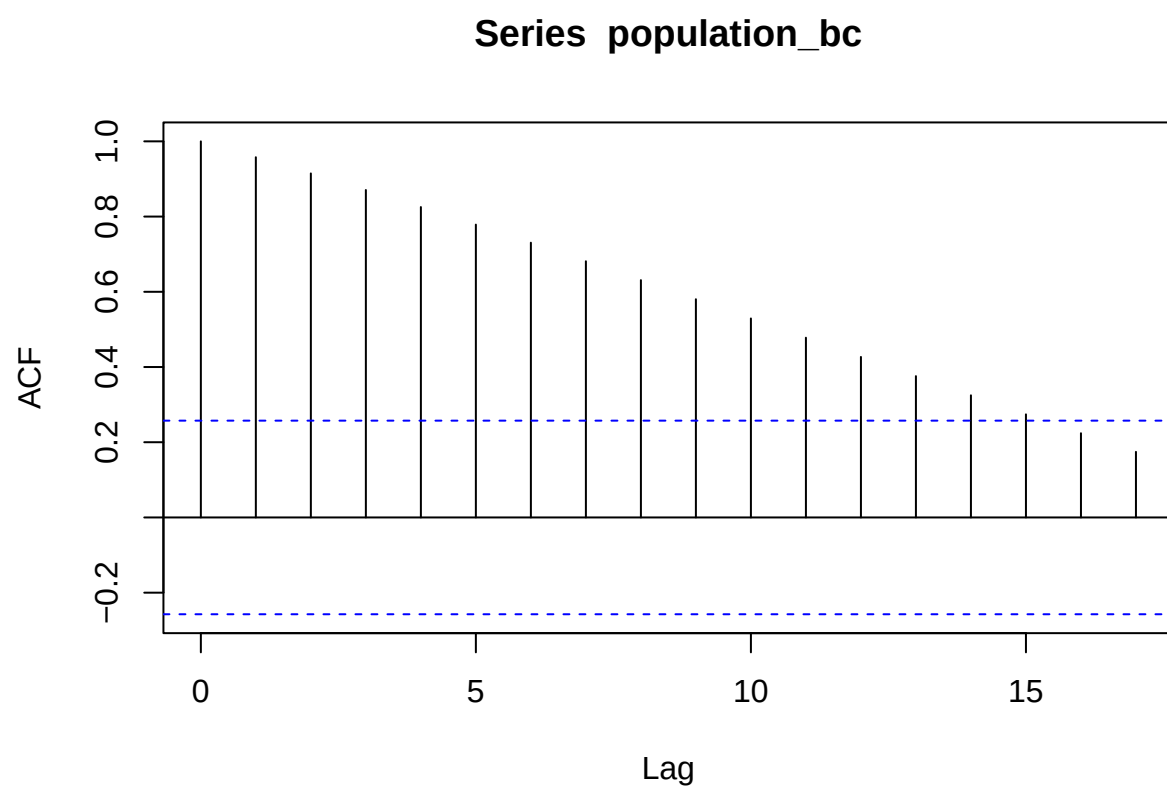


```
pacf(exports_bc)
```

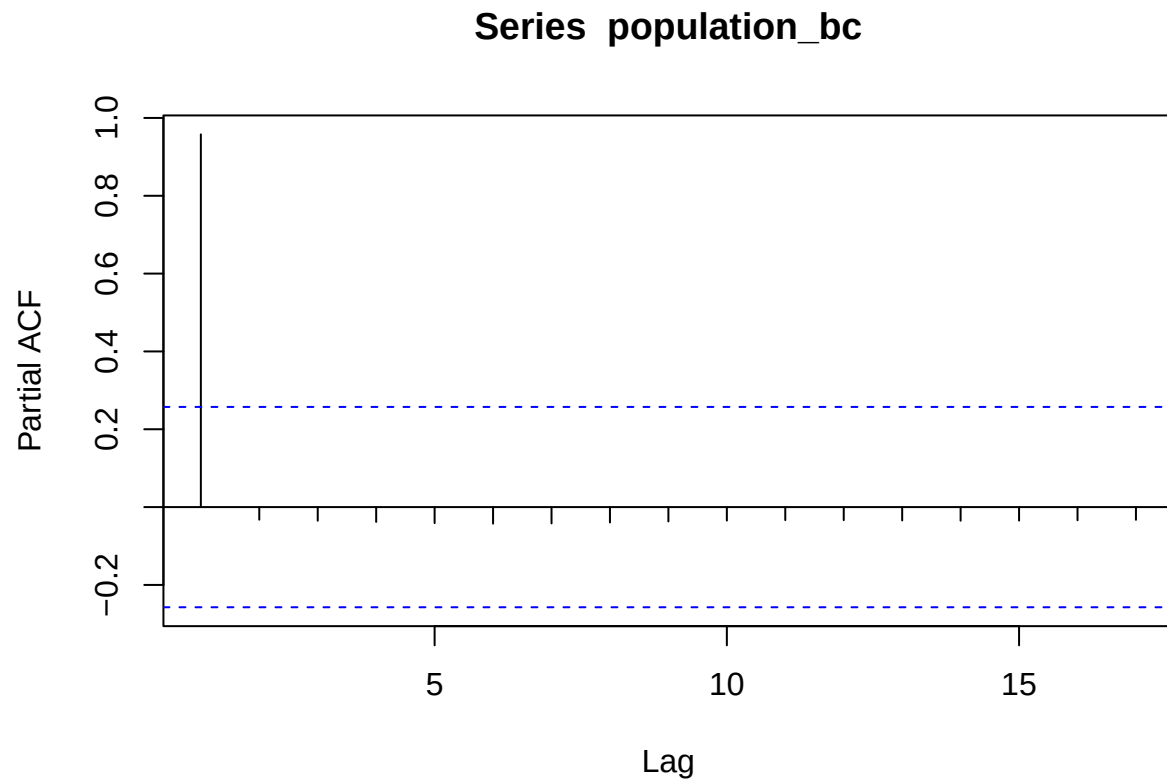
Series exports_bc



```
acf(population_bc)
```



```
pacf(population_bc)
```



Performs well but not as interpretable. This is expected because Box-Cox has better performance.

Log-Transform Data

```
log_gdp <- log(gdp)
log_exports <- log(exports)
log_population <- log(population)

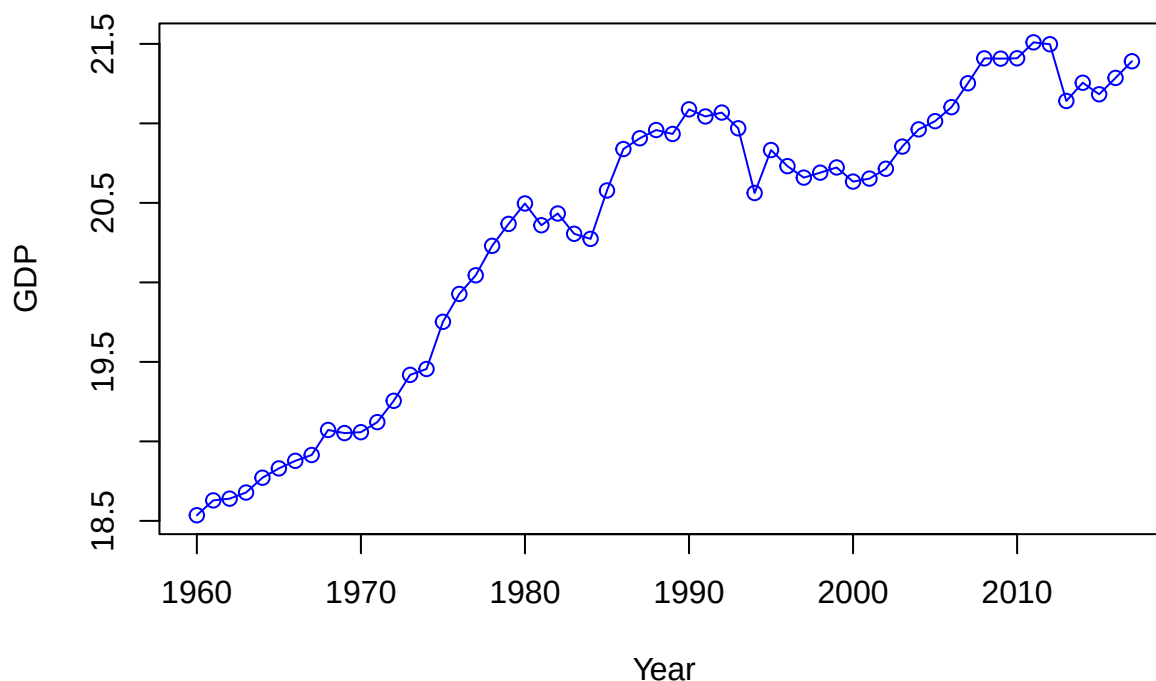
plot(log_gdp,
     type = "o",
     col = "blue",
     ylab = "GDP",
     xlab = "Year",
     main = "Transformed Central African Republic GDP (1960-2017)")
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Transformed Central African Republic GDP (1960-2017)' in
## 'mbcsToSbcs': dot substituted for <e2>
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Transformed Central African Republic GDP (1960-2017)' in
## 'mbcsToSbcs': dot substituted for <80>
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Transformed Central African Republic GDP (1960-2017)' in
## 'mbcsToSbcs': dot substituted for <93>
```

Transformed Central African Republic GDP (1960...2017)



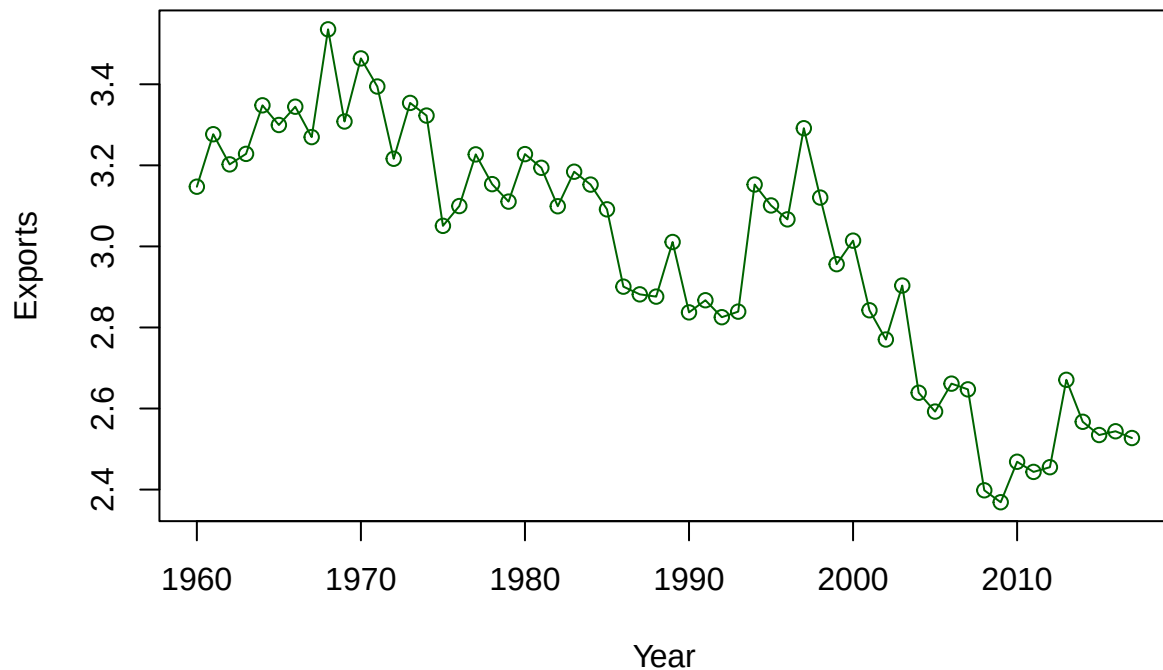
```
plot(log_exports,  
     type = "o",  
     col = "darkgreen",  
     ylab = "Exports",  
     xlab = "Year",  
     main = "Transformed Central African Republic Exports (1960-2017)")
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion  
## failure on 'Transformed Central African Republic Exports (1960-2017)' in  
## 'mbcsToSbcs': dot substituted for <e2>
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion  
## failure on 'Transformed Central African Republic Exports (1960-2017)' in  
## 'mbcsToSbcs': dot substituted for <80>
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion  
## failure on 'Transformed Central African Republic Exports (1960-2017)' in  
## 'mbcsToSbcs': dot substituted for <93>
```

Transformed Central African Republic Exports (1960...2017)



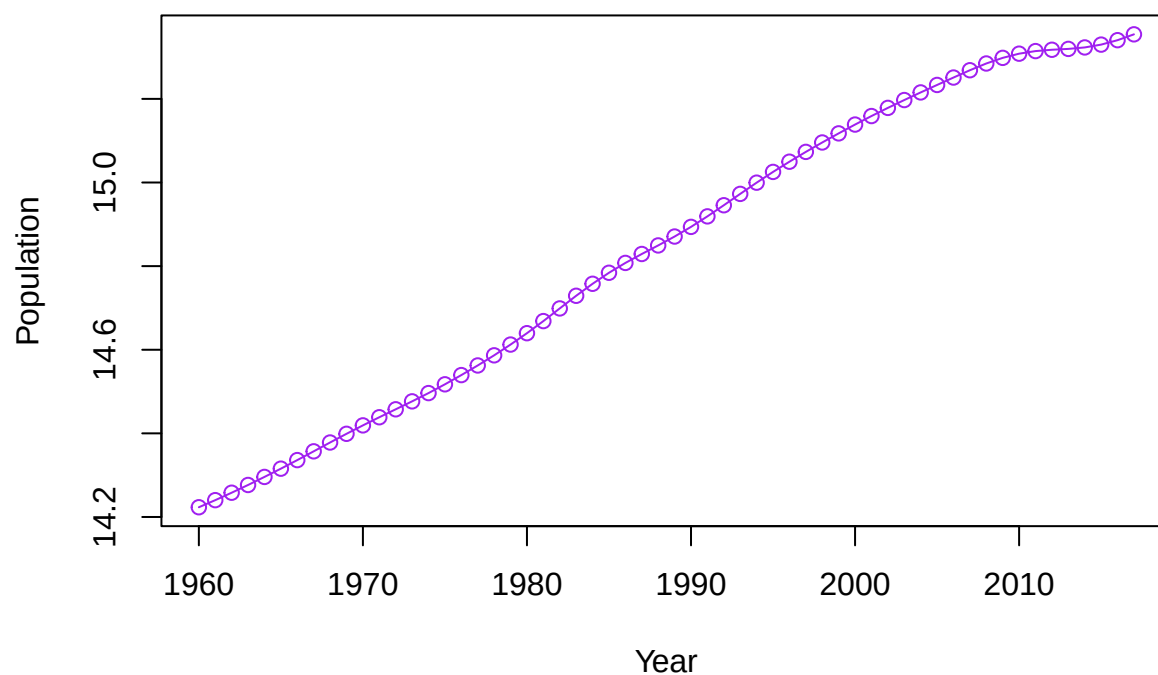
```
plot(log_population,
     type = "o",
     col = "purple",
     ylab = "Population",
     xlab = "Year",
     main = "Transformed Central African Republic Population (1960-2017)")
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Transformed Central African Republic Population (1960-2017)' in
## 'mbcsToSbcs': dot substituted for <e2>
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Transformed Central African Republic Population (1960-2017)' in
## 'mbcsToSbcs': dot substituted for <80>
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Transformed Central African Republic Population (1960-2017)' in
## 'mbcsToSbcs': dot substituted for <93>
```

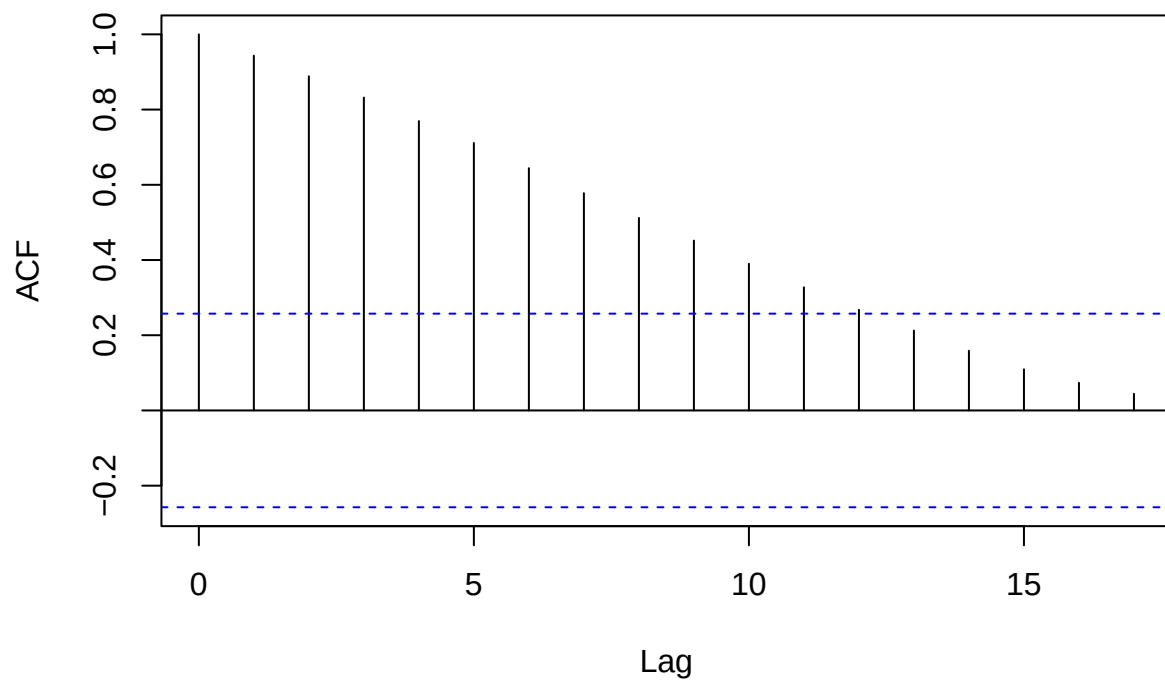
Transformed Central African Republic Population (1960...2017)



ACF/PACF - Log Transformed Data

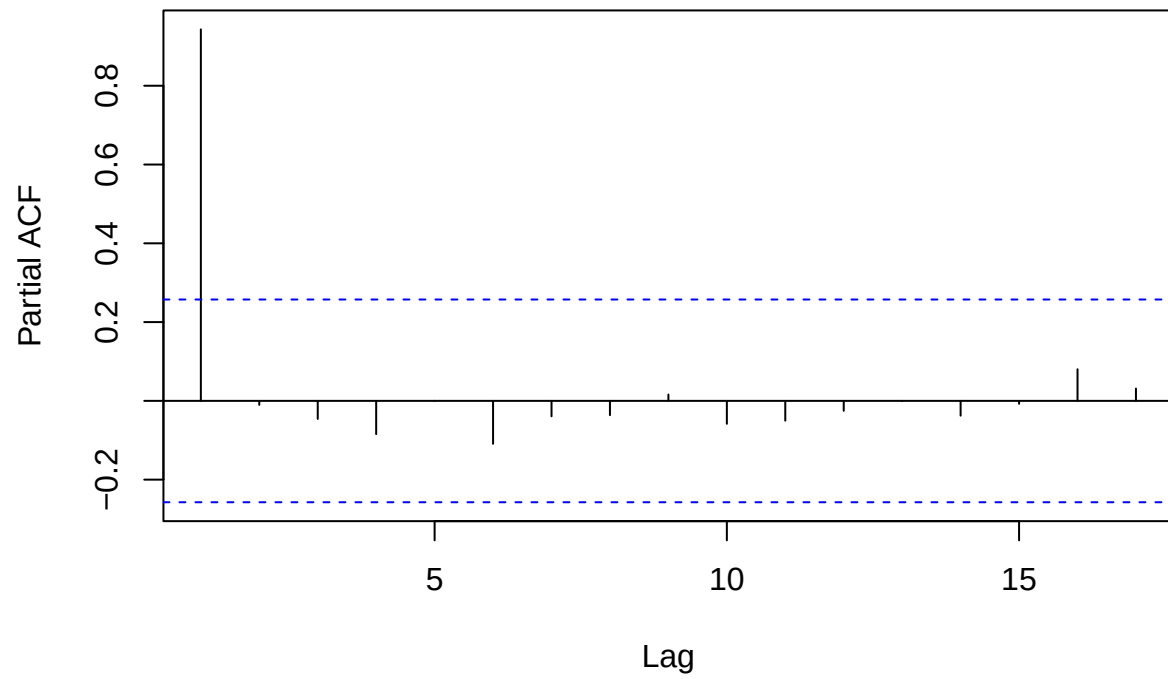
```
acf(log_gdp, main = "ACF of Log GDP")
```

ACF of Log GDP



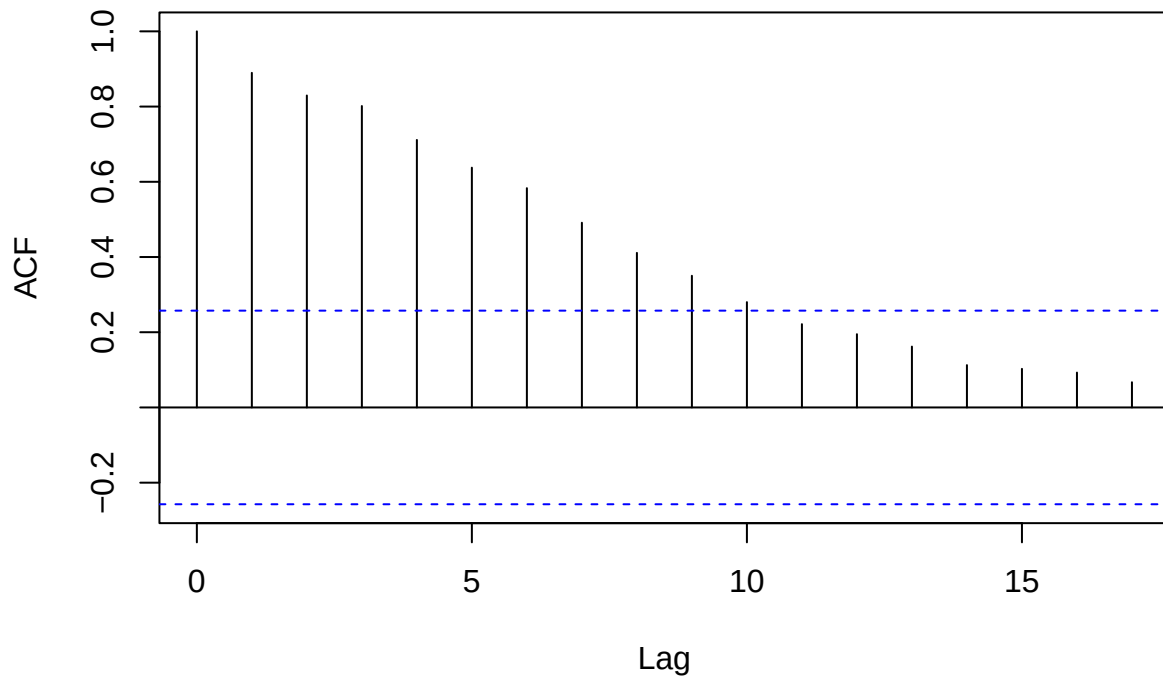
```
pacf(log_gdp, main = "PACF of Log GDP")
```

PACF of Log GDP



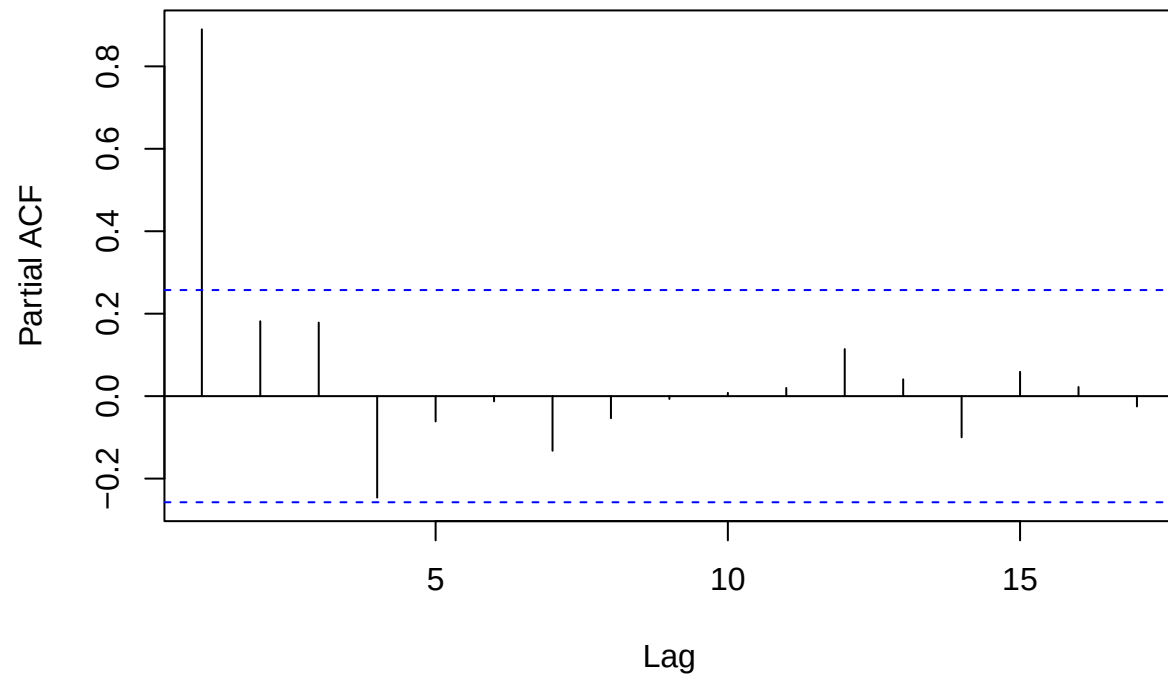
```
acf(log_exports, main = "ACF of Log Exports")
```

ACF of Log Exports



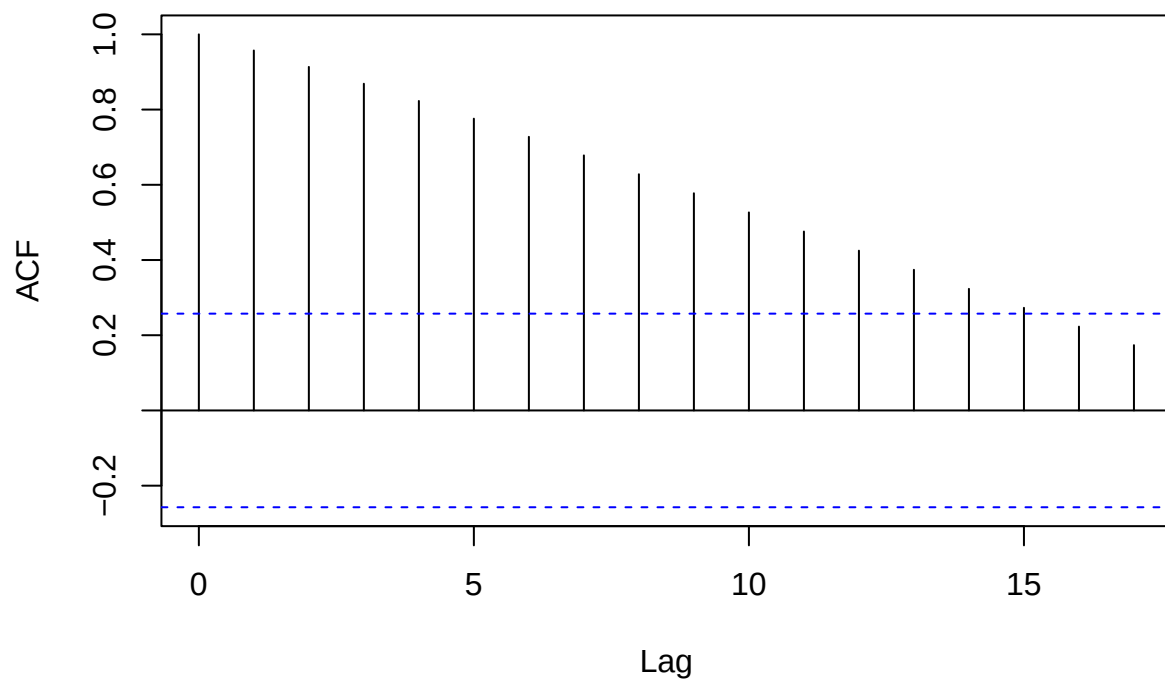
```
pacf(log_exports, main = "PACF of Log Exports")
```

PACF of Log Exports

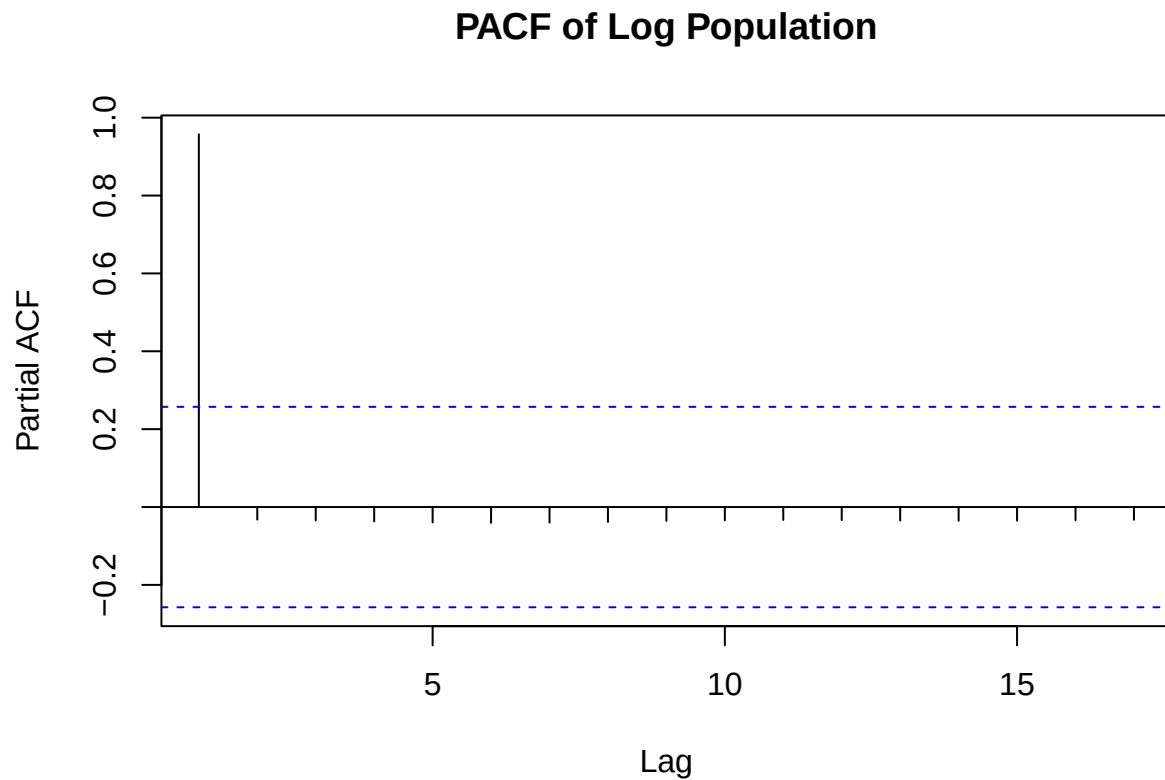


```
acf(log_population, main = "ACF of Log Population")
```

ACF of Log Population



```
pacf(log_population, main = "PACF of Log Population")
```



Log transformation improves plots slightly, but still needs differencing since log transformation plot is non-stationary. The transformed Central African Republic GDP and Population both have upwards trend and the transformed exports data has a downwards trend. This means we have non-stationary data and will still need to do differencing.

Log-Outliers

```
log_gdp_z <- scale(log_gdp)
which(abs(log_gdp_z) > 3)
```

```
## integer(0)
```

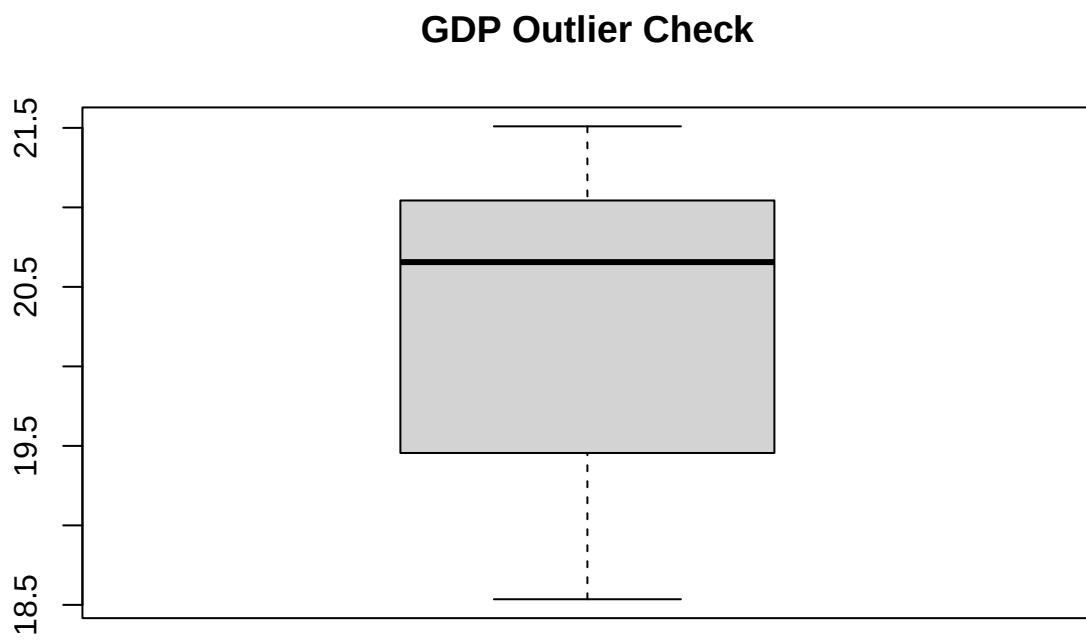
```
log_exports_z <- scale(log_exports)
which(abs(log_exports_z) > 3)
```

```
## integer(0)
```

```
log_pop_z <- scale(log_population)
which(abs(log_pop_z) > 3)
```

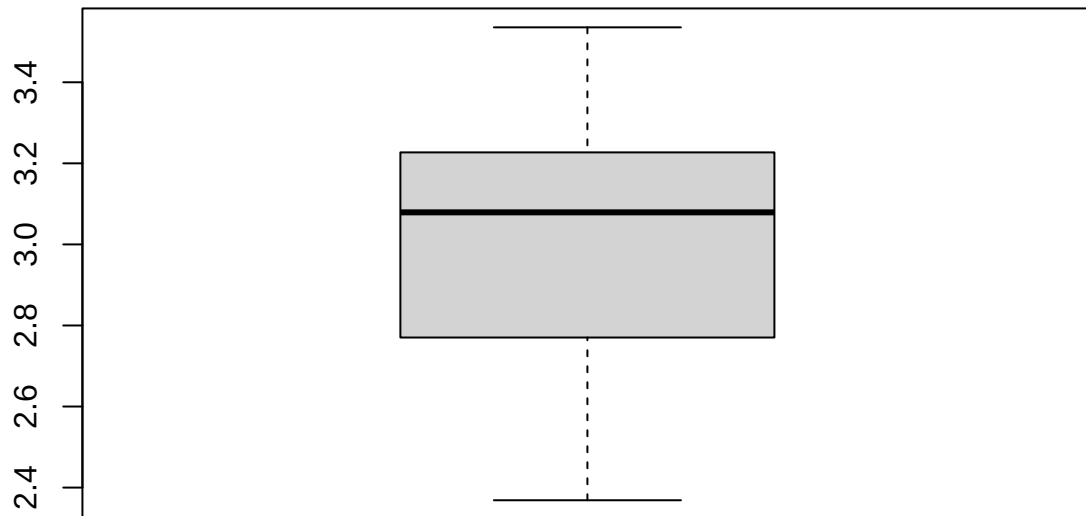
```
## integer(0)
```

```
boxplot(log_gdp, main="GDP Outlier Check")
```



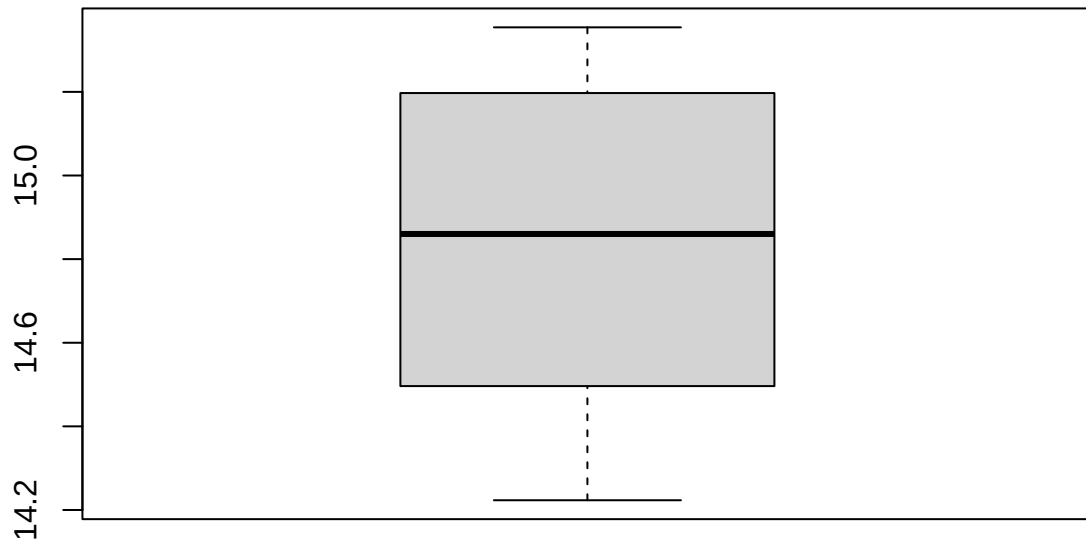
```
boxplot(log_exports, main="GDP Outlier Check")
```

GDP Outlier Check



```
boxplot(log_population, main="GDP Outlier Check")
```

GDP Outlier Check



```
tsoutliers(log_gdp)
```

```
## $index  
## integer(0)  
##  
## $replacements  
## numeric(0)
```

```
tsoutliers(log_exports)
```

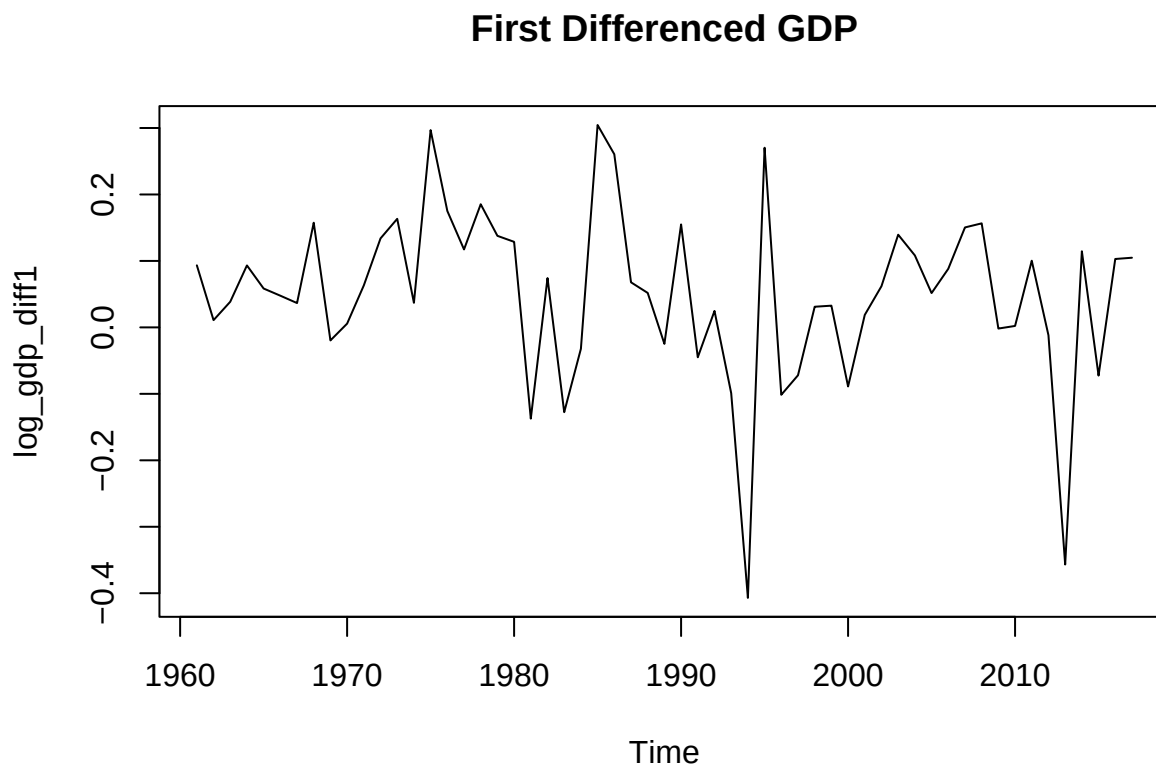
```
## $index  
## integer(0)  
##  
## $replacements  
## numeric(0)
```

```
tsoutliers(log_population)
```

```
## $index  
## integer(0)  
##  
## $replacements  
## numeric(0)
```

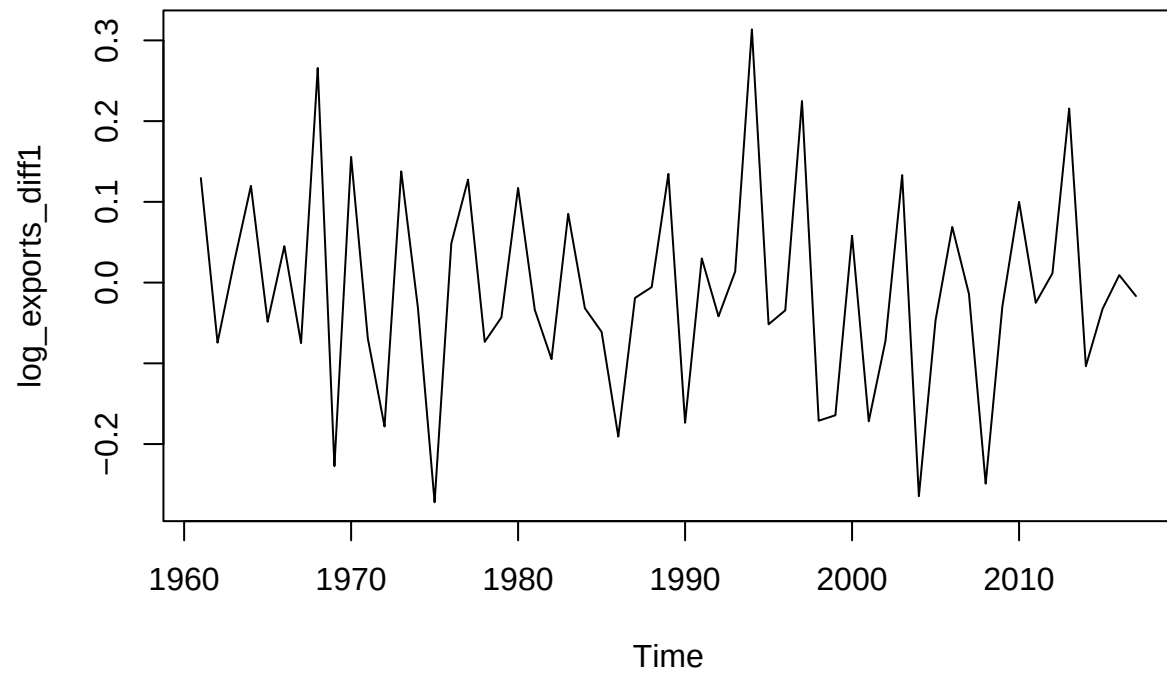
Stationary - (Differencing) Log Transformed Data Differencing can help stabilize the mean of a time series by removing changes in the level of a time series, and therefore eliminating (or reducing) trend and seasonality.

```
log_gdp_diff1 <- diff(log_gdp, differences = 1)
plot(log_gdp_diff1, type = "l", main = "First Differenced GDP")
```



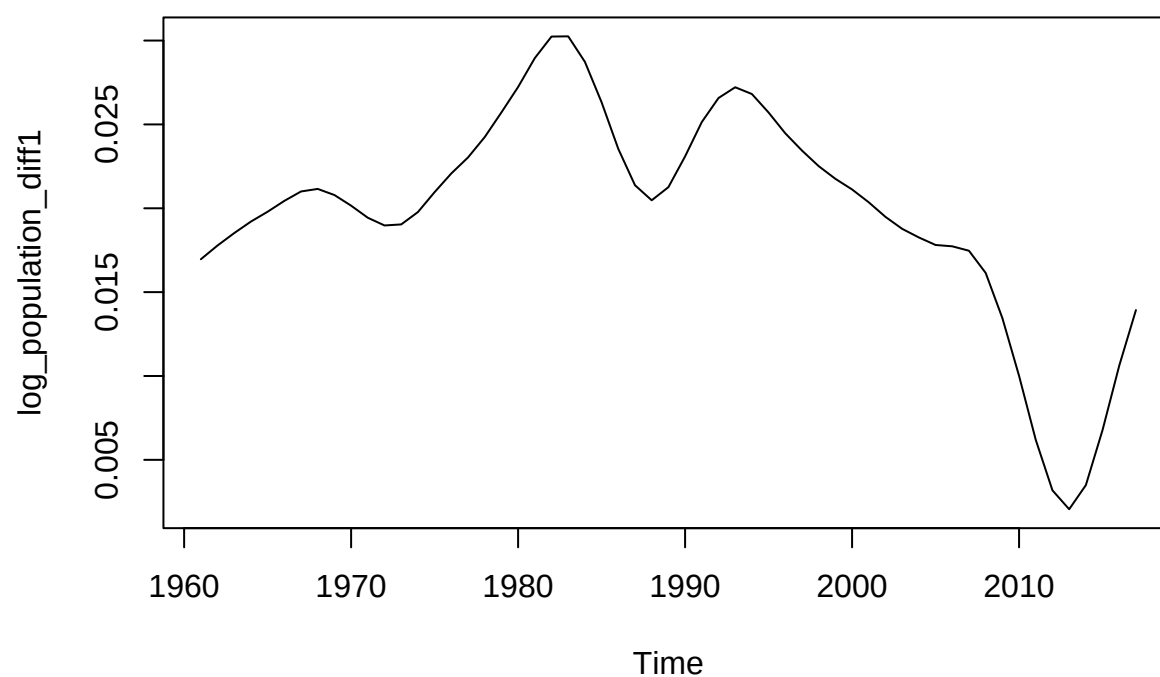
```
log_exports_diff1 <- diff(log_exports, differences = 1)
plot(log_exports_diff1, type = "l", main = "First Differenced Exports")
```

First Differenced Exports



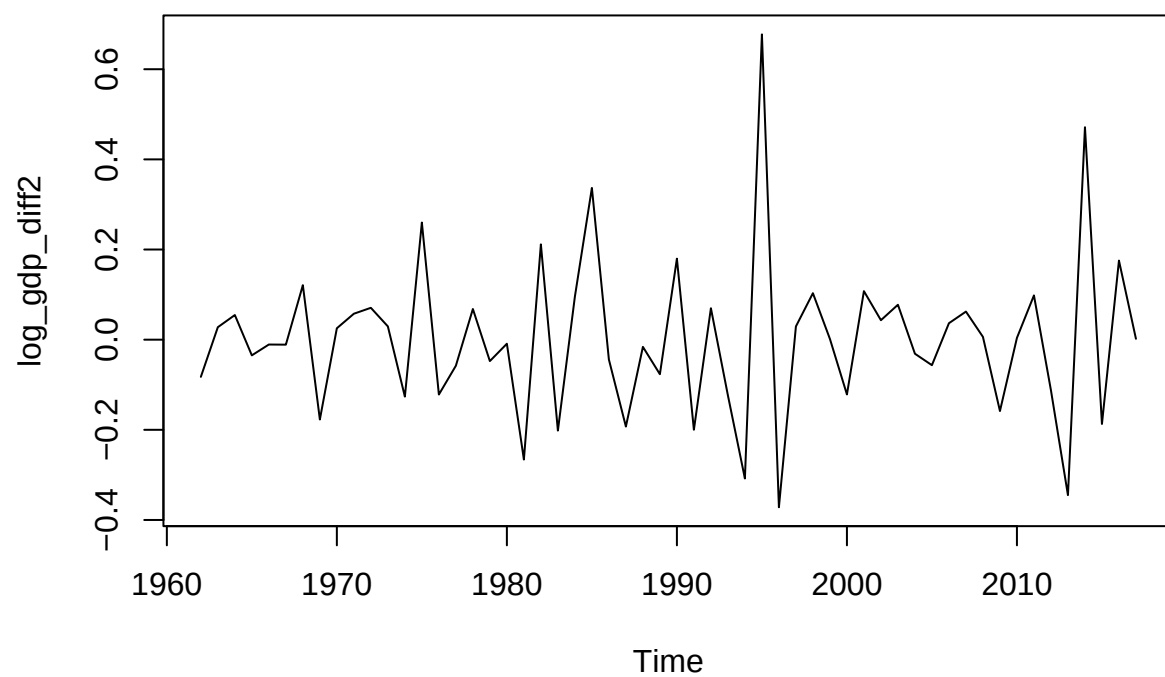
```
log_population_diff1 <- diff(log_population, differences = 1)
plot(log_population_diff1, type = "l", main = "First Differenced Population")
```

First Differenced Population



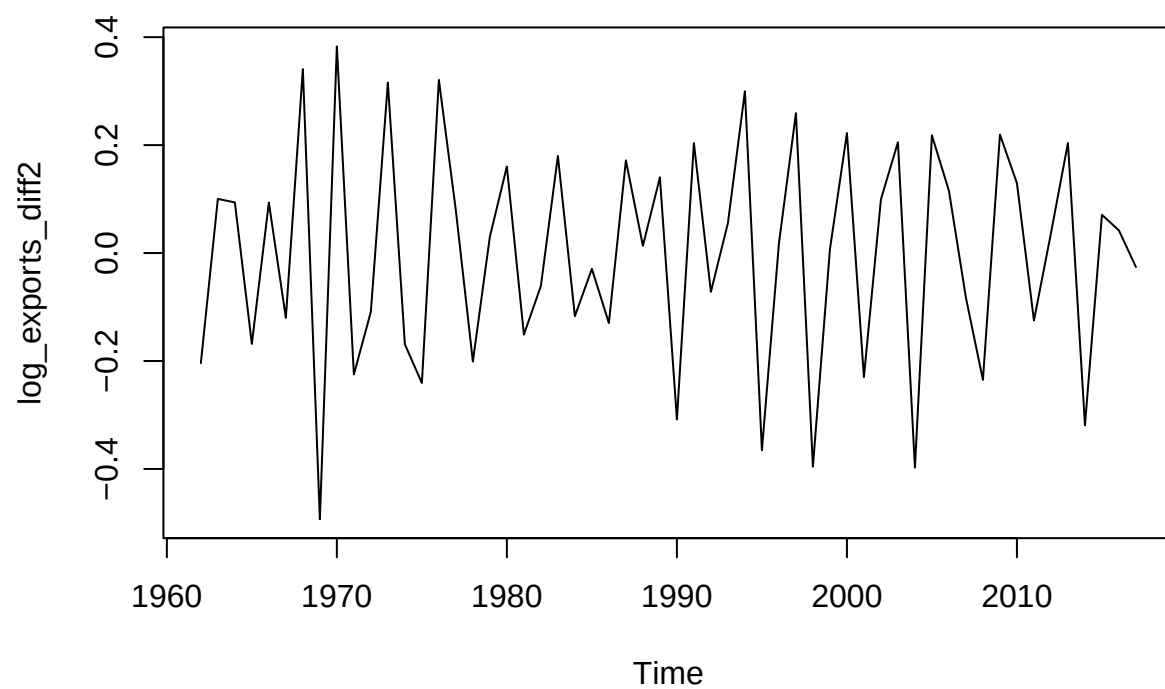
```
log_gdp_diff2 <- diff(log_gdp, differences = 2)
plot(log_gdp_diff2, type = "l", main = "Second Differenced GDP")
```

Second Differenced GDP

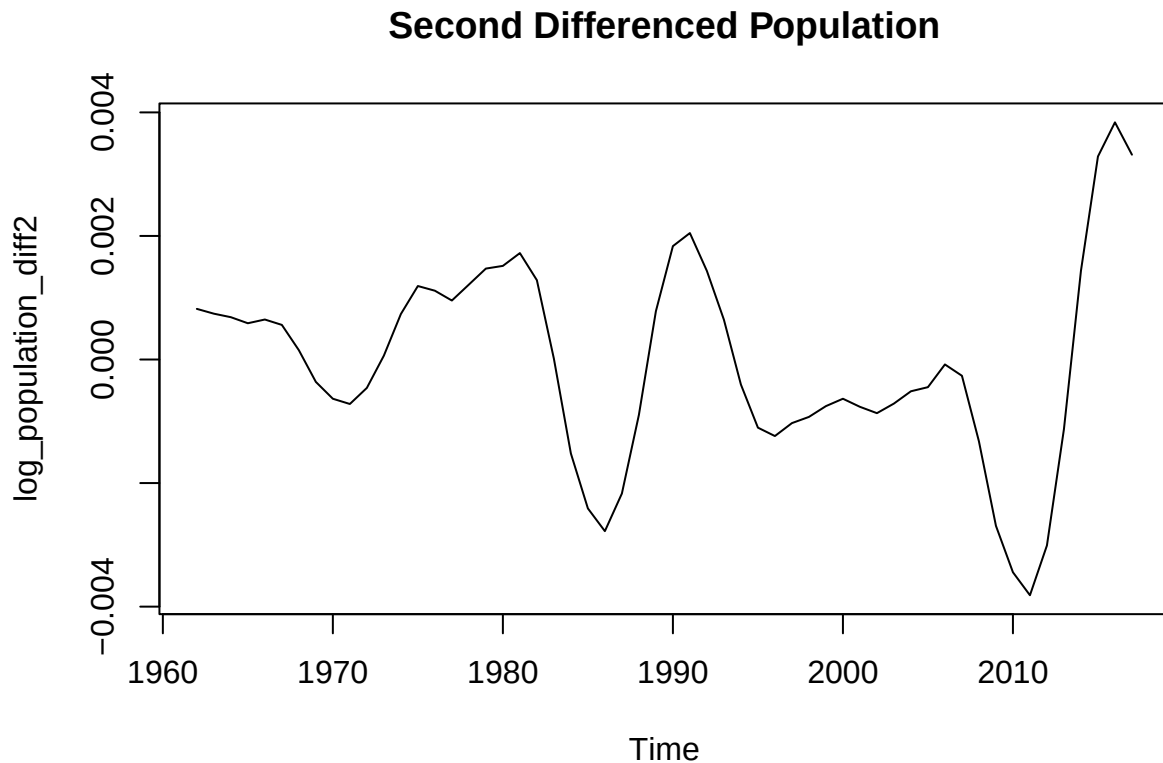


```
log_exports_diff2 <- diff(log_exports, differences = 2)
plot(log_exports_diff2, type = "l", main = "Second Differenced Exports")
```

Second Differenced Exports



```
log_population_diff2 <- diff(log_population, differences = 2)
plot(log_population_diff2, type = "l", main = "Second Differenced Population")
```



Log Transformed Difference 1:

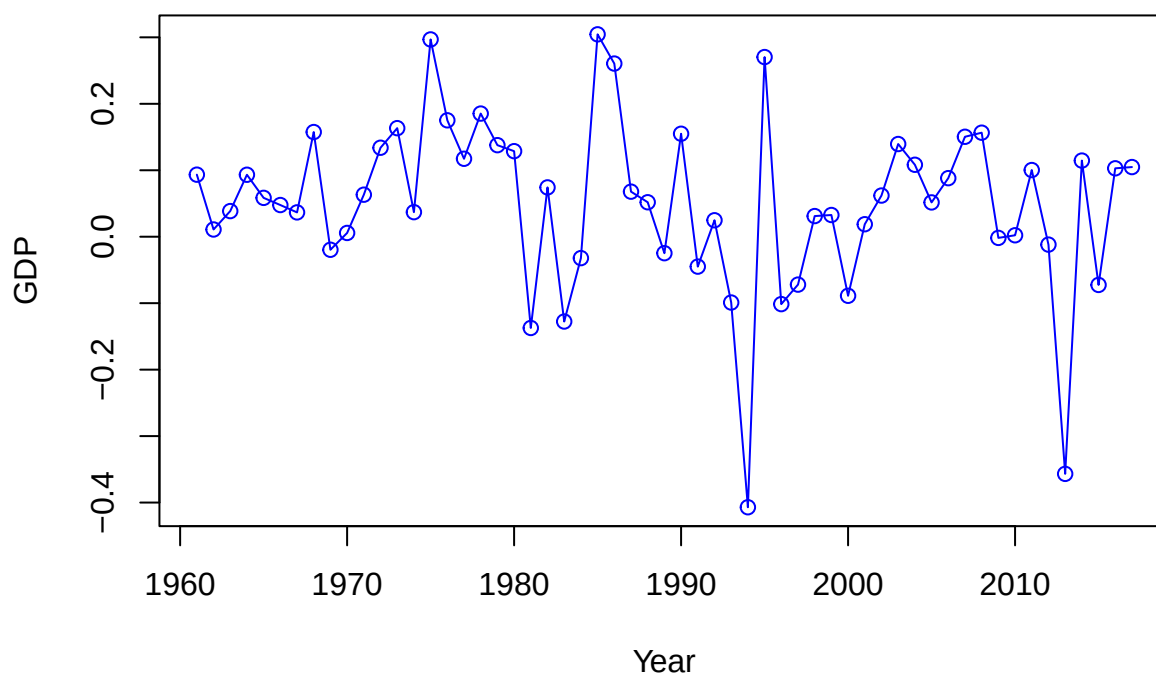
```
plot(log_gdp_diff1,
     type = "o",
     col = "blue",
     ylab = "GDP",
     xlab = "Year",
     main = "Transformed Central African Republic GDP (1960-2017)")
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Transformed Central African Republic GDP (1960-2017)' in
## 'mbcsToSbcs': dot substituted for <e2>
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Transformed Central African Republic GDP (1960-2017)' in
## 'mbcsToSbcs': dot substituted for <80>
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Transformed Central African Republic GDP (1960-2017)' in
## 'mbcsToSbcs': dot substituted for <93>
```

Transformed Central African Republic GDP (1960...2017)



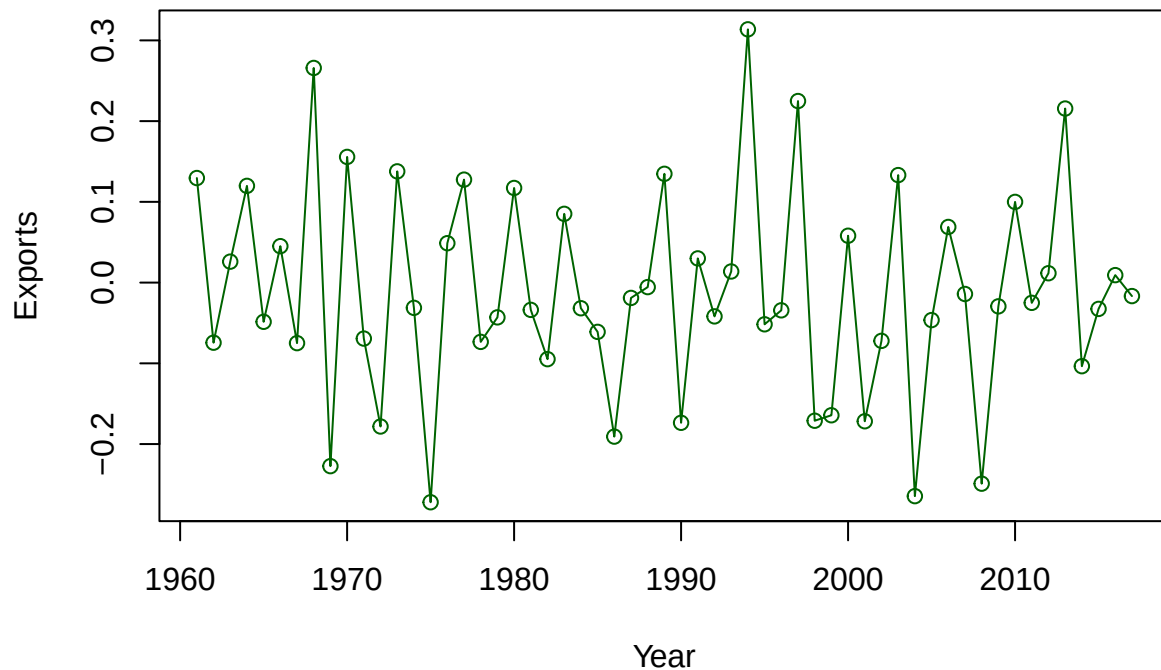
```
plot(log_exports_diff1,  
     type = "o",  
     col = "darkgreen",  
     ylab = "Exports",  
     xlab = "Year",  
     main = "Transformed Central African Republic Exports (1960-2017)")
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion  
## failure on 'Transformed Central African Republic Exports (1960-2017)' in  
## 'mbcsToSbcs': dot substituted for <e2>
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion  
## failure on 'Transformed Central African Republic Exports (1960-2017)' in  
## 'mbcsToSbcs': dot substituted for <80>
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion  
## failure on 'Transformed Central African Republic Exports (1960-2017)' in  
## 'mbcsToSbcs': dot substituted for <93>
```

Transformed Central African Republic Exports (1960...2017)



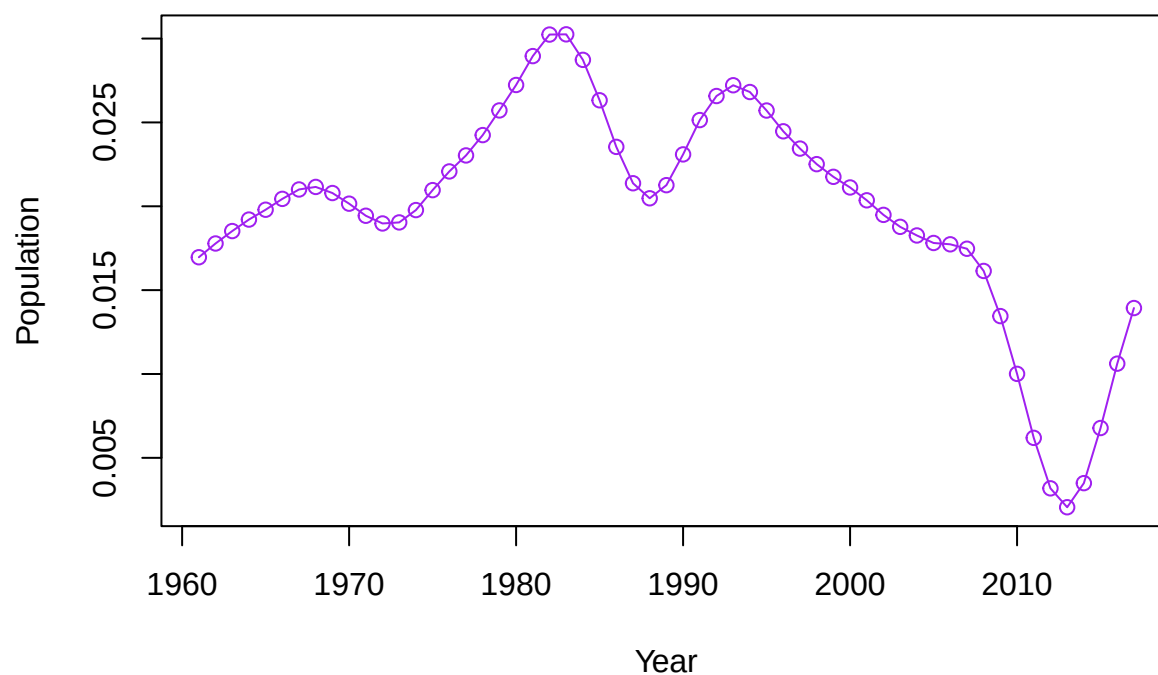
```
plot(log_population_diff1,
     type = "o",
     col = "purple",
     ylab = "Population",
     xlab = "Year",
     main = "Transformed Central African Republic Population (1960-2017)")
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Transformed Central African Republic Population (1960-2017)' in
## 'mbcsToSbcs': dot substituted for <e2>
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Transformed Central African Republic Population (1960-2017)' in
## 'mbcsToSbcs': dot substituted for <80>
```

```
## Warning in title(main = main, xlab = xlab, ylab = ylab, ...): conversion
## failure on 'Transformed Central African Republic Population (1960-2017)' in
## 'mbcsToSbcs': dot substituted for <93>
```

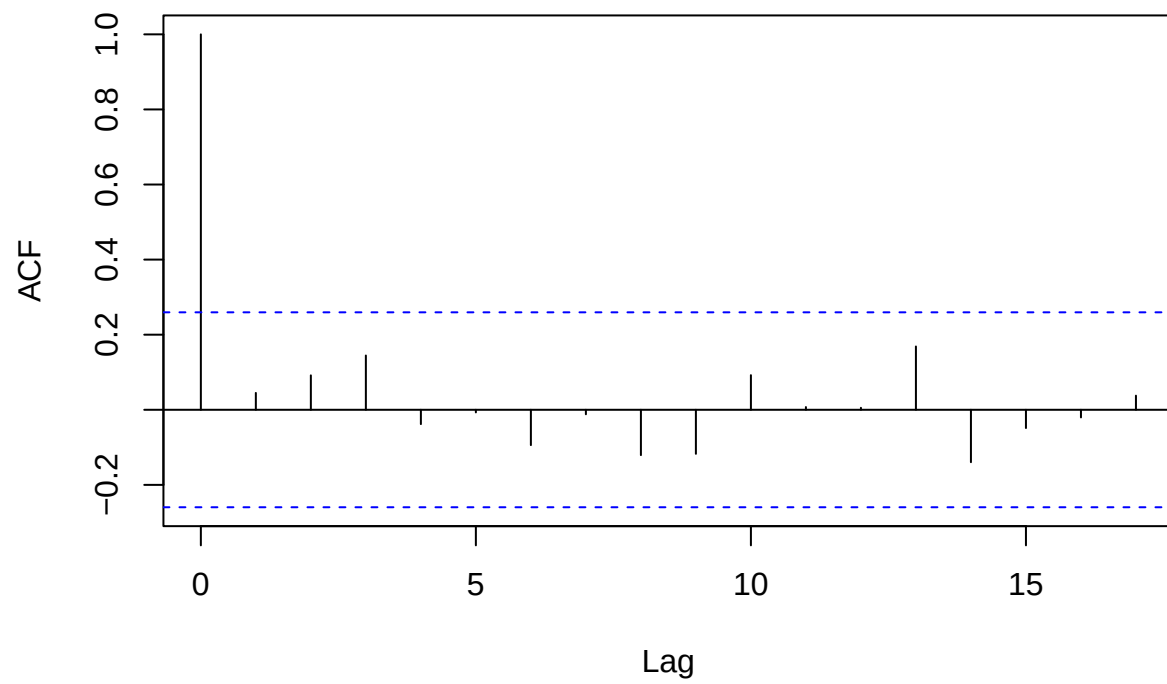
Transformed Central African Republic Population (1960...2017)



ACF/PACF Log Transformation:

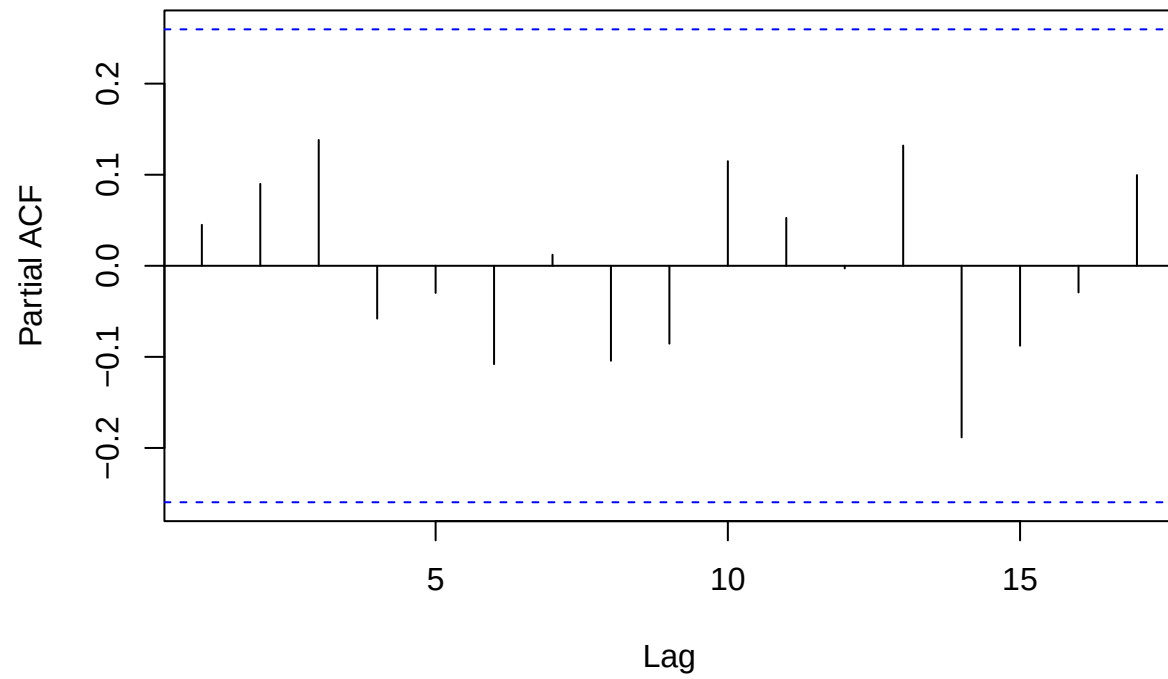
```
acf(log_gdp_diff1, main = "ACF of Log-Differenced GDP (1st Difference)")
```

ACF of Log-Differenced GDP (1st Difference)



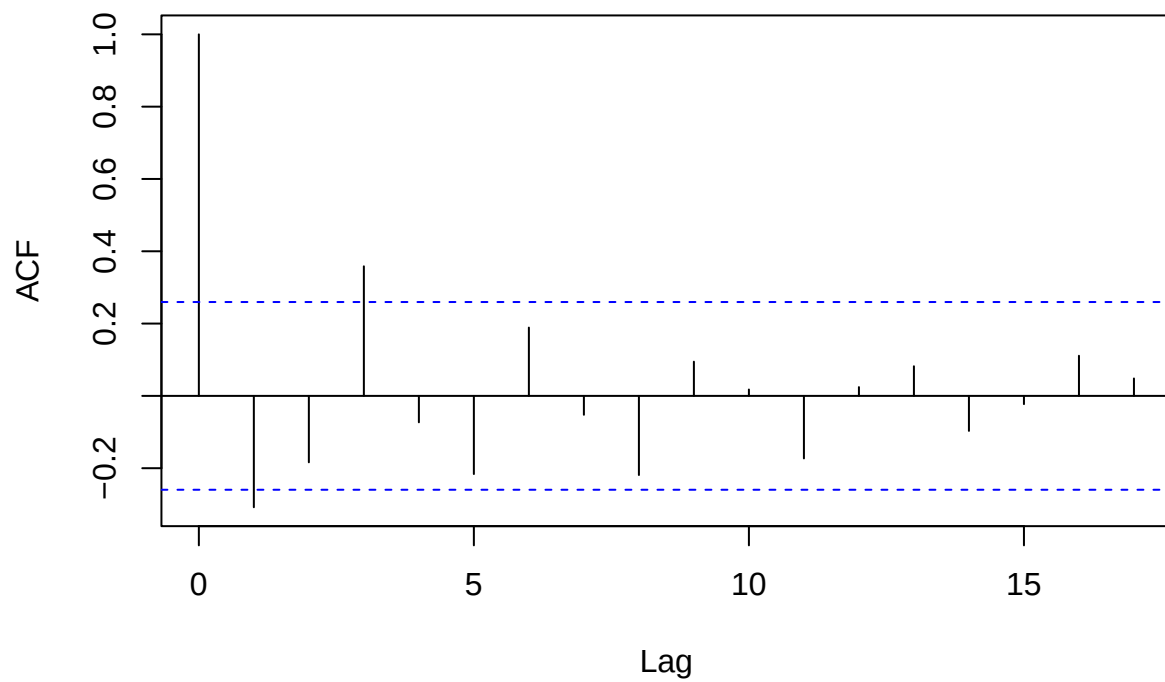
```
pacf(log_gdp_diff1, main = "PACF of Log-Differenced GDP (1st Difference)")
```

PACF of Log-Differenced GDP (1st Difference)



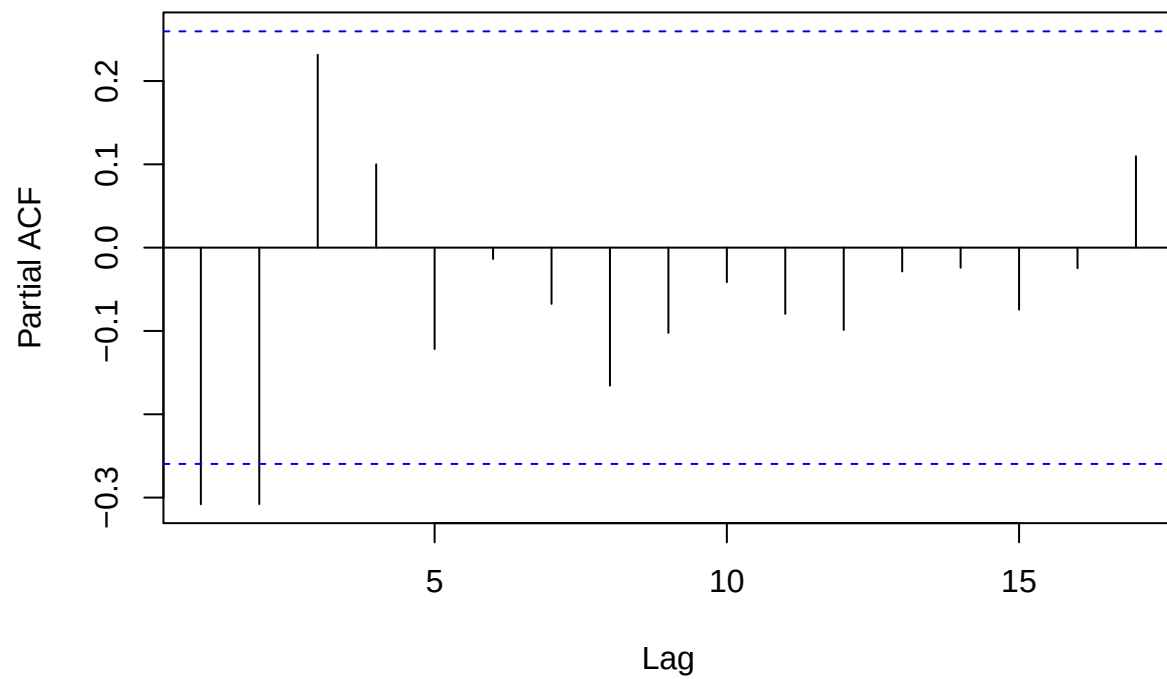
```
acf(log_exports_diff1, main = "ACF of Log-Differenced Exports (1st Difference)")
```

ACF of Log-Differenced Exports (1st Difference)



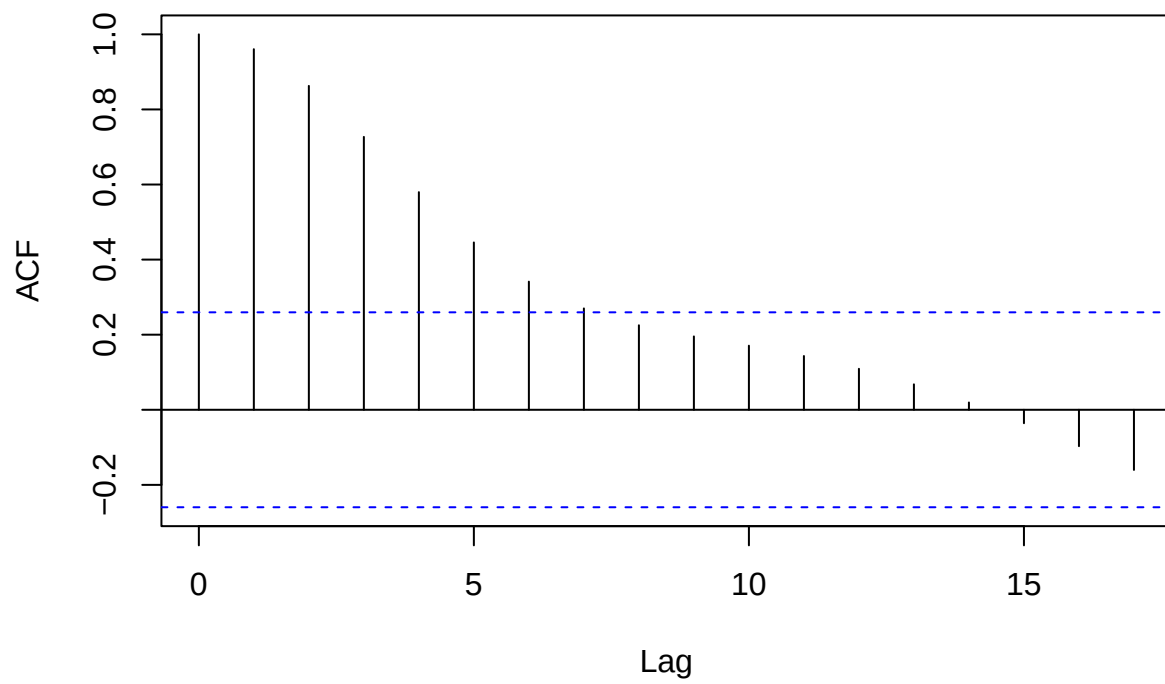
```
pacf(log_exports_diff1, main = "PACF of Log-Differenced Exports (1st Difference)")
```

PACF of Log-Differenced Exports (1st Difference)



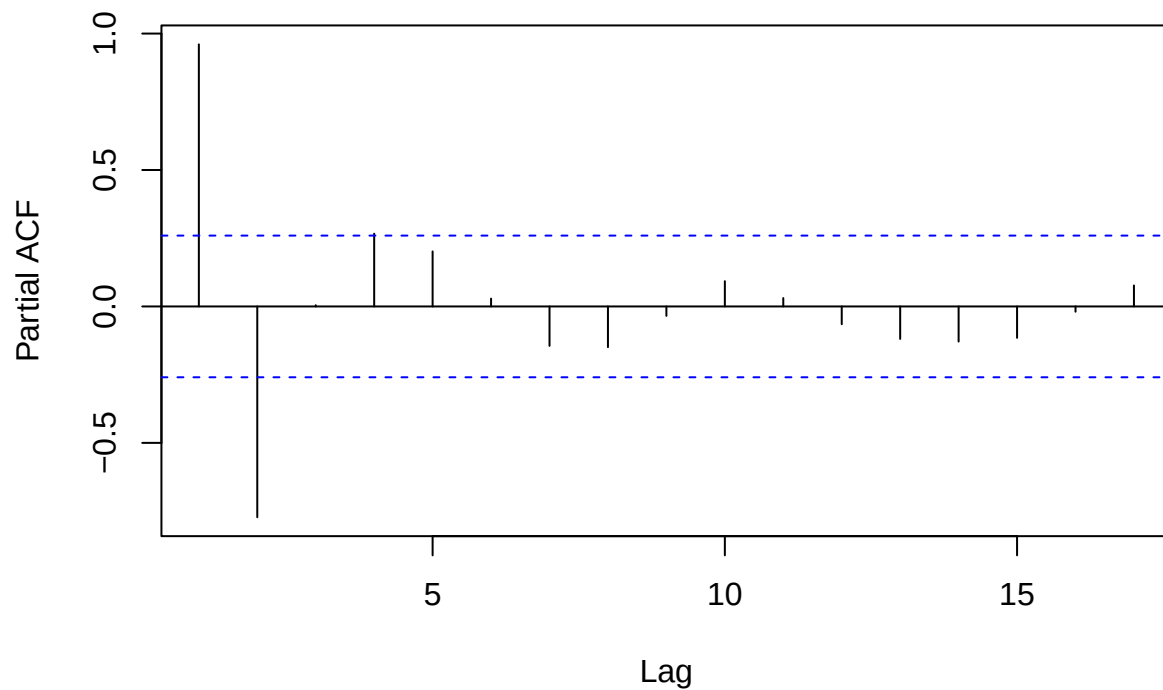
```
acf(log_population_diff1, main = "ACF of Log-Differenced Population (1st Difference)")
```

ACF of Log-Differenced Population (1st Difference)



```
pacf(log_population_diff1, main = "PACF of Log-Differenced Population (1st Difference)")
```

PACF of Log-Differenced Population (1st Difference)



```
# acf(log_gdp_diff2)
# pacf(log_gdp_diff2)
#
# acf(log_exports_diff2)
# pacf(log_exports_diff2)
#
# acf(log_population_diff2)
# pacf(log_population_diff2)
```

With difference of 1: GDP: MA(1), MA(2) or later ARMA(1,1), ARMA(2,1)

Exports: MA(2), AR(2), or later MA(1), AR(1), ARMA(1,1), ARMA(1,2), ARMA(2,1), ARMA(2,2)

Population: AR(1), AR(2), or later ARMA(1,1), ARMA(1,2)

Prefer simpler models

Just for reference:

```
library(tseries)
```

```
## Warning: package 'tseries' was built under R version 4.3.3
```

```
adf.test(log_gdp)
```

```
##
```

```
## Augmented Dickey-Fuller Test
##
## data: log_gdp
## Dickey-Fuller = -1.9149, Lag order = 3, p-value = 0.609
## alternative hypothesis: stationary
```

```
adf.test(log_exports)
```

```
##
## Augmented Dickey-Fuller Test
##
## data: log_exports
## Dickey-Fuller = -2.9724, Lag order = 3, p-value = 0.1821
## alternative hypothesis: stationary
```

```
adf.test(log_population)
```

```
##
## Augmented Dickey-Fuller Test
##
## data: log_population
## Dickey-Fuller = -2.1218, Lag order = 3, p-value = 0.5255
## alternative hypothesis: stationary
```

```
adf.test(log_gdp_diff1)
```

```
##
## Augmented Dickey-Fuller Test
##
## data: log_gdp_diff1
## Dickey-Fuller = -3.3099, Lag order = 3, p-value = 0.07899
## alternative hypothesis: stationary
```

```
adf.test(log_exports_diff1)
```

```
##
## Augmented Dickey-Fuller Test
##
## data: log_exports_diff1
## Dickey-Fuller = -3.263, Lag order = 3, p-value = 0.08637
## alternative hypothesis: stationary
```

```
adf.test(log_population_diff1)
```

```
##
## Augmented Dickey-Fuller Test
##
## data: log_population_diff1
## Dickey-Fuller = -1.4801, Lag order = 3, p-value = 0.7844
## alternative hypothesis: stationary
```

```
adf.test(log_gdp_diff2)
```

```
## Warning in adf.test(log_gdp_diff2): p-value smaller than printed p-value
```

```
##  
## Augmented Dickey-Fuller Test  
##  
## data: log_gdp_diff2  
## Dickey-Fuller = -5.1059, Lag order = 3, p-value = 0.01  
## alternative hypothesis: stationary
```

```
adf.test(log_exports_diff2)
```

```
## Warning in adf.test(log_exports_diff2): p-value smaller than printed p-value
```

```
##  
## Augmented Dickey-Fuller Test  
##  
## data: log_exports_diff2  
## Dickey-Fuller = -5.6121, Lag order = 3, p-value = 0.01  
## alternative hypothesis: stationary
```

```
adf.test(log_population_diff2)
```

```
##  
## Augmented Dickey-Fuller Test  
##  
## data: log_population_diff2  
## Dickey-Fuller = -3.2152, Lag order = 3, p-value = 0.09399  
## alternative hypothesis: stationary
```

Finding a Candidate Model GDP:

```
#arma(p, d, q)
```

```
#GDP
```

```
gdp_m1 <- Arima(log_gdp, order = c(1,1,0), include.mean = TRUE) # MA(1)  
gdp_m2 <- Arima(log_gdp, order = c(0,1,2), include.mean = TRUE) # MA(2)
```

```
#Exports
```

```
exports_m1 <- Arima(log_exports, order = c(0,1,2), include.mean = TRUE) # MA(2)  
exports_m2 <- Arima(log_exports, order = c(2,1,0), include.mean = TRUE) # AR(2)  
exports_m3 <- Arima(log_exports, order = c(2,1,2), include.mean = TRUE) # ARMA(2,2)
```

```
#Population
```

```
population_m1 <- Arima(log_population, order = c(2,1,0), include.mean = TRUE) # AR(2)  
population_m2 <- Arima(log_population, order = c(1,1,0), include.mean = TRUE) # AR(1)
```

AIC/BIC Choose BIC because it penalizes model complexity more.

```
summary(gdp_m1)
```

```
## Series: log_gdp
## ARIMA(1,1,0)
##
## Coefficients:
##      ar1
##      0.1642
## s.e.  0.1306
##
## sigma^2 = 0.01885: log likelihood = 32.8
## AIC=-61.61  AICc=-61.38  BIC=-57.52
##
## Training set error measures:
##              ME      RMSE      MAE      MPE      MAPE      MASE
## Training set 0.04174331 0.1349037 0.102239 0.2077372 0.499746 0.963075
##              ACF1
## Training set -0.1324105
```

```
summary(gdp_m2)
```

```
## Series: log_gdp
## ARIMA(0,1,2)
##
## Coefficients:
##      ma1      ma2
##      0.0827 0.1508
## s.e.  0.1416 0.1327
##
## sigma^2 = 0.01889: log likelihood = 33.24
## AIC=-60.48  AICc=-60.03  BIC=-54.35
##
## Training set error measures:
##              ME      RMSE      MAE      MPE      MAPE      MASE
## Training set 0.04070831 0.1338413 0.09857835 0.2025987 0.4821589 0.9285919
##              ACF1
## Training set -0.05876644
```

```
summary(exports_m1)
```

```
## Series: log_exports
## ARIMA(0,1,2)
##
## Coefficients:
##      ma1      ma2
##      -0.4281 0.1207
## s.e.  0.1836 0.2222
##
## sigma^2 = 0.01515: log likelihood = 39.46
## AIC=-72.92  AICc=-72.46  BIC=-66.79
##
```

```
## Training set error measures:
##           ME           RMSE           MAE           MPE           MAPE           MASE
## Training set -0.01552691 0.1198489 0.08964931 -0.6554892 3.038706 0.8866345
##           ACF1
## Training set 0.01643604
```

```
summary(exports_m2)
```

```
## Series: log_exports
## ARIMA(2,1,0)
##
## Coefficients:
##           ar1           ar2
##          -0.3816        -0.2829
## s.e.    0.1269    0.1255
##
## sigma^2 = 0.01437:  log likelihood = 40.92
## AIC=-75.84  AICc=-75.39  BIC=-69.71
##
## Training set error measures:
##           ME           RMSE           MAE           MPE           MAPE           MASE
## Training set -0.01794603 0.1167409 0.08800061 -0.7426934 2.961359 0.8703288
##           ACF1
## Training set 0.04665926
```

```
summary(exports_m3)
```

```
## Series: log_exports
## ARIMA(2,1,2)
##
## Coefficients:
##           ar1           ar2           ma1           ma2
##          -0.6924        -0.8006         0.3985         0.5571
## s.e.    0.1455    0.1879    0.2067    0.2985
##
## sigma^2 = 0.01354:  log likelihood = 43.49
## AIC=-76.98  AICc=-75.81  BIC=-66.77
##
## Training set error measures:
##           ME           RMSE           MAE           MPE           MAPE           MASE
## Training set -0.01395682 0.1112344 0.08445742 -0.5805899 2.849815 0.8352866
##           ACF1
## Training set -0.03795421
```

```
summary(population_m1)
```

```
## Series: log_population
## ARIMA(2,1,0)
##
## Coefficients:
##           ar1           ar2
##          1.8788        -0.8846
```

```
## s.e.  0.0561  0.0566
##
## sigma^2 = 4.275e-06:  log likelihood = 324.38
## AIC=-642.77  AICc=-642.32  BIC=-636.64
##
## Training set error measures:
##           ME           RMSE           MAE           MPE           MAPE
## Training set 0.0003968695 0.002013452 0.0007831319 0.002742284 0.005319975
##           MASE           ACF1
## Training set 0.03946759 0.1138815
```

```
summary(population_m2)
```

```
## Series: log_population
## ARIMA(1,1,0)
##
## Coefficients:
##          ar1
##          0.9954
## s.e.  0.0054
##
## sigma^2 = 6.199e-06:  log likelihood = 283.93
## AIC=-563.87  AICc=-563.65  BIC=-559.78
##
## Training set error measures:
##           ME           RMSE           MAE           MPE           MAPE
## Training set 0.0003061378 0.002446392 0.001494217 0.002199244 0.01007951
##           MASE           ACF1
## Training set 0.07530424 0.4145039
```

Analysis: GDP

ARIMA(1,1,0)

Coefficients: ar1 0.1642 s.e. 0.1306

sigma^2 = 0.01885: log likelihood = 32.8 AIC=-61.61 AICc=-61.38 BIC=-57.52

Exports Series: log_exports ARIMA(2,1,0)

Coefficients: ar1 ar2 -0.3816 -0.2829 s.e. 0.1269 0.1255

sigma^2 = 0.01437: log likelihood = 40.92 AIC=-75.84 AICc=-75.39 BIC=-69.71

Population Series: log_population ARIMA(2,1,0)

Coefficients: ar1 ar2 1.8788 -0.8846 s.e. 0.0561 0.0566

sigma^2 = 4.275e-06: log likelihood = 324.38 AIC=-642.77 AICc=-642.32 BIC=-636.64

Coeffs:

```
get_arma_coefs <- function(model) {
  if (is.null(model$var.coef)) stop("Model does not have variance-covariance matrix")
  coefs <- coef(model)
  se <- sqrt(diag(model$var.coef))
  tval <- coefs / se
  pval <- 2 * (1 - pnorm(abs(tval)))
}
```

```
data.frame(
  Estimate = coefs,
  SE = se,
  t.value = tval,
  p.value = pval,
  row.names = names(coefs)
)
```

```
# Example for GDP
get_arima_coefs(gdp_m1)
```

```
##      Estimate      SE t.value  p.value
## ar1 0.1641636 0.1305829 1.25716 0.2086958
```

```
get_arima_coefs(exports_m2)
```

```
##      Estimate      SE  t.value  p.value
## ar1 -0.3815858 0.1268795 -3.007467 0.002634349
## ar2 -0.2829136 0.1254718 -2.254798 0.024146001
```

```
get_arima_coefs(population_m1)
```

```
##      Estimate      SE  t.value p.value
## ar1  1.8788207 0.05613192 33.47152      0
## ar2 -0.8846124 0.05656786 -15.63808      0
```

Model Diagnostics (Checking residual)

```
library(astsa)
```

```
## Warning: package 'astsa' was built under R version 4.3.3
```

```
##
## Attaching package: 'astsa'
```

```
## The following object is masked _by_ 'GlobalEnv':
##
##      gdp
```

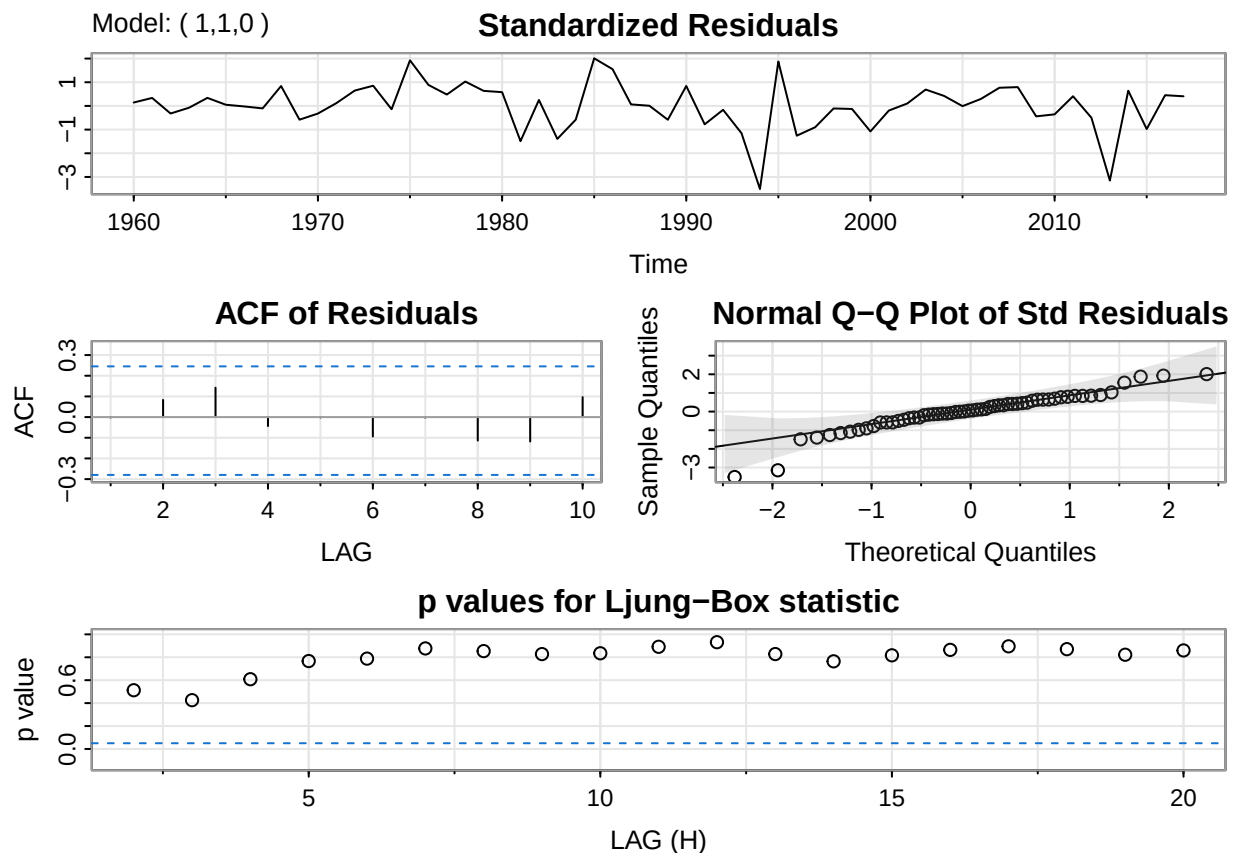
```
## The following object is masked from 'package:forecast':
##
##      gas
```

```
#GDP:
arma_110 <- sarima(log(gdp), 1,1,0)
```

```

## initial value -2.043604
## iter 2 value -2.044628
## iter 3 value -2.044635
## iter 4 value -2.044638
## iter 4 value -2.044638
## iter 4 value -2.044638
## final value -2.044638
## converged
## initial value -2.052438
## iter 2 value -2.052450
## iter 3 value -2.052458
## iter 3 value -2.052458
## iter 3 value -2.052458
## final value -2.052458
## converged
## <><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1           0.0444 0.1315  0.3380  0.7367
## constant      0.0502 0.0178  2.8207  0.0067
##
## sigma^2 estimated as 0.01649082 on 55 degrees of freedom
##
## AIC = -1.161776  AICc = -1.157878  BIC = -1.054247
##

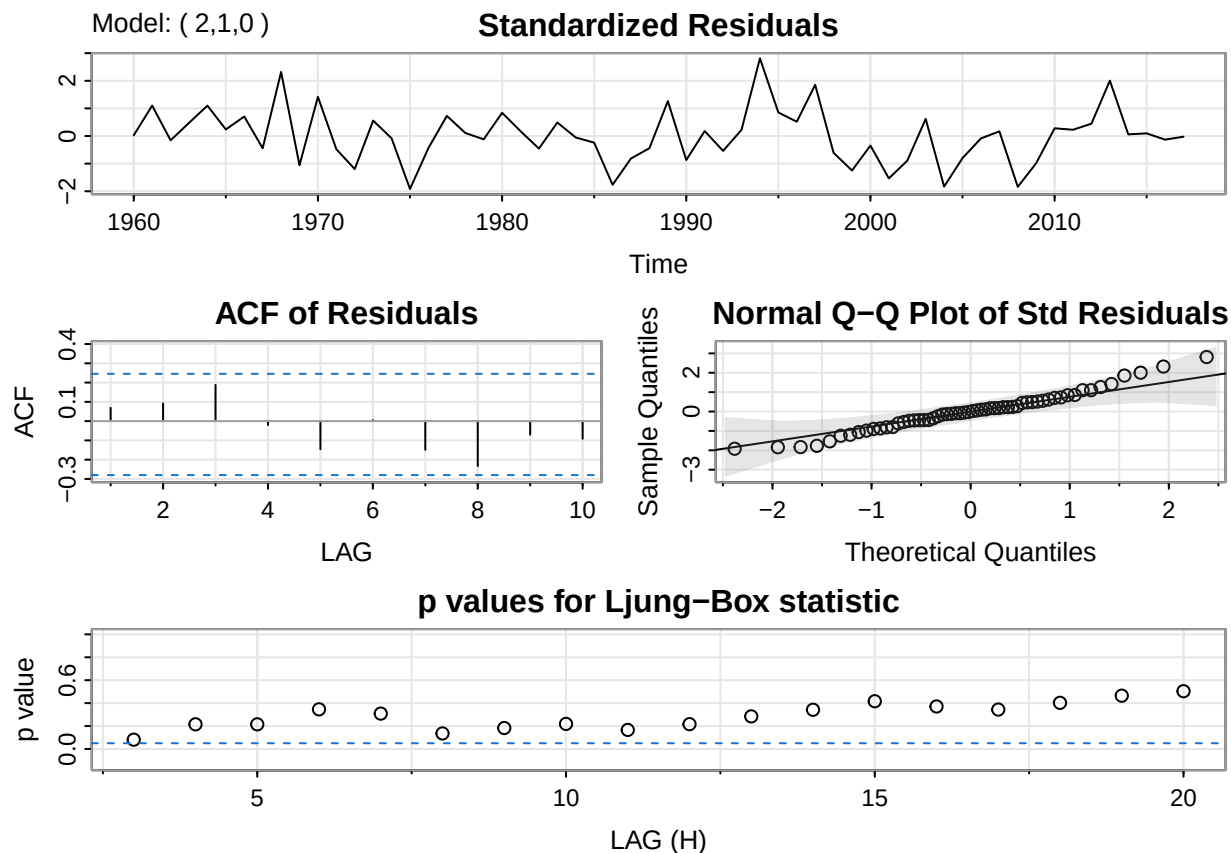
```



#Exports:

```
arima_210_exports <- sarima(log(exports), 2,1,0)
```

```
## initial value -2.046549
## iter 2 value -2.136489
## iter 3 value -2.140824
## iter 4 value -2.143878
## iter 5 value -2.145964
## iter 6 value -2.146586
## iter 7 value -2.146587
## iter 8 value -2.146587
## iter 8 value -2.146587
## final value -2.146587
## converged
## initial value -2.150777
## iter 2 value -2.150816
## iter 3 value -2.150825
## iter 4 value -2.150826
## iter 5 value -2.150826
## iter 5 value -2.150826
## iter 5 value -2.150826
## final value -2.150826
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      -0.4035 0.1263 -3.1957 0.0023
## ar2      -0.3037 0.1247 -2.4351 0.0182
## constant -0.0117 0.0091 -1.2895 0.2027
##
## sigma^2 estimated as 0.01347641 on 54 degrees of freedom
##
## AIC = -1.323425 AICc = -1.31548 BIC = -1.180053
##
```



Exports: AR(1) (-0.4035) p-value is 0.0023 so that suggests that coefficient is significant AR(2) (-0.3037) p-value is 0.0182 so that suggests that coefficient is significant constant (-0.0117) p-value is 0.2027, so not significant

#Population:

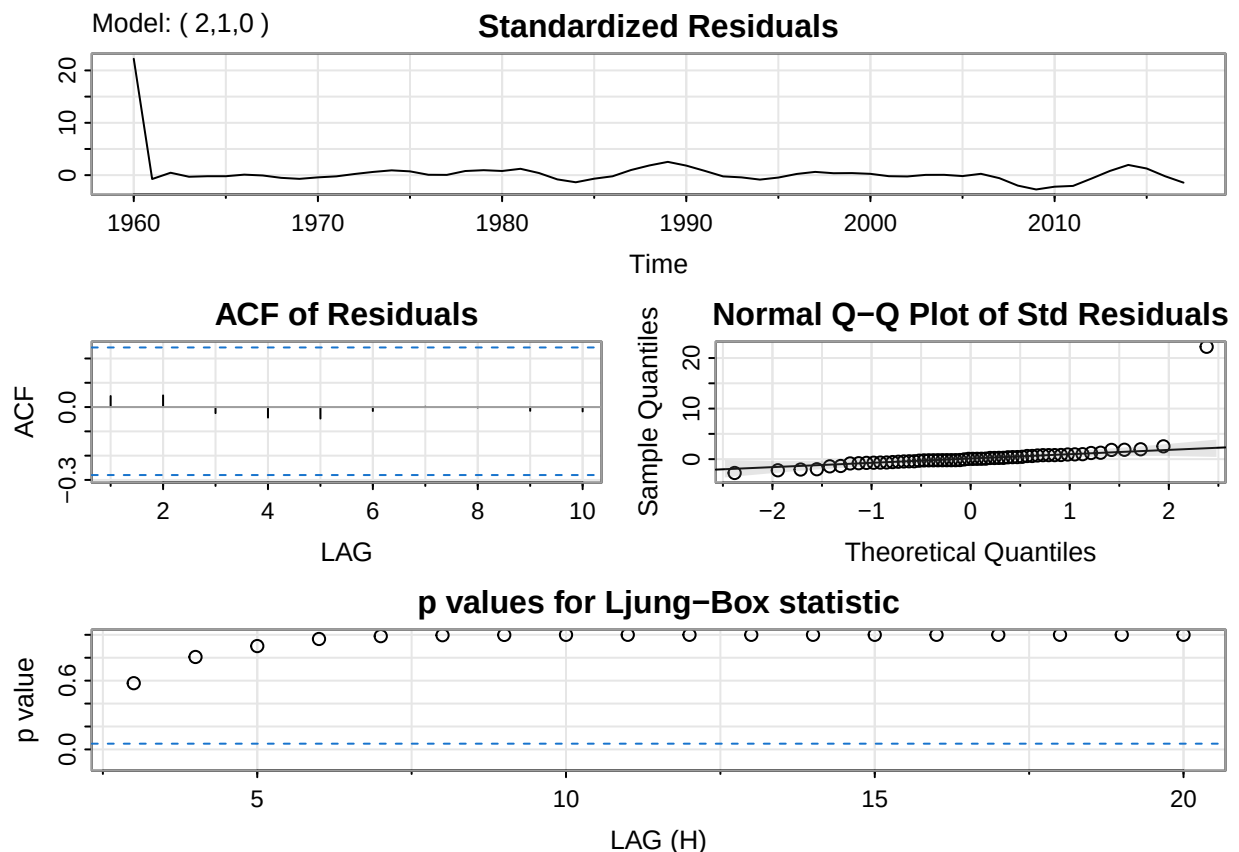
```
arima_210_pop <- sarima(log(population), 2,1,0)
```

```
## initial value -5.030781
## iter 2 value -5.114043
## iter 3 value -6.094349
## iter 4 value -6.144149
## iter 5 value -6.557846
## iter 6 value -7.233697
## iter 7 value -7.273859
## iter 8 value -7.332591
## iter 9 value -7.333733
## iter 10 value -7.343975
## iter 11 value -7.348005
## iter 12 value -7.348103
## iter 13 value -7.348105
## iter 13 value -7.348105
## iter 13 value -7.348105
## final value -7.348105
## converged
## initial value -7.289435
## iter 2 value -7.290530
## iter 3 value -7.293845
```

```

## iter    4 value -7.296074
## iter    5 value -7.296795
## iter    6 value -7.296905
## iter    7 value -7.296906
## iter    7 value -7.296906
## iter    7 value -7.296906
## final   value -7.296906
## converged
## <><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE  t.value p.value
## ar1           1.8622 0.0443  42.0719     0
## ar2           -0.9296 0.0444 -20.9352     0
## constant      0.0204 0.0013  16.0619     0
##
## sigma^2 estimated as 4.084694e-07 on 54 degrees of freedom
##
## AIC = -11.61558  AICc = -11.60764  BIC = -11.47221
##

```



Population: AR(1) (1.8622) p-value is 0 so that suggests that coefficient is significant AR(2) (-0.9296) p-value is 0 so that suggests that coefficient is significant constant (0.0204) p-value is 0, so significant

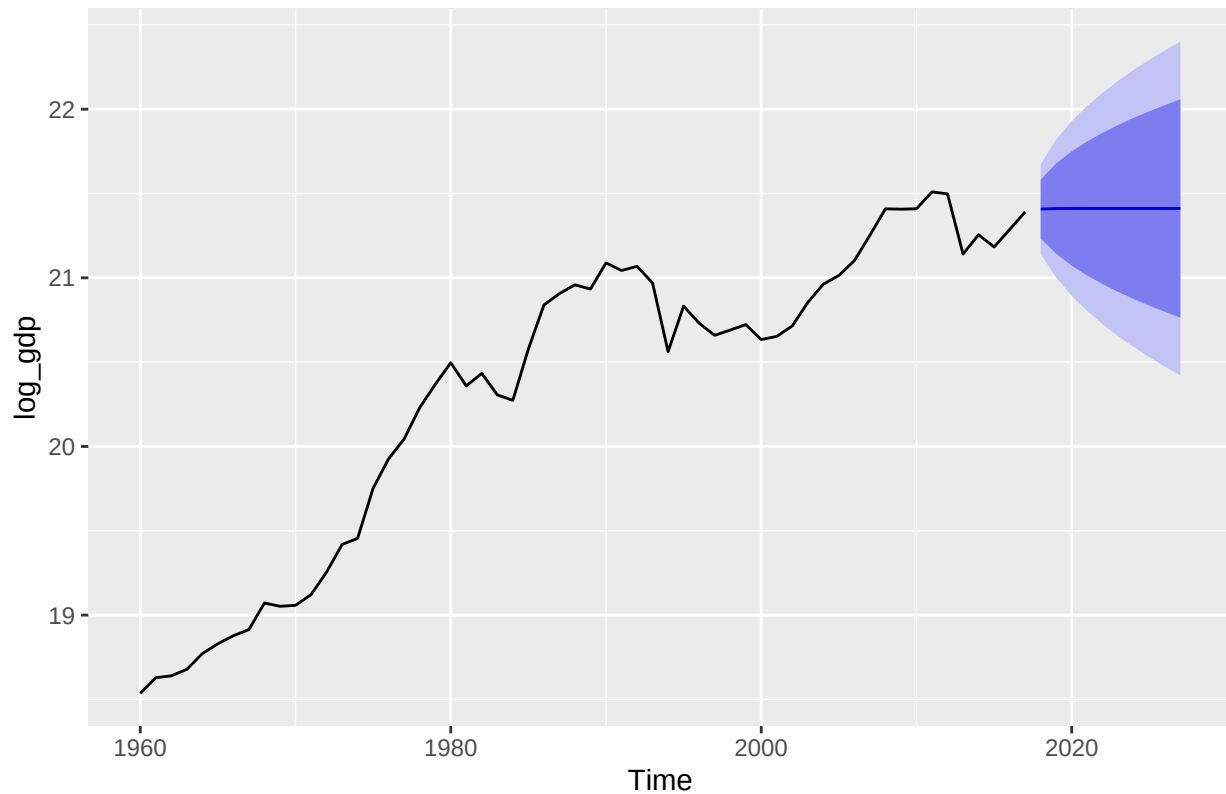
The final models- GDP: ARIMA(1,1,0) - m1 Exports: ARIMA(2,1,0) - m2 Population: ARIMA(2,1,0) - m1

Forecasts:

```
#change to arima
log_gdp_arima110 <- arima(log_gdp, order= c(1,1,0))
log_exports_arima210 <- arima(log_exports, order= c(2,1,0))
log_population_arima210 <- arima(log_population, order= c(2,1,0))
```

```
library(forecast)
forecast_gdp <- forecast(log_gdp_arima110, h=2, level = 95)
autoplot(forecast(log_gdp_arima110))
```

Forecasts from ARIMA(1,1,0)



```
forecast_gdp$mean
```

```
## Time Series:
## Start = 2018
## End = 2019
## Frequency = 1
## [1] 21.40800 21.41082
```

```
forecast_gdp$lower
```

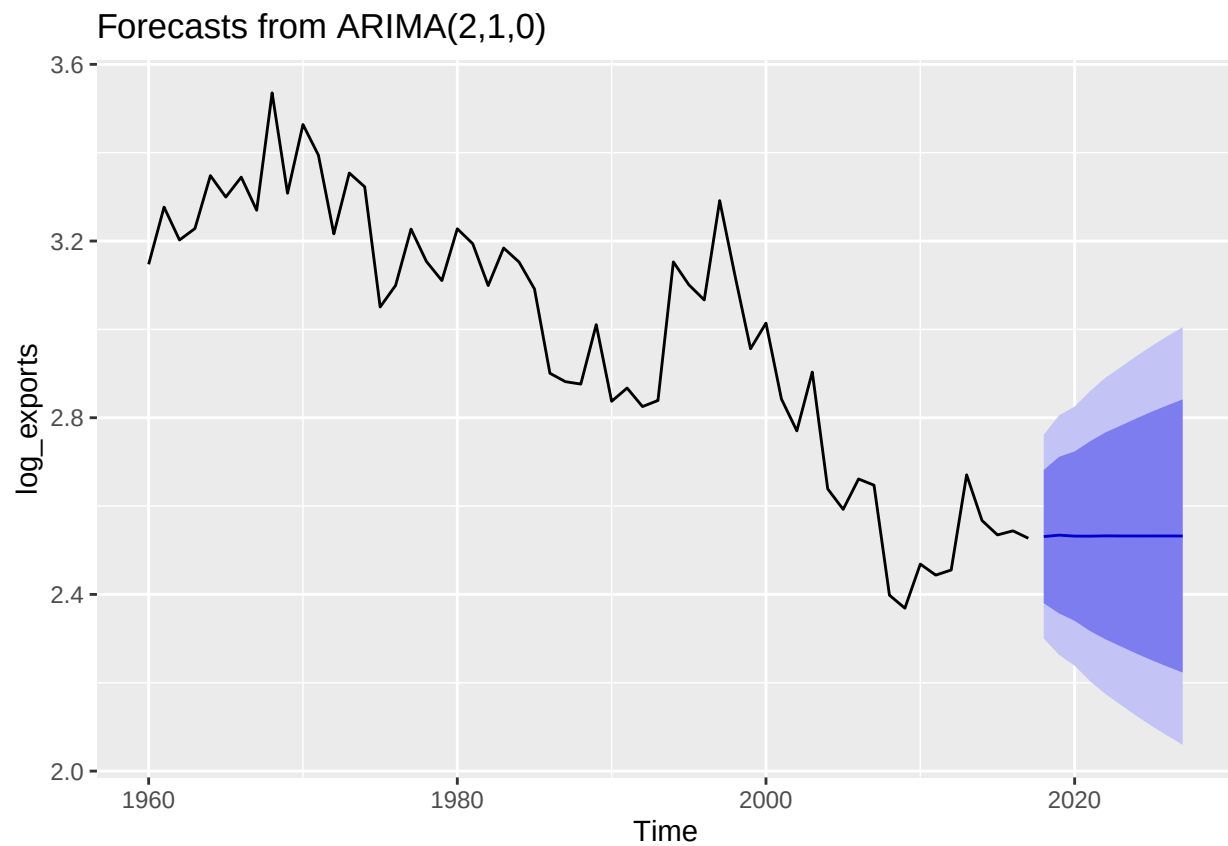
```
## Time Series:
## Start = 2018
## End = 2019
## Frequency = 1
```

```
##          95%
## [1,] 21.14132
## [2,] 21.00156
```

```
forecast_gdp$upper
```

```
## Time Series:
## Start = 2018
## End = 2019
## Frequency = 1
##          95%
## [1,] 21.67467
## [2,] 21.82008
```

```
forecast_exports <- forecast(log_exports_arima210, h=2, level = 95)
autoplot(forecast(log_exports_arima210))
```



```
forecast_exports$mean
```

```
## Time Series:
## Start = 2018
## End = 2019
## Frequency = 1
## [1] 2.530936 2.534229
```

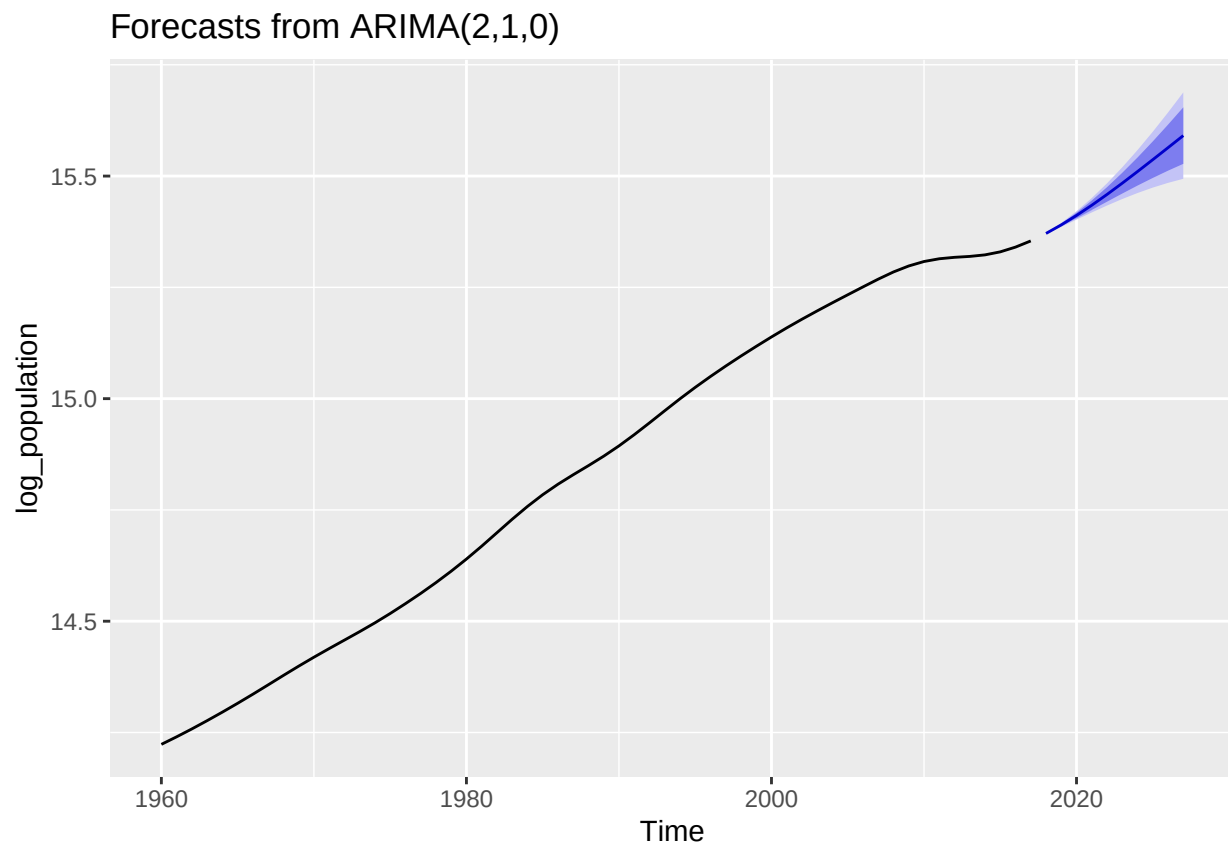
```
forecast_exports$lower
```

```
## Time Series:  
## Start = 2018  
## End = 2019  
## Frequency = 1  
##          95%  
## [1,] 2.300132  
## [2,] 2.262855
```

```
forecast_exports$upper
```

```
## Time Series:  
## Start = 2018  
## End = 2019  
## Frequency = 1  
##          95%  
## [1,] 2.761741  
## [2,] 2.805602
```

```
forecast_population <- forecast(log_population_arma210, h=2, level = 95)  
autoplot(forecast(log_population_arma210))
```



```
forecast_population$mean
```

```
## Time Series:  
## Start = 2018  
## End = 2019  
## Frequency = 1  
## [1] 15.37111 15.39032
```

```
forecast_population$lower
```

```
## Time Series:  
## Start = 2018  
## End = 2019  
## Frequency = 1  
##           95%  
## [1,] 15.36962  
## [2,] 15.38578
```

```
forecast_population$upper
```

```
## Time Series:  
## Start = 2018  
## End = 2019  
## Frequency = 1  
##           95%  
## [1,] 15.37260  
## [2,] 15.39487
```

Back transformed:

```
library(ggplot2)
```

```
gdp_mean_back <- exp(forecast_gdp$mean)  
gdp_mean_back
```

```
## Time Series:  
## Start = 2018  
## End = 2019  
## Frequency = 1  
## [1] 1983237590 1988846394
```

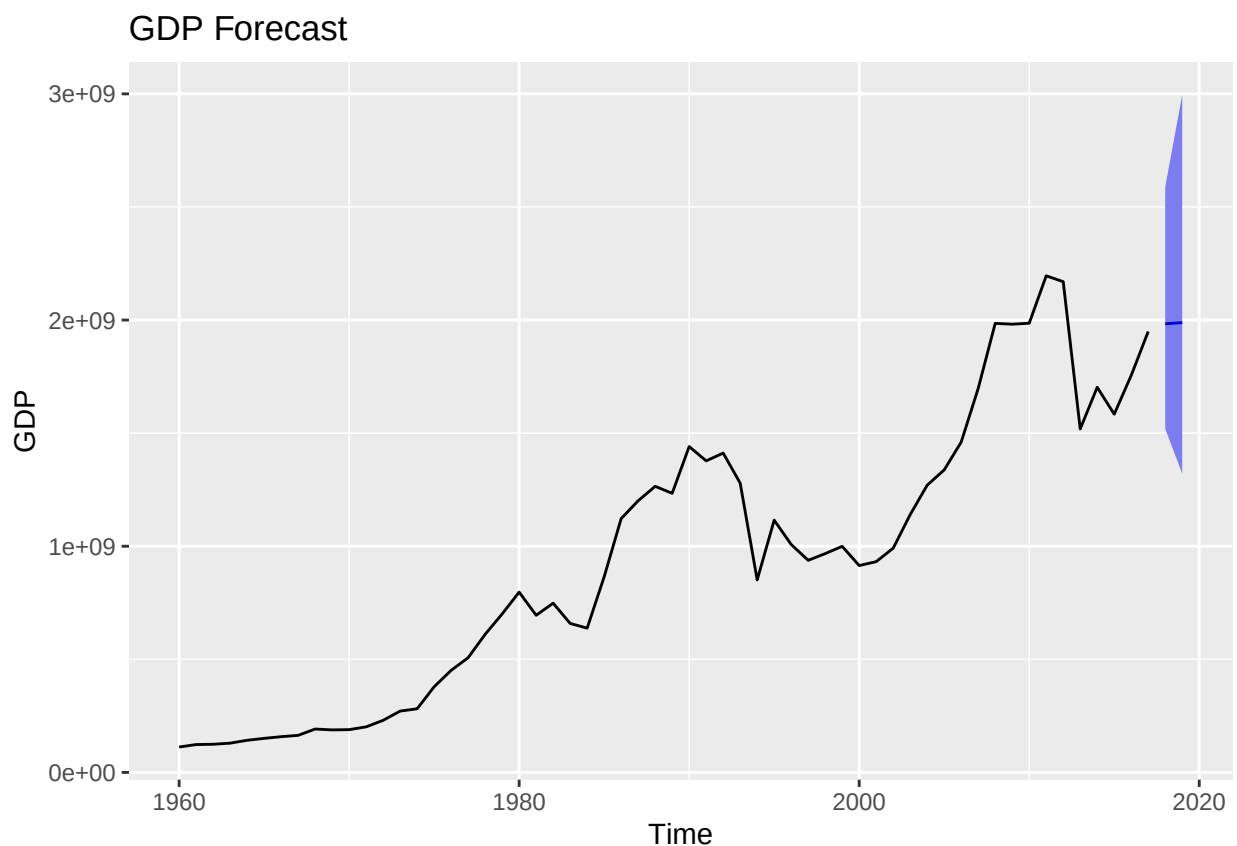
```
gdp_lower_back <- exp(forecast_gdp$lower)  
gdp_lower_back
```

```
## Time Series:  
## Start = 2018  
## End = 2019  
## Frequency = 1  
##           95%  
## [1,] 1519009254  
## [2,] 1320875915
```

```
gdp_upper_back <- exp(forecast_gdp$upper)
gdp_upper_back
```

```
## Time Series:
## Start = 2018
## End = 2019
## Frequency = 1
##          95%
## [1,] 2589339944
## [2,] 2994611328
```

```
forecast_gdp$mean = exp(forecast_gdp$mean)
forecast_gdp$lower = exp(forecast_gdp$lower)
forecast_gdp$upper = exp(forecast_gdp$upper)
forecast_gdp$x = exp(forecast_gdp$x)
autoplot(forecast_gdp) +
  ggtitle("GDP Forecast") +
  xlab("Time") +
  ylab("GDP")
```



```
forecast_gdp$mean
```

```
## Time Series:
## Start = 2018
```

```
## End = 2019
## Frequency = 1
## [1] 1983237590 1988846394
```

```
forecast_gdp$lower
```

```
## Time Series:
## Start = 2018
## End = 2019
## Frequency = 1
##           95%
## [1,] 1519009254
## [2,] 1320875915
```

```
forecast_gdp$upper
```

```
## Time Series:
## Start = 2018
## End = 2019
## Frequency = 1
##           95%
## [1,] 2589339944
## [2,] 2994611328
```

```
exports_mean_back <- exp(forecast_exports$mean)
exports_mean_back
```

```
## Time Series:
## Start = 2018
## End = 2019
## Frequency = 1
## [1] 12.56527 12.60671
```

```
exports_lower_back <- exp(forecast_exports$lower)
exports_lower_back
```

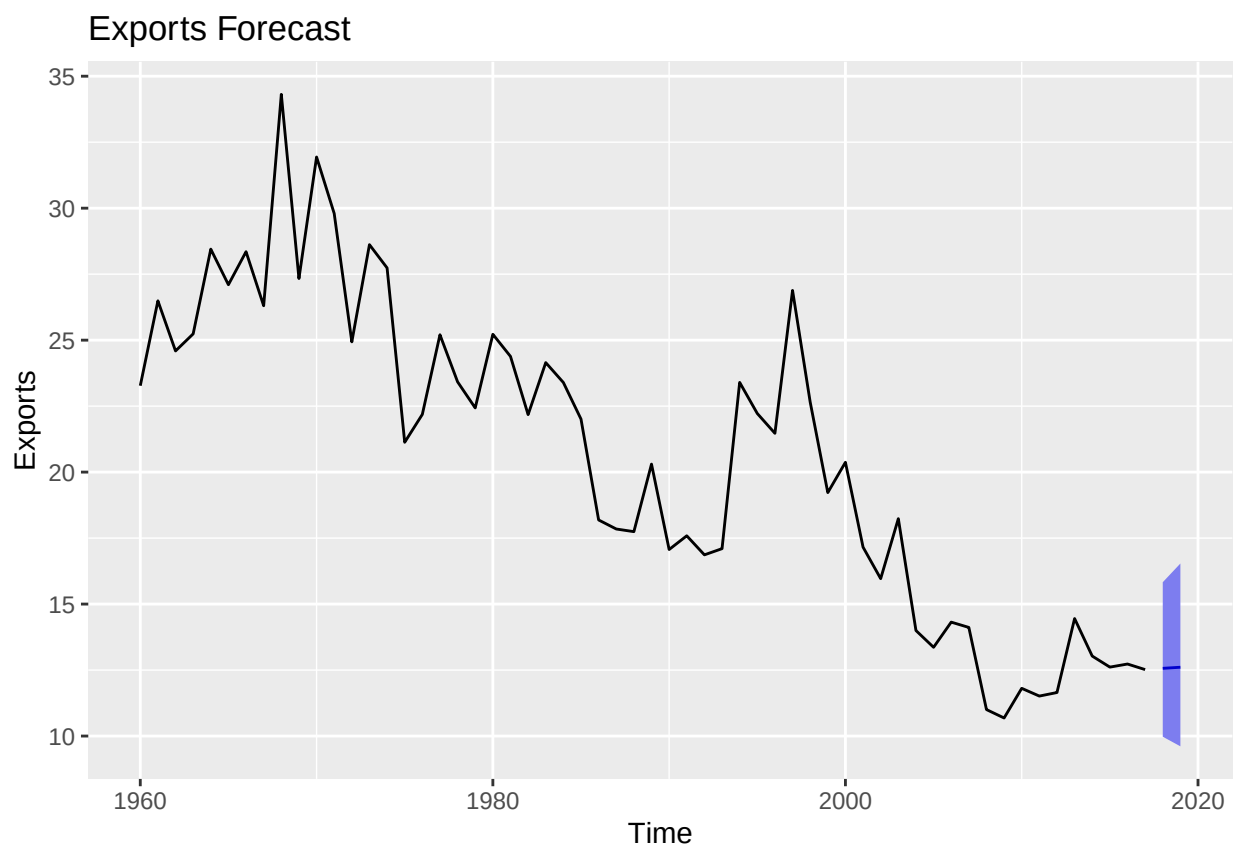
```
## Time Series:
## Start = 2018
## End = 2019
## Frequency = 1
##           95%
## [1,] 9.975494
## [2,] 9.610491
```

```
exports_upper_back <- exp(forecast_exports$upper)
exports_upper_back
```

```
## Time Series:
## Start = 2018
## End = 2019
```

```
## Frequency = 1
##      95%
## [1,] 15.82738
## [2,] 16.53704
```

```
forecast_exports$mean = exp(forecast_exports$mean)
forecast_exports$lower = exp(forecast_exports$lower)
forecast_exports$upper = exp(forecast_exports$upper)
forecast_exports$x = exp(forecast_exports$x)
autoplot(forecast_exports) +
  ggtitle("Exports Forecast") +
  xlab("Time") +
  ylab("Exports")
```



```
forecast_exports$mean
```

```
## Time Series:
## Start = 2018
## End = 2019
## Frequency = 1
## [1] 12.56527 12.60671
```

```
forecast_exports$lower
```

```
## Time Series:
## Start = 2018
## End = 2019
## Frequency = 1
##      95%
## [1,] 9.975494
## [2,] 9.610491
```

```
forecast_exports$upper
```

```
## Time Series:
## Start = 2018
## End = 2019
## Frequency = 1
##      95%
## [1,] 15.82738
## [2,] 16.53704
```

```
population_mean_back <- exp(forecast_population$mean)
population_mean_back
```

```
## Time Series:
## Start = 2018
## End = 2019
## Frequency = 1
## [1] 4737936 4829829
```

```
population_lower_back <- exp(forecast_population$lower)
population_lower_back
```

```
## Time Series:
## Start = 2018
## End = 2019
## Frequency = 1
##      95%
## [1,] 4730878
## [2,] 4807935
```

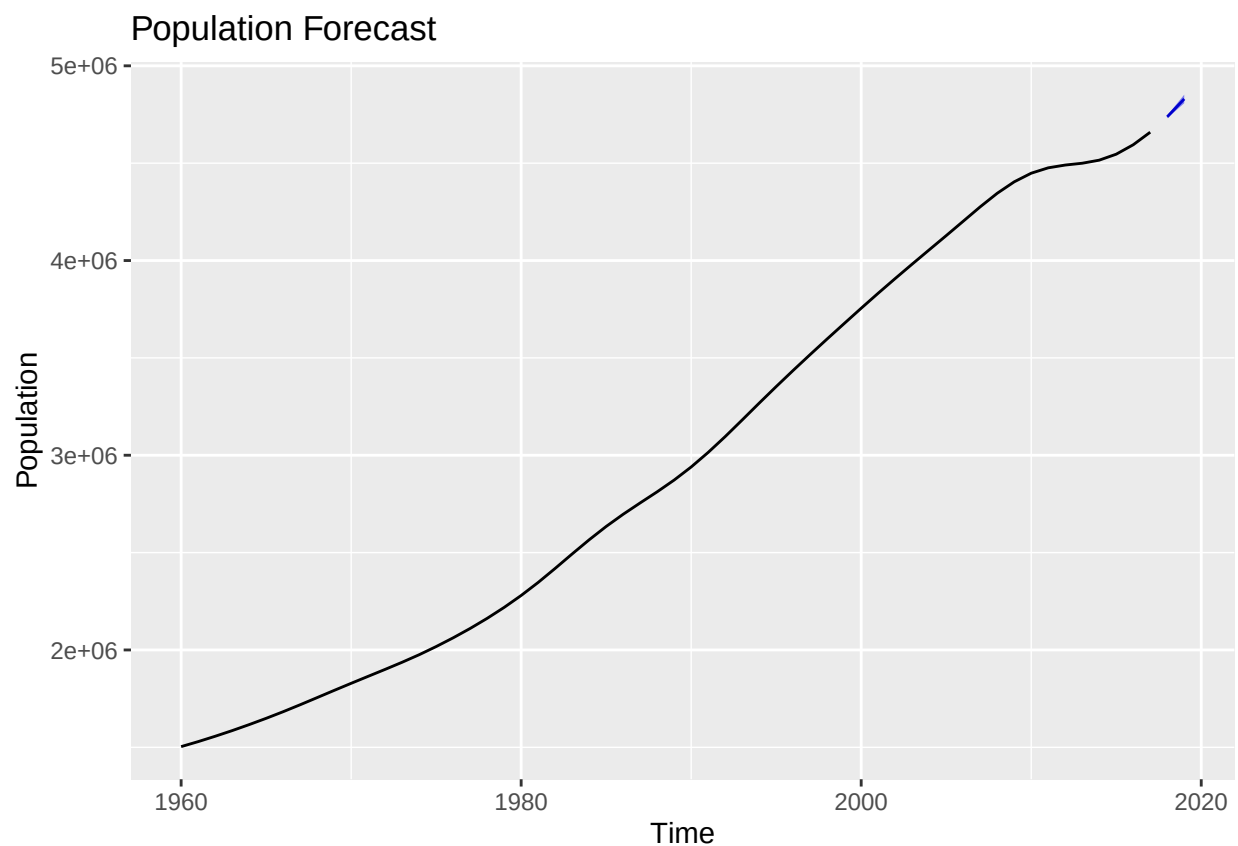
```
population_upper_back <- exp(forecast_population$upper)
population_upper_back
```

```
## Time Series:
## Start = 2018
## End = 2019
## Frequency = 1
##      95%
## [1,] 4745005
## [2,] 4851823
```

```

forecast_population$mean = exp(forecast_population$mean)
forecast_population$lower = exp(forecast_population$lower)
forecast_population$upper = exp(forecast_population$upper)
forecast_population$x = exp(forecast_population$x)
autoplot(forecast_population) +
  ggtitle("Population Forecast") +
  xlab("Time") +
  ylab("Population")

```



```
forecast_population$mean
```

```

## Time Series:
## Start = 2018
## End = 2019
## Frequency = 1
## [1] 4737936 4829829

```

```
forecast_population$lower
```

```

## Time Series:
## Start = 2018
## End = 2019
## Frequency = 1
##          95%

```

```
## [1,] 4730878
## [2,] 4807935
```

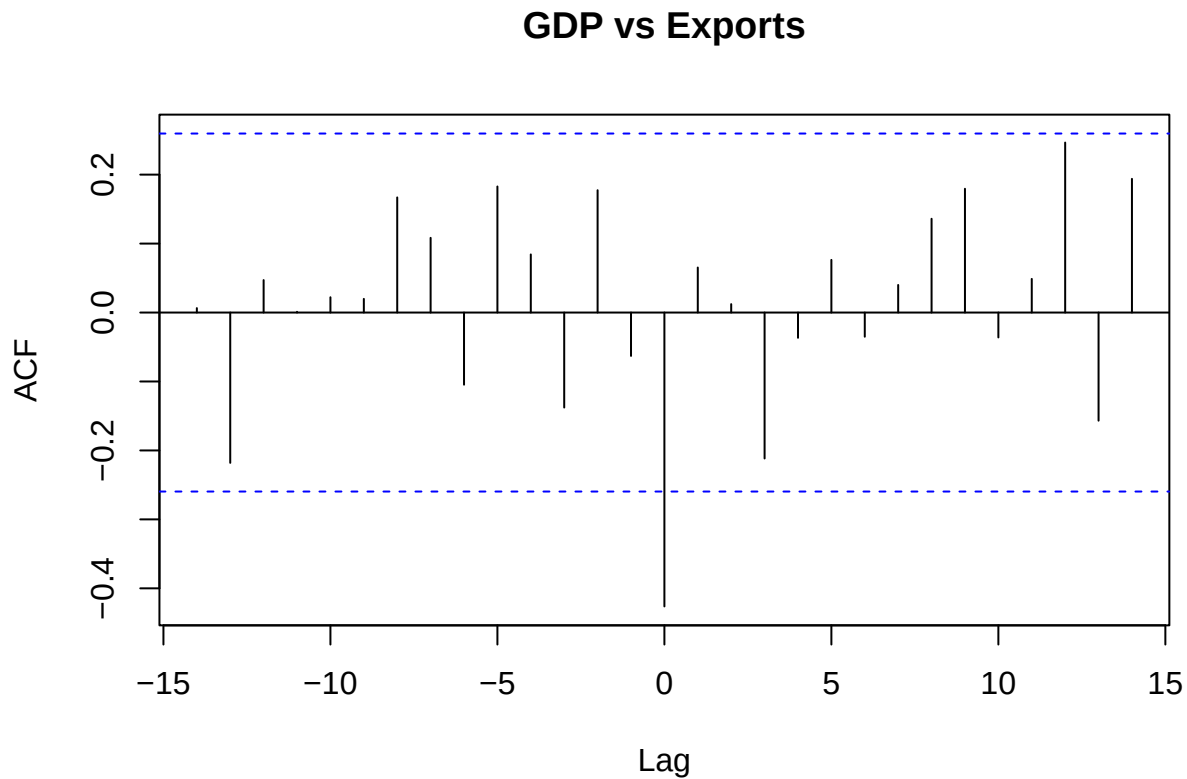
```
forecast_population$upper
```

```
## Time Series:
## Start = 2018
## End = 2019
## Frequency = 1
##      95%
## [1,] 4745005
## [2,] 4851823
```

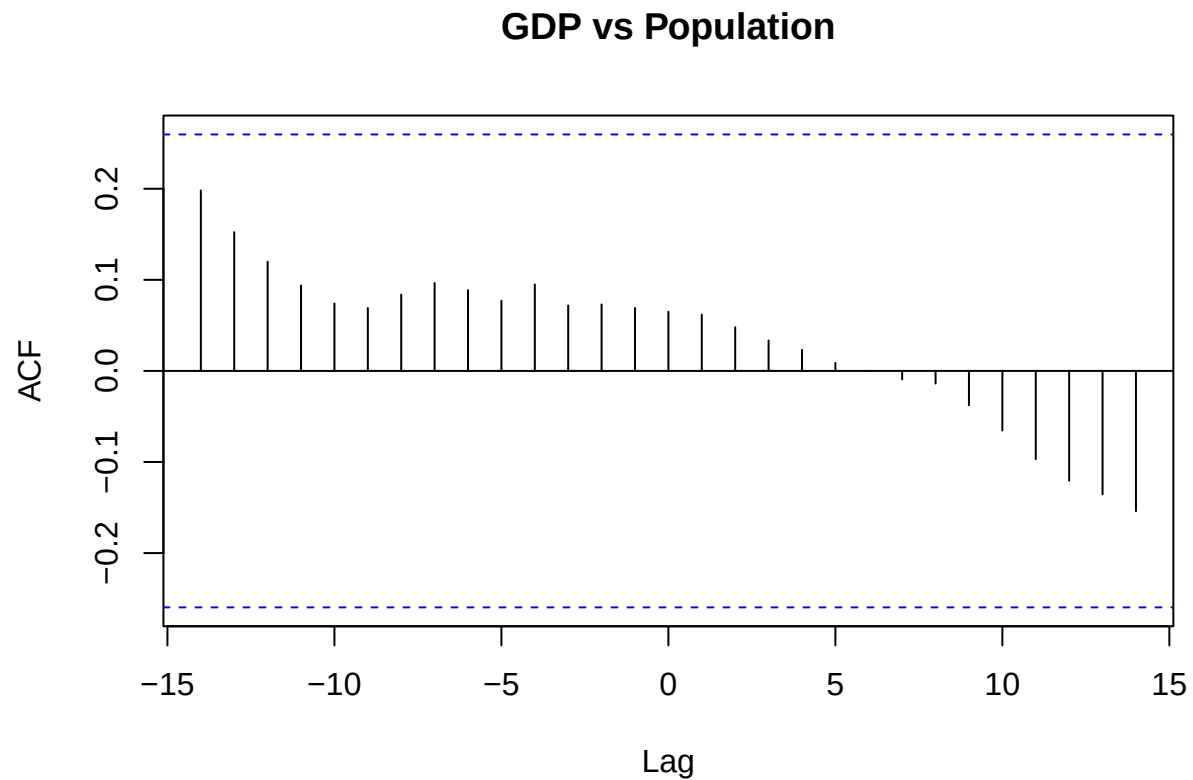
Comparative Analysis/Joint Analysis: Variables don't move on their own independently and usually move together, so we analyze the cross correlation. Earlier we only analyzed GDP, Exports, and Population separately, so we do joint analysis to see how these variables interact with each other. Cross-correlation is how you quantify relationships between economic variables over time.

Cross-Correlation:

```
#GDP vs. Exports
ccf(log_gdp_diff1, log_exports_diff1, main = "GDP vs Exports")
```

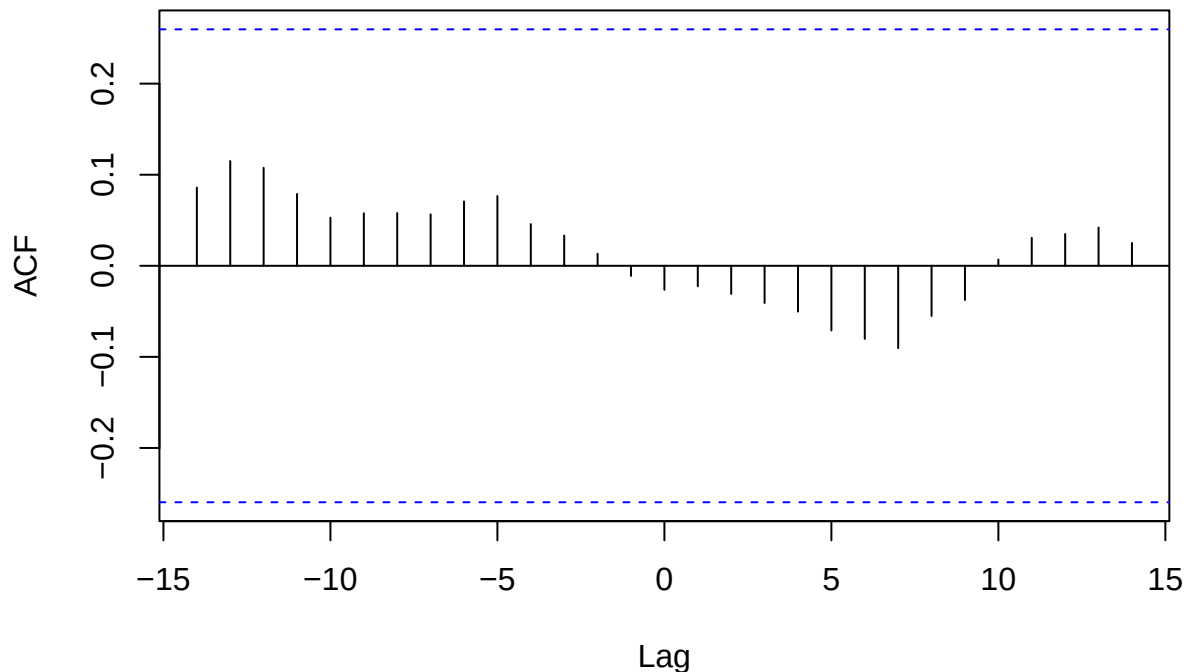


```
#GDP vs. Population  
ccf(log_gdp_diff1, log_population_diff1, main = "GDP vs Population")
```



```
#Exports vs. Population  
ccf(log_exports_diff1, log_population_diff1, main = "Exports vs. Population")
```

Exports vs. Population



GDP vs. Exports Analysis: Peak at lag 0 meaning that GDP and Exports move together, meaning that when exports increase, GDP tends to increase in the same year.

GDP vs. Population Analysis: Since there are no peaks above the dashed line, this means there is no statistically significant correlation at any lag, CHanges in population growth doesn't directly lead or lag GDP changes in this dataset. In the Central African Republic, GDP fluctuations seem to be driven by other factors (like exports) rather than population changes.

Exports vs. Population Analysis: There are no peaks above the dashed lines, meaning there is no statistically significant correlation at any lag. Changes in population growth do not appear to lead or lag changes in exports. Export performance seems independent of population growth over this period; other factors (like global demand, production capacity, or policy) likely drive exports more than demographic changes. Both GDP vs Population and Exports vs Population show no significant cross-correlation. This suggests that population growth is not a strong driver of either economic output or exports in the Central African Republic for this dataset.

Multivariate Analysis: We can capture dynamic interactions between variables. We can see how one variable affects others over time. Vector Autoregression (VAR) model.

```
library(vars)
```

```
## Warning: package 'vars' was built under R version 4.3.3
```

```
## Loading required package: MASS
```

```
## Loading required package: strucchange
```

```
## Warning: package 'strucchange' was built under R version 4.3.3
```

```
## Loading required package: zoo
```

```
##
```

```
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      as.Date, as.Date.numeric
```

```
## Loading required package: sandwich
```

```
## Loading required package: urca
```

```
## Warning: package 'urca' was built under R version 4.3.3
```

```
## Loading required package: lmtest
```

```
multi_ts <- cbind(log_gdp_diff2, log_exports_diff2, log_population_diff2)
colnames(multi_ts) <- c("GDP", "Exports", "Population")

lag_selection <- VARselect(multi_ts, lag.max = 10, type = "const")
lag_selection$selection
```

```
## AIC(n)  HQ(n)  SC(n) FPE(n)
##      10      2      2      2
```

Pick the lag suggested by BIC for parsimonious model. Therefore 3.

VAR Model:

```
var_model <- VAR(multi_ts, p = 2, type = "const")
summary(var_model)
```

```
##
```

```
## VAR Estimation Results:
```

```
## =====
```

```
## Endogenous variables: GDP, Exports, Population
```

```
## Deterministic variables: const
```

```
## Sample size: 54
```

```
## Log Likelihood: 429.742
```

```
## Roots of the characteristic polynomial:
```

```
## 0.9866 0.9866 0.8005 0.8005 0.5729 0.5729
```

```
## Call:
```

```
## VAR(y = multi_ts, p = 2, type = "const")
```

```
##
```

```
##
```

```
## Estimation results for equation GDP:
```

```
## =====
```

```
## GDP = GDP.l1 + Exports.l1 + Population.l1 + GDP.l2 + Exports.l2 + Population.l2 + const
```

```
##
```

```
##           Estimate Std. Error t value Pr(>|t|)
```

```

## GDP.l1      -0.687424    0.152822   -4.498 4.48e-05 ***
## Exports.l1   0.089142    0.136014    0.655  0.5154
## Population.l1 2.757448   26.330165    0.105  0.9170
## GDP.l2      -0.293814    0.156752   -1.874  0.0671 .
## Exports.l2   0.163783    0.132307    1.238  0.2219
## Population.l2 3.132830   27.944917    0.112  0.9112
## const       0.002805    0.020715    0.135  0.8929
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.1507 on 47 degrees of freedom
## Multiple R-Squared: 0.4026, Adjusted R-squared: 0.3263
## F-statistic: 5.279 on 6 and 47 DF, p-value: 0.0003155
##
##
## Estimation results for equation Exports:
## =====
## Exports = GDP.l1 + Exports.l1 + Population.l1 + GDP.l2 + Exports.l2 + Population.l2 + const
##
##              Estimate Std. Error t value Pr(>|t|)
## GDP.l1      -2.227e-01  1.393e-01  -1.599   0.117
## Exports.l1   -9.894e-01  1.240e-01  -7.982 2.71e-10 ***
## Population.l1 -2.310e+01  2.400e+01  -0.963   0.341
## GDP.l2       -7.156e-02  1.429e-01  -0.501   0.619
## Exports.l2   -6.775e-01  1.206e-01  -5.619 1.01e-06 ***
## Population.l2  2.181e+01  2.547e+01   0.856   0.396
## const       -8.303e-05  1.888e-02  -0.004   0.997
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.1373 on 47 degrees of freedom
## Multiple R-Squared: 0.6294, Adjusted R-squared: 0.5821
## F-statistic: 13.3 on 6 and 47 DF, p-value: 9.614e-09
##
##
## Estimation results for equation Population:
## =====
## Population = GDP.l1 + Exports.l1 + Population.l1 + GDP.l2 + Exports.l2 + Population.l2 + const
##
##              Estimate Std. Error t value Pr(>|t|)
## GDP.l1      -1.910e-04  3.360e-04  -0.568   0.572
## Exports.l1   1.572e-04  2.990e-04   0.526   0.602
## Population.l1 1.705e+00  5.789e-02  29.460 <2e-16 ***
## GDP.l2      -6.565e-05  3.446e-04  -0.190   0.850
## Exports.l2   1.743e-04  2.909e-04   0.599   0.552
## Population.l2 -9.745e-01  6.144e-02 -15.861 <2e-16 ***
## const       -4.213e-05  4.554e-05  -0.925   0.360
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.0003313 on 47 degrees of freedom

```

```
## Multiple R-Squared: 0.9636, Adjusted R-squared: 0.9589
## F-statistic: 207.2 on 6 and 47 DF, p-value: < 2.2e-16
##
##
##
## Covariance matrix of residuals:
##           GDP      Exports Population
## GDP      2.271e-02 -8.615e-03 -5.795e-06
## Exports  -8.615e-03  1.886e-02  9.116e-06
## Population -5.795e-06  9.116e-06  1.098e-07
##
## Correlation matrix of residuals:
##           GDP Exports Population
## GDP      1.0000 -0.4163   -0.1161
## Exports  -0.4163  1.0000    0.2004
## Population -0.1161  0.2004    1.0000
```

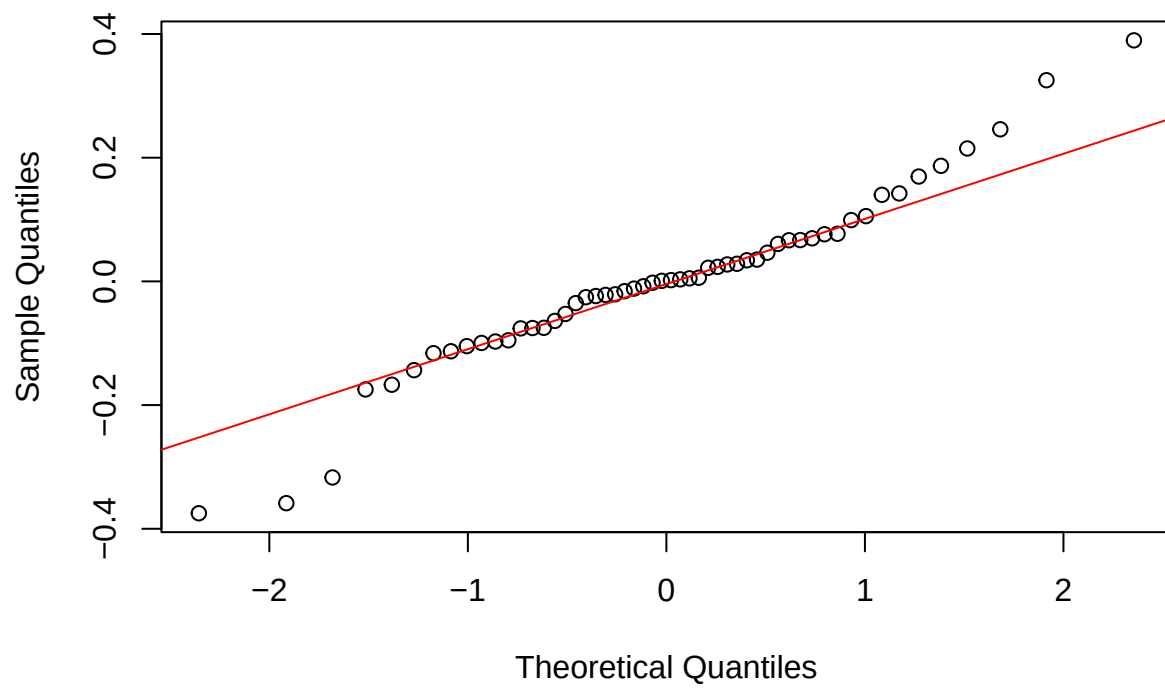
Checking Residual

```
serial.test(var_model, lags.pt = 10, type = "PT.asymptotic")
```

```
##
## Portmanteau Test (asymptotic)
##
## data: Residuals of VAR object var_model
## Chi-squared = 69.213, df = 72, p-value = 0.5712
```

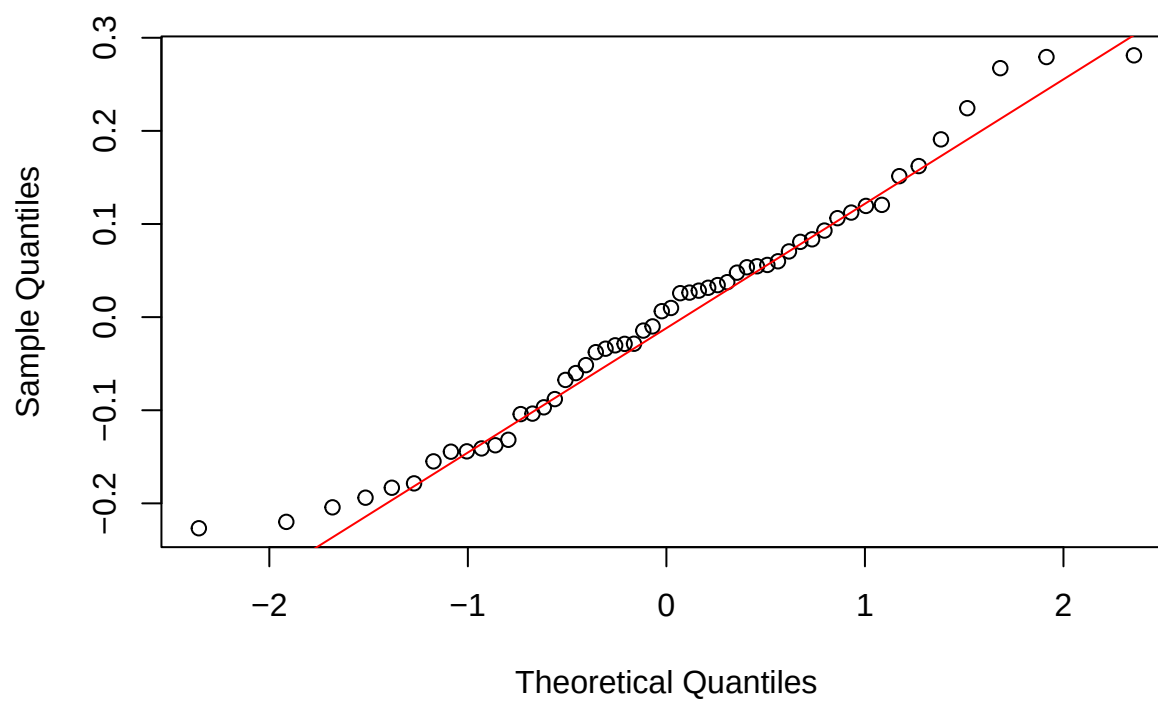
```
qqnorm(resid(var_model)[,1], main = "Normal Q-Q Plot: GDP Residuals")
qqline(resid(var_model)[,1], col = "red")
```

Normal Q-Q Plot: GDP Residuals



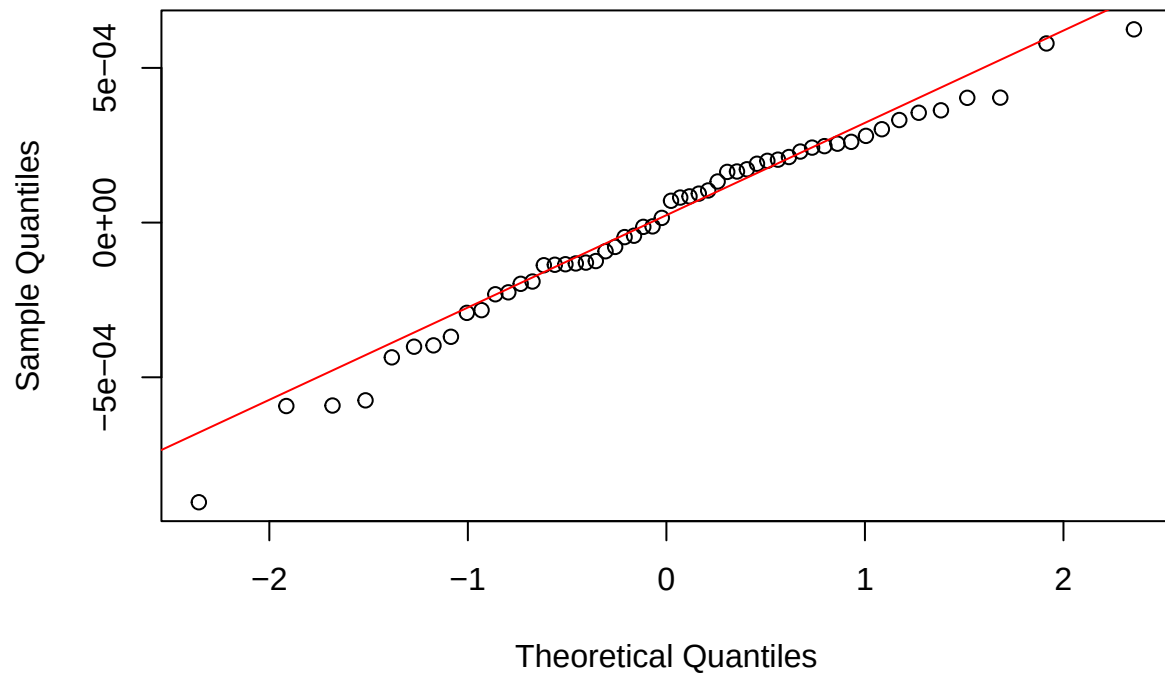
```
qqnorm(resid(var_model)[,2], main = "Normal Q-Q Plot: Exports Residuals")  
qqline(resid(var_model)[,2], col = "red")
```

Normal Q-Q Plot: Exports Residuals

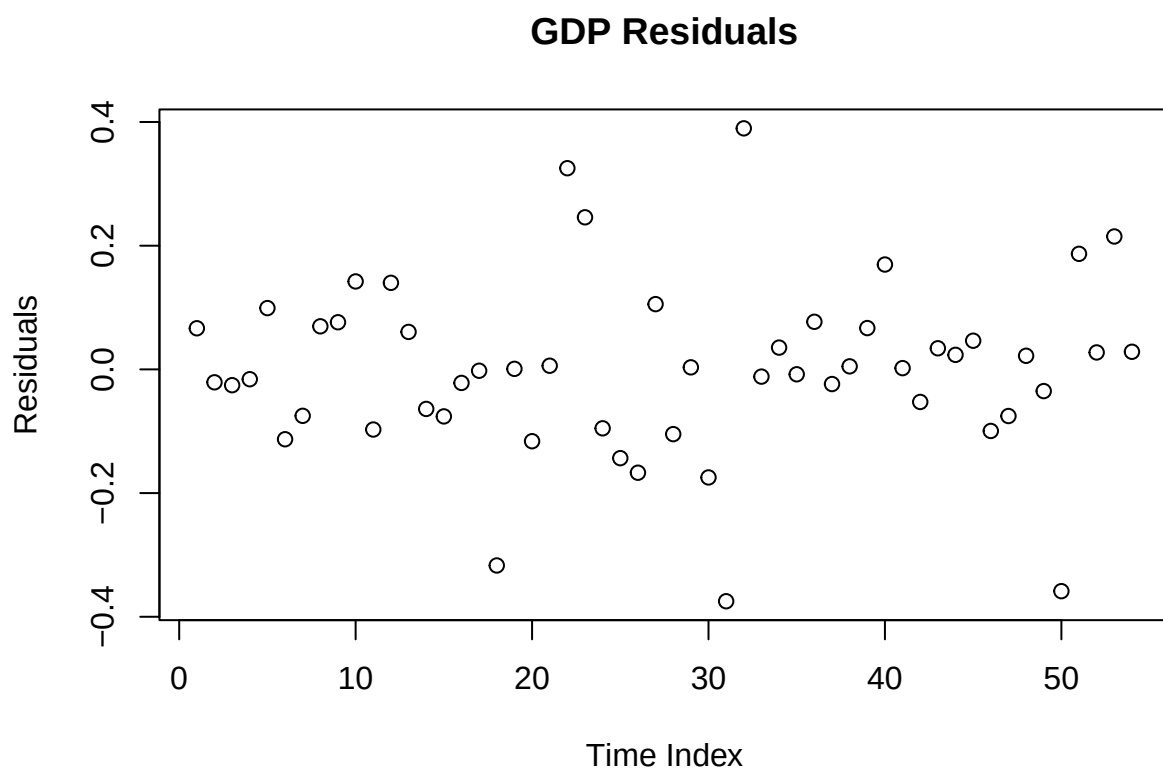


```
qqnorm(resid(var_model)[,3], main = "Normal Q-Q Plot: Population Residuals")  
qqline(resid(var_model)[,3], col = "red")
```

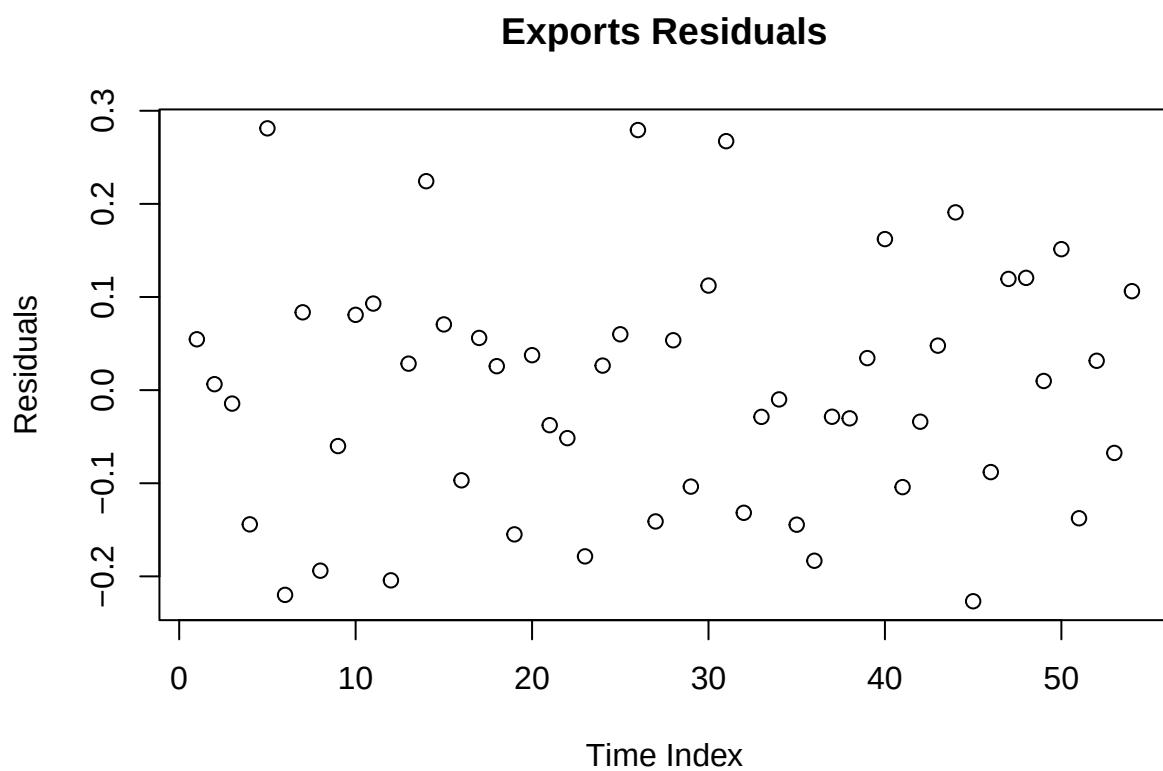
Normal Q-Q Plot: Population Residuals



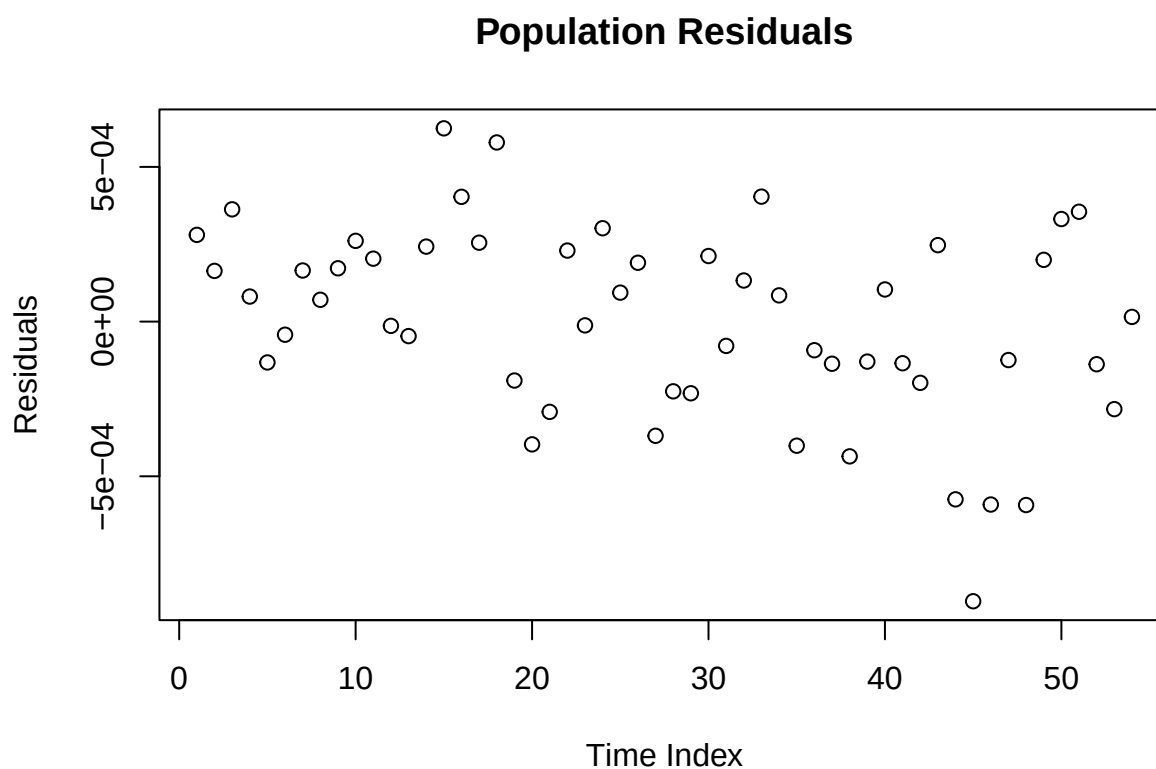
```
plot(resid(var_model)[,1],  
     main = "GDP Residuals",  
     ylab = "Residuals",  
     xlab = "Time Index")
```



```
plot(resid(var_model)[,2],  
     main = "Exports Residuals",  
     ylab = "Residuals",  
     xlab = "Time Index")
```



```
plot(resid(var_model)[,3],  
     main = "Population Residuals",  
     ylab = "Residuals",  
     xlab = "Time Index")
```



```
normality.test(var_model)
```

```
## $JB
##
## JB-Test (multivariate)
##
## data: Residuals of VAR object var_model
## Chi-squared = 7.362, df = 6, p-value = 0.2887
##
##
## $Skewness
##
## Skewness only (multivariate)
##
## data: Residuals of VAR object var_model
## Chi-squared = 2.9838, df = 3, p-value = 0.3941
##
##
## $Kurtosis
##
## Kurtosis only (multivariate)
##
## data: Residuals of VAR object var_model
## Chi-squared = 4.3783, df = 3, p-value = 0.2234
```

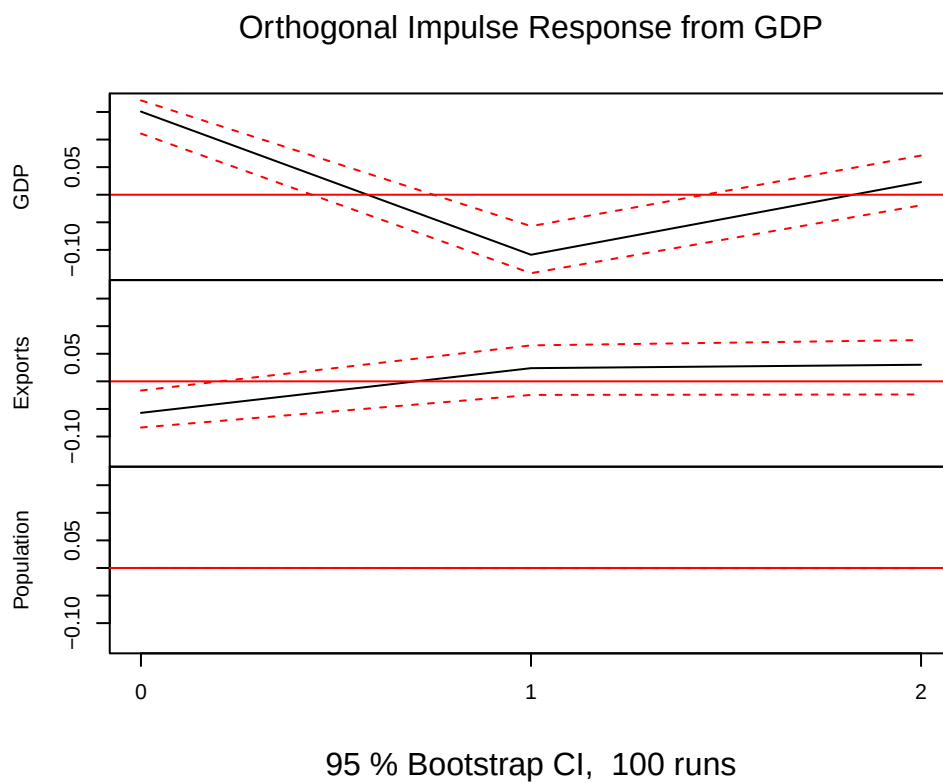
```
arch.test(var_model)
```

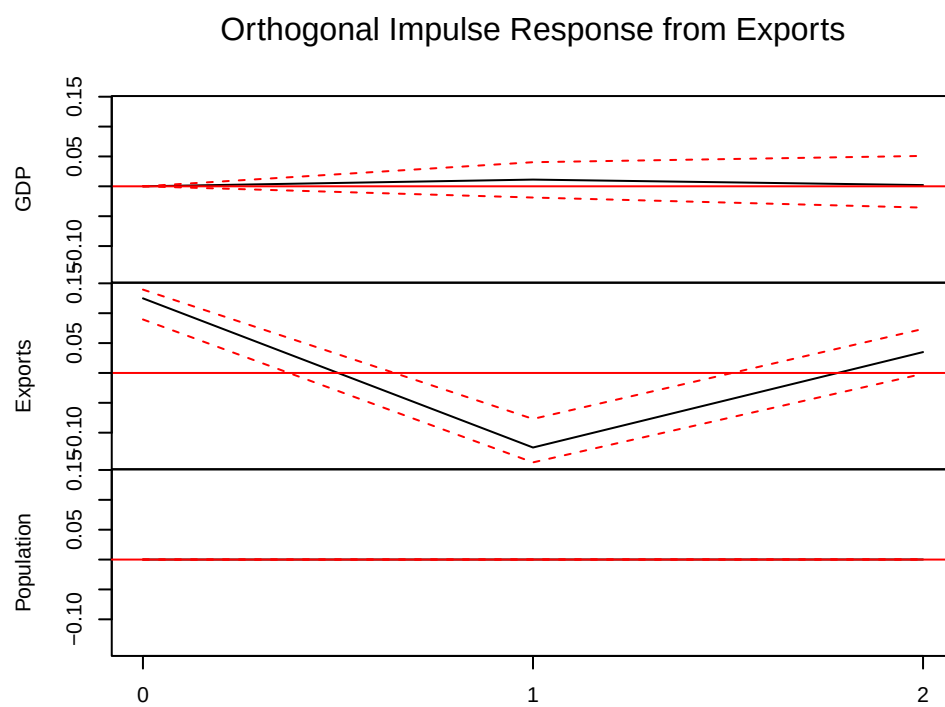
```
##  
## ARCH (multivariate)  
##  
## data: Residuals of VAR object var_model  
## Chi-squared = 174.58, df = 180, p-value = 0.5999
```

Greater than 0.05, therefore no residual autocorrelation.

Impulse Response Functions:

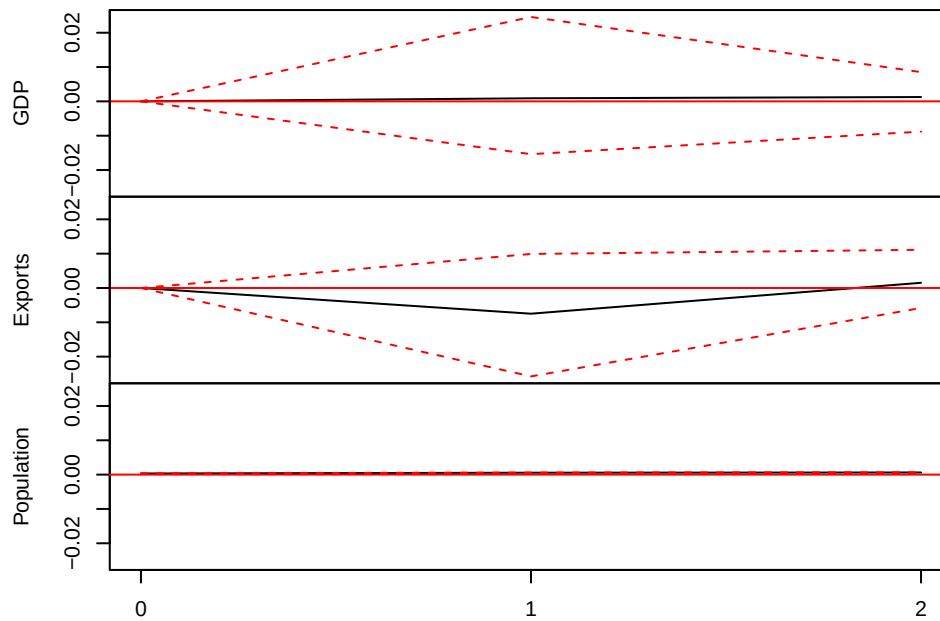
```
irf_all <- irf(var_model, impulse = NULL, response = NULL, n.ahead = 2  
              , boot = TRUE)  
  
plot(irf_all)
```





95 % Bootstrap CI, 100 runs

Orthogonal Impulse Response from Population



95 % Bootstrap CI, 100 runs

Forecast:

```
forecast_var <- predict(var_model, n.ahead = 2, ci = 0.95)
```

```
forecast_var$fcst
```

```
## $GDP
```

```
##          fcst      lower      upper      CI
```

```
## [1,] -0.02431654 -0.3196656 0.2710326 0.2953491
```

```
## [2,] 0.02947463 -0.3354757 0.3944249 0.3649503
```

```
##
```

```
## $Exports
```

```
##          fcst      lower      upper      CI
```

```
## [1,] -0.008596438 -0.2777543 0.2605614 0.2691578
```

```
## [2,] 0.060579537 -0.3064308 0.4275899 0.3670104
```

```
##
```

```
## $Population
```

```
##          fcst      lower      upper      CI
```

```
## [1,] 1.861926e-03 0.001212605 0.002511246 0.0006493206
```

```
## [2,] -9.880374e-05 -0.001398033 0.001200426 0.0012992295
```

```
mean_back <- exp(forecast_var$fcst$GDP[, "fcst"])
```

```
mean_back
```

```
## [1] 0.9759767 1.0299133
```

```
lower_back <- exp(forecast_var$fcst$GDP[, "lower"])
lower_back
```

```
## [1] 0.7263919 0.7149979
```

```
upper_back <- exp(forecast_var$fcst$GDP[, "upper"])
upper_back
```

```
## [1] 1.311318 1.483531
```

```
mean_back_exports <- exp(forecast_var$fcst$Exports[, "fcst"])
mean_back_exports
```

```
## [1] 0.9914404 1.0624521
```

```
lower_back_exports <- exp(forecast_var$fcst$Exports[, "lower"])
lower_back_exports
```

```
## [1] 0.7574829 0.7360694
```

```
upper_back_exports <- exp(forecast_var$fcst$Exports[, "upper"])
upper_back_exports
```

```
## [1] 1.297658 1.533557
```

```
mean_back_pop <- exp(forecast_var$fcst$Population[, "fcst"])
mean_back_pop
```

```
## [1] 1.0018637 0.9999012
```

```
lower_back_pop <- exp(forecast_var$fcst$Population[, "lower"])
lower_back_pop
```

```
## [1] 1.0012133 0.9986029
```

```
upper_back_pop <- exp(forecast_var$fcst$Population[, "upper"])
upper_back_pop
```

```
## [1] 1.002514 1.001201
```